

Matching Forward Dynamics to Kinematic Data

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Publication and Evidence Status

Local research manuscript, 2026-10-05. This book has been assembled for AffineDrift publication. It has not yet been deployed to affinedrift.com, rendered through the repository's Quarto pipeline, or independently peer reviewed. External link validation remains incomplete because GitHub returned an API quota error and the repository requires stopping network activity. The reference and verification ledger records these limits.

The source review pins UpstreamDrift to revision `d7c7d07add0d7407a5a3376d5fe4a97aee9112a0`. It separates implemented algorithms, stored test and experiment evidence, historical failures, and proposed improvements. Native engines and empirical subject measurements were not executed in this review. A proposed adaptive model is a research design, not a delivered matcher.

The Central Question

Can a declared mechanical model, admissible actuation, and one initial state reproduce recorded motion while satisfying physical constraints? A convincing answer requires more than a close animation: observation semantics, numerical consistency, uninterrupted replay, uncertainty, and independently tested prediction all matter.

The book develops that question from measurement and inverse dynamics through shooting, collocation, state estimation, differentiable simulation, muscle-driven tracking, learned proposals, and adaptive model selection. Its focus is improving model-constrained matching without confusing fitted shape, physical degrees of freedom, latent memory, and control representation.

How to Read the Book

Start with the problem formulation and the two UpstreamDrift review chapters for the shortest route to current capabilities and gaps. Read the classical methods before the adaptive chapters if trajectory estimation is unfamiliar. The final design and synthesis chapters combine those foundations into a proposed implementation and research program.

Each chapter includes assumptions, failure modes, worked arguments, and exercises with solution sketches. Mathematical examples are synthetic or derived. Historical numerical values are attributed to checked-in upstream receipts and remain conditional on their recorded model and run.

Foundations and the Existing Program

1. The Forward Dynamics Matching Problem
2. The UpstreamDrift Method Inventory
3. Lessons From Historical Fits
4. Measurements and Preprocessing
5. Inverse Kinematics and Inverse Dynamics

Classical Matching and Estimation

6. Shooting Methods
7. Direct Collocation
8. Control Parameterizations and Continuation
9. Constraints, Contact, and Hybrid Events
10. Differentiable Simulation and Gradient Checking
11. Muscle-Driven Matching
12. Filtering, Smoothing, and Data Assimilation
13. Uncertainty and Validation
14. Bayesian Parameter and Model Inference

Learning and Adaptive Dimensions

15. Identifiability Before Dimension Selection
16. Physical Parameters and Anatomical Dimension
17. Sparse Discovery and Model Order
18. An Adaptive Model-Constrained Matching Design
19. Research Synthesis and Priority Gaps
20. Latent Learned Dynamics
21. Experimental Design and Model Comparison
22. Learning-Based Tracking and Iterative Control

The appendix, Forward Surrogates and Multimodal Inverse Proposals, develops learned acceleration and native reranking in detail.

Engineering and Publication

23. Software Architecture and Test Contracts
24. References and Verification Ledger
25. Benchmarks, Reproducibility, and Maintenance

What the Adaptive Design Adds

The proposed learner adapts a restricted physical geometry before interpreting dynamics, profiles out compensating controls and calibration variables, and compares registered structures on independent trials. It reports data information separately from prior information and retains compatible alternatives when the observations cannot distinguish them.

No new production code is introduced in this publication change. The engineering chapter specifies tests-first implementation, design by contract, the Law of Demeter, and shared logic for future code. The source audit does not certify existing upstream code as uniformly compliant with those principles.

Related Articles

- Degrees of Freedom and Dimensionality—Distinguishes configuration, tangent, and task dimensions before interpreting a learned model.
- UpstreamDrift Educational Integration—Connects the executable engine platform with the textbook library.
- The Model Ladder—Organizes mechanical assumptions and the predictions each model can represent.
- Books and Textbooks—Returns to AffineDrift’s long-form library.

The Forward Dynamics Matching Problem

What the Match Must Establish

Matching a forward dynamics model to recorded motion asks whether a declared mechanical system, initial state, and admissible input can reproduce specified observations. This question differs from asking whether a skeleton can pass through measured points, or whether a controller can keep a simulation near a reference. Each result answers a different inference problem, even when the displayed trajectories look similar.

This book reviews methods for fitting rigid-body, constrained multibody, muscle-driven, compliant, and learned dynamics to kinematic observations. Its application is golf motion, but its equations also apply to gait, robotic manipulation, and other articulated movements. The mathematical arguments below are derivations under declared assumptions; they are not measurements of golfers. Published methods provide external context (Kelly 2017; Seth et al. 2018; Dembia et al. 2020; Uchida and Delp 2021).

A match should identify its observation channels, fitting interval, physical constraints, numerical tolerances, and replay protocol. It should also state which parameters were fixed, estimated, or selected by a prior. A low positional error without this information cannot establish dynamically feasible motion, unique actuation, or anatomically correct internal states.

Consider three outputs from the same target recording. In the first, inverse kinematics independently chooses joint angles at each frame. In the second, a feedback controller follows those angles while applying whatever generalized forces its actuator limits allow. In the third, an optimizer chooses a bounded control trajectory and then integrates it from one initial state without resetting the simulation. Only the third directly demonstrates the declared open-loop forward reproduction, although the other two can provide valuable initialization and diagnostics.

An open-loop match is still conditional on the model. If the model uses ideal torque actuators, its controls are net generalized inputs rather than identified muscle activations. If it uses prescribed ground motion, the work of that moving boundary belongs in the accounting. If it uses a rigid shaft, its apparent wrist motion may compensate for shaft flexibility that the equations cannot represent.

A Common Mathematical Language

Let q denote configuration coordinates and v generalized velocities. Configuration storage dimension need not equal tangent velocity dimension, particularly when orientations use unit quaternions. Write the kinematic evolution as

$$\dot{q} = N(q)v.$$

A quaternion model must use its declared configuration update or retraction. Adding a tangent displacement directly to configuration storage is generally not a valid integration rule. Matching packages must therefore distinguish the coordinate representation from the physical degrees of freedom.

A constrained mechanical model may satisfy

$$M(q; \theta)\dot{v} + b(q, v; \theta) = B(q; \theta)u + J(q; \theta)^T \lambda + f_{\text{ext}}(q, v, t; \theta), \quad c(q; \theta) = 0.$$

Here M is the mass matrix, b includes the declared velocity-dependent, gravity, and passive terms, B maps control channels to generalized forces, and λ collects reactions. Parameters θ may include segment geometry, inertia, damping, contact properties, and sensor calibration. Muscle activation or elastic deformation requires additional states rather than being hidden inside an unexplained torque curve.

For smooth bilateral constraints, differentiating $c(q) = 0$ yields a velocity condition and an acceleration condition. If J is the constraint velocity Jacobian, these take the schematic forms

$$Jv = 0, \quad J\dot{v} + \dot{J}v = 0.$$

Solving the mechanical and acceleration equations together gives the saddle-point system

$$\begin{bmatrix} M & -J^T \\ J & 0 \end{bmatrix} \begin{bmatrix} \dot{v} \\ \lambda \end{bmatrix} = \begin{bmatrix} Bu + f_{\text{ext}} - b \\ -Jv \end{bmatrix}.$$

The sign convention for reactions must match the engine adapter. Rank-deficient constraint rows require an explicit independent representation or a documented regularization strategy. A solver that silently changes the rows changes the mechanical problem.

Define the complete state $x = (q, v, a, z)$, where a denotes optional activation states and z optional compliant or latent states. Forward integration produces

$$x(t) = \Phi(t, t_0; x_0, u, \theta, k),$$

where k identifies a model structure: a joint topology, contact formulation, or selected state dimension. Keeping k explicit prevents a smooth parameter optimizer from being mistaken for a method that changes topology.

The Observation Model Is Part of the Dynamics Match

A marker recording observes transformed body attachment points, not generalized coordinates directly. For marker j attached to body $b(j)$, a common rigid attachment model is

$$y_{ij} = p_{b(j)}(q_i; \theta) + R_{b(j)}(q_i; \theta)s_j + \epsilon_{ij},$$

where s_j is the attachment offset in the body frame. Camera observations additionally require projection, lens calibration, and time alignment. Soft-tissue motion means that a constant rigid offset can be an imperfect observation model even if the dynamics are correct.

Let ψ collect observation nuisance parameters, including attachment offsets, camera poses, and synchronization. A general recorded output satisfies

$$y_i = h(x(t_i), \theta, \psi) + \epsilon_i.$$

Estimating ψ jointly with mechanical parameters can reduce systematic measurement error. It can also create confounding: a camera scale and a segment length may change together without changing the projected image. The fit must report these ambiguities instead of assigning them biological meaning.

Orientation errors require geometry consistent with the measurement. For a measured rotation R_i^{obs} and prediction R_i , use a local rotation-vector residual such as

$$r_i^R = \text{Log}((R_i^{\text{obs}})^T R_i).$$

Quaternion sign equivalence must be handled before interpreting component errors. A raw quaternion difference can penalize two representations of the same rotation. Near rotation-log branch boundaries, local derivatives and residual interpretation need additional care.

Missing markers define a selection operator S_i . The residual is

$$r_i = S_i(h(x(t_i), \theta, \psi) - y_i).$$

A missing channel contributes no information; replacing it with zero invents an observation. Interpolating missing measurements also creates correlated uncertainty, so the interpolated frames should not automatically receive the same statistical weight as independently measured samples.

The Estimation Objective

For known covariance Σ_i on observed channels, define W_i so that $W_i^T W_i = \Sigma_i^{-1}$. A likelihood-based trajectory loss is

$$J_{\text{obs}} = \frac{1}{2} \sum_i \|W_i r_i\|^2.$$

Positions, orientations, forces, and pixel coordinates have different units. Whitening or explicitly declared scales prevent the largest numerical unit from dominating the fit. Engineering weights may intentionally emphasize delivery, but they should then be described as preferences rather than measurement covariance.

A broad constrained objective is

$$\min_{x_0, u, \theta, \psi, k} J_{\text{obs}} + \alpha J_{\text{effort}} + \beta J_{\text{roughness}} + J_{\text{prior}} + \gamma \mathcal{C}(k)$$

subject to dynamics, contact laws, admissible geometry, control limits, and path constraints. The discrete structure k makes this a family of estimation problems rather than one ordinary smooth nonlinear program. Comparing structures requires a separate selection protocol.

The effort term needs a declared interpretation. An integral of squared torque measures one engineering regularizer, not automatically metabolic energy or neural effort. A muscle

activation objective requires a recruitment model and does not uniquely identify physiology from kinematics.

Control roughness can penalize $\int \|\dot{u}\|^2 dt$, while a low-dimensional basis can regularize the trajectory implicitly. Those mechanisms impose different priors. A smooth polynomial input and a piecewise-constant input with a roughness penalty can achieve similar marker errors while predicting different reaction loads.

Robust losses can reduce the influence of isolated observation outliers. They do not repair a wrong marker identity, a timestamp error, or a structurally inadequate model. Correlated residuals and repeated failure around a particular event should trigger model or acquisition review before another loss function is introduced.

Four Meanings of Learned Dimensions

Anatomical dimensions are continuous physical quantities such as link lengths, joint-center locations, and marker offsets. Their estimation keeps the topology fixed while changing geometry. Inertial parameters must change according to an explicit physical scaling law or an independent identification model.

Mechanical degrees of freedom describe independent allowable motions after constraints. Adding a scapular coordinate, changing a wrist joint, or releasing a bilateral constraint changes the admissible configuration space. This operation is a structural hypothesis that needs a new coordinate map and model fingerprint.

Latent state dimension specifies the size of an unobserved state used to capture memory or unresolved dynamics. A latent coordinate may improve prediction without corresponding to any anatomical joint. Latent coordinates can also rotate or rescale without changing their decoded observations.

Representation dimension describes the number of control coefficients, basis functions, retained modes, or sparse dictionary terms. Increasing the polynomial order changes actuation flexibility without necessarily changing the mechanical system. A method should not describe that increase as discovering an additional human degree of freedom.

These dimensions interact. A longer arm can reduce marker error that previously required extra shoulder motion. More flexible controls can compensate for wrong inertia. More latent states can absorb synchronization mistakes. Consequently, dimension adaptation should follow calibration and observability analysis rather than replace them.

A Derived Ambiguity Example

Consider a single-link planar pendulum with unknown inertia I , damping d , gravitational moment coefficient G , and unrestricted torque $u(t)$:

$$I\ddot{q} + d\dot{q} + G \sin q = u(t).$$

Assume the complete trajectory $q(t)$ is known exactly. For every admissible parameter triple, define the torque by the left-hand side. Every such pair of parameters and torque reproduces the same observed motion. Perfect kinematics therefore does not identify inertia or damping when actuation is unrestricted.

If torque is measured independently, the equation becomes a regression in the parameters:

$$u(t) = [\ddot{q} \quad \dot{q} \quad \sin q] \begin{bmatrix} I \\ d \\ G \end{bmatrix}.$$

Independent columns across the experiment can identify the three coefficients locally in this simplified model. Near-static data supplies little acceleration information, while a narrow motion interval can make columns nearly dependent.

A torque prior does not create an independent measurement. Minimizing squared torque chooses one member of the family according to the objective. It may be a useful candidate, but its inertia estimate depends on the chosen regularization and bounds.

A free-decay experiment with known zero input provides stronger information about coefficient ratios. Even then, multiplying I , d , and G by a common factor leaves the zero-input equation unchanged. Absolute load scale requires another measurement or a fixed physical parameter.

What Counts as a Result

A fitting result should pass three logically distinct screens. Observation agreement evaluates the measured channels. Mechanical feasibility evaluates dynamics, constraints, contacts, controls, and admissible parameters. Predictive evaluation tests observations that did not select the model or tune its weights.

The numerical optimizer contributes another status: termination. A solver can terminate successfully at a poor or physically rejected fit. It can also stop at its budget with a useful but uncertified candidate. Reports should preserve these statuses separately.

A fitting interval is not a validation interval. Holding out isolated adjacent frames provides weak protection against leakage because trajectories and smoothing create strong temporal dependence. Holding out complete trials, subjects, sensor channels, or motion phases asks a more informative generalization question.

The strongest computational replay for an open-loop fit integrates the accepted controls from one declared initial state without resets. Multiple-shooting nodes are optimization variables, not permission to teleport the accepted model through the recording. Contact mode changes and prescribed boundaries must still follow the declared plant.

Exercises and Solution Sketches

1. **Parameter and Input Ambiguity.** Show why unknown torque prevents unique pendulum inertia identification from perfect angle observations. **Solution:** choose any admissible inertia and evaluate the torque needed by the known angle curve; the observation remains identical.
2. **Observation Units.** Explain why summing squared marker distances and squared orientation angles without scales has no invariant statistical meaning. **Solution:** changing position units changes one term relative to the other; a covariance or declared engineering scale is required.
3. **Dimension Interpretation.** A fitting program raises control polynomial degree and improves marker error. What dimension has changed? **Solution:** control representation dimension; mechanical state dimension and anatomical geometry may remain fixed.
4. **Replay Contract.** A segmented trajectory has small marker errors but nonzero boundary defects. What can be concluded? **Solution:** its independently initialized pieces approximate observations; uninterrupted dynamic reproduction remains unestablished.

The UpstreamDrift Method Inventory

Evidence snapshot: UpstreamDrift commit `d7c7d07add0d7407a5a3376d5fe4a97aee9112a0`. This chapter is a source and evidence inventory, not a claim that every method has been trained or physically qualified. Links below pin code and documentation to that immutable commit. No network retrieval or test execution was performed for this chapter.

The phrase “match dynamics to kinematic data” hides several inverse problems with different unknowns. A motion-capture sequence may constrain joint angles, marker positions, club pose, or only a projected silhouette. A forward model maps a specified initial state, parameters, controls, contacts and environment to a trajectory. Matching asks which unknowns make that trajectory agree with measurements. Depending on what is unknown, the problem may be trajectory tracking, control reconstruction, parameter identification, shape calibration, or a combination. A solver can return a low residual while choosing a physically wrong combination of these variables. This chapter inventories the actual UpstreamDrift methods by these distinctions.

1. Classical Parameter Fitting Through a Forward Simulator

The foundational Simscape path parameterizes each actuated joint with a seven-coefficient polynomial and repeatedly executes the native model. The orchestration entry point is `fit_swing_full_pipeline`. It resolves a target, applies options, dispatches to a solver, renders diagnostics and writes a fit report. The forward oracle is `simulate_with_coefficients`. Cost terms are shared instead of embedded in each optimizer; see `COST_FUNCTION_SPEC.md`.

The optimizer choices span local smooth optimization (`fit_swing_fmincon`), randomized local starts (`fit_swing_multistart`), global derivative-free exploration (`fit_swing_surrogateopt`) and global-to-local hybrid search (`fit_swing_hybrid`). Each is executable under `option1_direct_optimization`. Synthetic recovery tests are present, including `fmincon` tests and `multistart` tests. Synthetic observability and optimizer wiring tests are valuable but do not answer whether the SLX model, its marker map, mass properties or ground contacts represent a recorded person.

The coefficient vector is convenient because it makes a simulator call a deterministic function of a compact vector. It is also numerically fragile: powers of physical seconds have very different scales, high-order monomial coefficients can be strongly correlated, and a fixed coefficient box does not imply a bound on the actual torque curve unless time interval and basis are fixed. The tour matching guide records the native polynomial $A \cdot t^6 + \dots + G$, while shared Python uses ascending powers; conversion must happen at the engine boundary. A more stable parameterization uses normalized time and a basis with better conditioning, while preserving a

tested native conversion and interval torque bound.

Cross-engine implementations include OpenSim's `fit_swing_opensim`, Drake's bounded SLSQP fitter and a Drake AutoDiff pathway. OpenSim's fitter consumes the shared cost; Drake includes both float simulation and an analytic AutoDiff integration route. The code and test paths are in the audit report. Their default coefficient bounds and regularizers are not yet identical: OpenSim uses a relatively narrow symmetric bound and a coefficient-norm default when torque logs are absent, while Drake/Simscape document wider per-power envelopes, and the autodiff path has its own degree-scaled bound convention. These are scientific differences because optimization outcome depends on parameter scale and regularization. A cross-engine parity result is meaningful only after model identity, coefficient order, time normalization, units, initial state, outputs and objective are identical.

2. Kinematic Fitting and Constrained Inverse Kinematics

Kinematic fitting solves pose variables against marker or body targets, often using forward kinematics, without asking whether a torque-driven model would produce those poses. The Simscape shared tools include skeleton FK and a pose seed fitter, and the OpenSim pathway includes document IK and address-pose fitting. These tools are appropriate for model registration, initial-state construction and geometric residual analysis. They are not substitutes for dynamic matching.

The more recent club-only package distinguishes measured, derived and unobserved components through an observation contract, carries time/resampling provenance and supports sign-invariant $SO(3)$ error. A measured club orientation and a derived orientation are not equivalent evidence; an unobserved twist should remain masked rather than set to zero. Constrained IK adds backend capability declarations and posture branches. Starting guesses can be retrieved or seeded, then body candidates can be generated under plausibility priors. The package explicitly retains ambiguity because club motion alone cannot uniquely determine internal whole-body posture. This is methodologically sound: a single inverse solution is not "the golfer's motion" absent additional observations and priors.

The club-only path adds null-space analysis for grip closure and alternative offsets. These are useful tools for understanding the family of kinematic solutions that produce a similar end-effector constraint. Their role is proposal generation; candidates still need observation scoring, plausibility checks and, for a dynamics claim, torque/control recovery plus an independent forward trajectory. The `club_only` turnover says its software-contract work packages completed while the epic remained open on native Fit/G1 gates. This nuance should be preserved in any textbook description.

3. Direct Control Recovery and Tracking

There are two broad approaches to dynamic matching. In shooting or single shooting, optimize a control sequence and integrate from one initial state; gradients or finite differences pass through the simulator. In direct collocation or multiple shooting, optimize state and control nodes together, enforce defects between nodes, and then validate with a separate integration. Direct transcription can handle constraints and long horizons more robustly, but its optimizer nodes are not by themselves a realized motion: the cold replay is authoritative.

OpenSim's full-swing tracking module defines explicit event, trajectory, tracking-receipt, replay-receipt and qualification-result records. It validates model hashes, capture completeness, control naming, grip closure, continuity and coordinate limits. Moco tracking can produce a dynamically constrained tracking solution; a separate open-loop replay checks

whether the proposed control from the actual initial state reproduces the path. A marker-scoring lane without model marker predictions is explicitly documented as incapable of qualifying a fit. This design prevents a solver objective or status flag from becoming an unsupported matching claim.

Simscape tour-average work provides a concrete direct-node continuation workflow. It first calibrates geometry and initial state, then optimizes controls over progressively longer capture prefixes. Prefix extension must preserve earlier samples and reintegrate from the same initial state; state resets at windows would leak measured trajectory information into the simulation. A candidate can be stitched from windows only if continuity and the actual native control path are verified. The native qualification protocol specifies independent replay at tighter tolerances and reports per-marker/per-phase errors, actuator limits and missing force/penetration evidence. The 2026-10-05 snapshot includes a two-window 0.85-second uninterrupted replay that passes four gates but fails the terminal full-marker RMS gate; it is an informative partial success, not whole-capture acceptance.

The same distinction applies to feedback-assisted tracking. A controller that continuously corrects state error may track well even if its underlying model and open-loop torques are poor. Report tracking-controller mode and independent replay separately. The Simscape Human experiments document fixed supported holds with feedback and prescribed neck support, yet explicitly leave full-swing forward dynamics unqualified. The reference contains failed force/pose gates as well as successes, preserving evidence instead of selecting only visually appealing frames.

4. Learned Forward and Inverse Models

The current shared Python code has a real neural method portfolio. The trajectory-level forward model in `surrogate/model.py` maps torque coefficients to butt/grip and clubhead paths, orientation and auxiliary joints. FiLM modulation conditions a temporal backbone on the control vector. Its trainer, `train_surrogate`, is implemented. `fit_swing_via_surrogate` differentiates through the learned model to propose controls. Because a surrogate can be exploited in regions where its gradients do not represent native mechanics, validation includes trust-region, directional derivative and contact-boundary checks, followed by native evaluation.

Separate compact-corpus and per-step models handle different data organizations and tasks. Compact sequence models predict a whole trajectory and have their own target/coefficient normalization. Per-step surrogate training predicts state dynamics from structured inputs and the sequence optimizer adds orientation, work and smoothness terms. They should not be collapsed into one “neural matcher” in a review: target representation, supervision and native use differ.

Inverse models estimate controls from a trajectory. The full-trial CVAE uses a conditional prior and posterior with coefficient decoder and bounded outputs. It acknowledges the many-to-one mapping by sampling candidate coefficient vectors. The deterministic inverse regressor is faster but tends toward conditional mean solutions when labels are multi-modal; historical source notes a plateau near mean predictions, a claim NM-00 quarantines pending original curves. A per-timestep inverse dynamics network maps $(q, \dot{q}, \ddot{q})$ to generalized torque. That is a local force inference conditioned on state derivatives and a given model/labeling scheme; it is not automatically a whole-trajectory control solution or independent causal system identification. Masked proposals address incomplete observations and preserve proposal diversity.

The purpose of all these networks should be to amortize candidate generation or provide gradients, not to replace physics. Strong deployment architecture is “learned proposal $\rightarrow$ bounded native polish $\rightarrow$ independent replay $\rightarrow$ physical acceptance,” with classical fallback under fixed budget. The current NM-07 receipt is diagnostic and specifically records that some earlier

ablations used hard-coded literals; NM-10 speed estimates are diagnostic/unmeasured. Do not repeat their numerical values as measured results.

5. Model Dimensions and Identification

The repo contains anthropometric scaling, segment scaling and dimension configuration structures. These can build a subject-scaled model or choose among registered topologies. That is not the same as learning dimensions from motion. A dimension-learning method jointly varies geometry with control or torque and may let one compensate for another: a longer segment can mimic different joint motion; an offset can absorb camera calibration error; mass, inertia, actuator gains and torque scale can trade off in the dynamics. If all are free, a lower residual may indicate compensation rather than improved anatomy.

A rigorous adaptive model should use a small structured dimension vector with informative anthropometric priors and hard mechanical constraints. Fit camera registration and marker attachments first, then dimensions against calibration observations, then hold geometry fixed for dynamic control fitting. If alternating refinement is needed, constrain each block, log sensitivity matrices, and report which directions are identifiable. Include positive lengths/inertias, mass consistency, topology constraints and model complexity penalties. A parameter with a flat profile likelihood or small Jacobian singular value is prior-dominated, not learned. Evaluate dimensions against withheld markers, views or phases and compare them with anthropometric measurements where available.

6. Evidence Levels and Practical Reading

For every method, distinguish source implementation, unit tests, synthetic recovery, learned artifact training, native candidate simulation, independent open-loop replay, and physical qualification. UpstreamDrift has meaningful code in each category, but they are not interchangeable. Existing unit tests validate shapes, costs, failure handling and sometimes synthetic recovery. The artifact audit quarantines known absent checkpoint paths and unsupported training claims. Neural roadmap receipts explicitly limit many results to synthetic fixtures and software contracts. At the same time, Simscape has real R2025b dynamic replay evidence for a partial horizon with a clearly failed gate, and extensive experimental receipts for failed Human holds and unqualified full-swing matching. These candid partial results are more valuable than a fabricated pass: they tell the next method what to solve.

The strongest overall workflow is therefore not a single algorithm. It is a layered system: calibrate observations and frames; use IK/retrieval to generate a valid initial state; assess observability; choose a fixed or explicitly adaptive model; fit controls with bounded, scaled variables and transparent objectives; preserve diverse proposals; replay through the exact native forward solver without state injection; audit contact, continuity and actuator behavior; evaluate held-out data; and publish provenance and rejected attempts. The method should return a candidate plus uncertainties and evidence status, not a bare residual or a single claim of a “matched” human.

Lessons From Historical Fits

Evidence snapshot: all repository citations pin to commit `d7c7d07add0d7407a5a3376d5fe4a97aee9112a0`. The numerical results below are transcribed from checked-in handoffs and receipts. They were not rerun for this chapter. Each result is labelled by what its evidence actually establishes. The purpose is to extract reusable lessons, including failures, rather than to rank methods by their best isolated number.

Forward-dynamics matching is experimental science as much as optimization. A fit is conditional on the capture, target markers, coordinate frame, model topology, body geometry, initial pose and velocity, control basis, integrator, contact model, observation mask, and acceptance metric. Changing one can change the apparent success. The repository’s historical runs make this concrete: a very accurate partial club trajectory can coexist with poor distal body markers; an optimizer can remove bound saturation and expose a structural model floor; a saved pose can look plausible while its support contact collapses; and a process that exits successfully can still fail every physical gate.

1. Start With the Evidence Labels

The most important lesson from the historical record is to name the driving mode. Upstream-Drift contains inverse-kinematic exports, feedback-assisted holds, prescribed-coordinate experiments, fitted torque candidates, and independent torque-driven replay. These answer different questions. An IK clip answers whether a chosen configuration can resemble target frames. A feedback-assisted hold asks whether a controller can stabilize a pose under given forces and support. A torque-driven replay asks whether one set of controls, from one initial state and without target-state resets, generates the measured trajectory. Only the last can support a forward-dynamics matching claim, and even it needs contact, force, continuity and model validity checks.

The Simscape refined-Human reference states the boundary plainly: delivered clips are IK and dynamics-unqualified, feedback-assisted fixed holds do not establish full-swing dynamics, and full Human forward dynamics remains unqualified. This is a central finding, not an editorial caveat. The clips are useful for visual inspection, frame mapping and pose residuals, while the open dynamic gates identify the work still required. The maintained status is in the Simscape matching reference README; the calculation-level source is its linked `simscape_matching_reference.tex`.

2. Tour-Average Simscape: Short Horizons, Then Continuation

The tour-average effort demonstrates why forward validation should be incremental. The maintained tour-matching README records that 22 native tests passed in MATLAB R2025b

Update 5 and that a 20 ms model simulation completed with finite states and club/grip positions. The same document records a later 0.4 s quadratic forward-dynamics candidate independently replayed with 3.119775 mm marker RMS, while warning that full-swing fitting was still unqualified at that checkpoint. A short-horizon match establishes local fit feasibility; it does not establish a full downswing, impact or follow-through.

The capture audit in that guide matters as much as the optimizer. The driver file has 654 frames at 360 Hz, source units in metres and no event group. Some marker tracks have substantially fewer valid samples, and an unlabeled cluster cannot be silently relabelled as a grip or clubhead. The guide also warns that skin markers are not anatomical joint centers and that the model's output extractor can construct a time grid or fill unavailable signals with NaNs. Accordingly, accepted simulation data must preserve actual solver timestamps, gaps and marker validity. A convenient array shape is not data provenance.

The incremental fitting protocol hashes the capture, SLX dependencies, input MAT and marker map; first qualifies registration and fixed geometry; and then increases the fitted time prefix while retaining all earlier observations. Each prefix is reintegrated from the same initial state. This avoids a hidden reset at the stitch point and makes earlier trajectory regression measurable. If a single global sixth-order polynomial cannot fit the complete interval, the protocol proposes a piecewise representation with explicit continuity constraints and a newly qualified native implementation. The final gate is a cold native replay with tighter tolerances and no kinematic prescription. The guide distinguishes the weighted optimization objective from unweighted physical-distance metrics and asks for per-marker and per-phase error, continuity, torque and velocity limits, failures, and sensitivity.

A later two-window direct-node result reaches 0.85 seconds of uninterrupted R2025b replay. The run 102 handoff reports whole-marker RMS 20.267 mm, early RMS 9.995 mm, club-cluster RMS 8.389 mm, pelvis-yaw error 0.610 percent, zero active coefficient or node bounds, and strong Pinocchio/Simscape parity. These are notable improvements over its cited preceding runs. Yet the exact Simscape replay qualification JSON says `is_qualified: false`. The terminal RMS is 40.303 mm versus a 35 mm gate, despite early and club gates passing. The receipt says contact-force and penetration measurements were unavailable. This is the right interpretation: a high-quality partial dynamic result with an explicit failed gate and missing physical quantities.

The run teaches three optimization lessons. First, active bounds can be numerical artifacts: run 101 had 22 controls and 11 node coordinates at bounds; recentering control bounds and anchoring the shooting chart at the integrated endpoint removed active bounds in run 102. A solver stuck at a box edge may need better coordinates, not a wider physical model. Second, global mean error can hide terminal failure; acceptance needs phase-specific and marker-family metrics. Third, cross-engine agreement verifies the equations and implementation only to the tested model contract; both engines can agree on the same incomplete topology or absent contact measurement.

3. Human Ellipsoid Experiments: Geometry and Observability

The Human ellipsoid research history shows why geometry, orientation, mass and contact cannot be treated as cosmetic switches. The maintained Simscape reference compares tour-average Capture A and owner Capture O as separate sources. It warns that appearance does not establish skill, and that marker-protocol differences, offsets and geometry can explain apparent posture distortion. Refinement to subject-specific longitudinal dimensions improved geometric fit, but transverse artistic proportions, head geometry and shoe size remained fixed. A native subject-physics adapter applied declared mass and selected lengths, but the

selected IK videos continued to use baseline physical parameters. Thus visual adaptation did not amount to a subject-specific dynamic model.

Position-only fits exposed orientation ambiguity: on three sample frames, position RMS was around 12–13 mm while the left-foot angular discrepancy approached 165–167 degrees. Adding an orientation objective at one tested low weight reduced angular error but worsened position RMS to roughly 39–51 mm, and that candidate was rejected. A fresh calibration on those same frames subsequently achieved approximately 1.2 mm position error and sub-degree foot angles, but the record correctly says those were calibration—not held-out validation—frames. The experiments show both that orientation data can resolve a weakly observed direction and that a selected weight can disturb other terms. They do not establish that a lower training-frame residual generalizes.

The body topology audit found 37 independent fit coordinates, correcting earlier hard-coded counts and indexing. That correction is an important model-contract repair: optimization over wrong dimensions can report failures or silently map residuals to the wrong joints. The same experiments identified contact assumptions as a separate problem. Treating a flat sole at address as an orientation calibration is not a measurement of a ground reaction force. A read-only FK audit found contact-sphere penetrations relative to the ground plane, but this was geometry evidence only; it was not a simulated force measurement and no dynamics were run. The reference warns that ground placement, gravity and consistent dynamic initialization require separate validation.

The supported-hold campaign supplies especially useful negative evidence. Balance-enabled tour and owner holds moved the pelvis about 20.0 and 14.8 mm and rotated about 5.6 and 4.4 degrees, exceeding 5 mm and 1-degree pose limits. Increased stiffness reduced movement but exceeded a peak-force limit. Ramped feedforward improved force gates in tested windows but still failed pose gates. Longer stance simulations lost normal force, tilted substantially and dropped the pelvis below the plane; increased damping alone also failed. These tests demonstrate that a plausible initial pose, strong servo and clean exit status do not establish stable support physics. The associated source and receipts retain rejected experiments and state that the full Human swing remains unqualified.

4. Historical Neural Fits: Code and Experiment Are Different Objects

Older design records proposed four alternatives: repeated native optimization, a learned forward map with differentiable inversion, a direct inverse CVAE, and a Python-to-Simscape bridge. Current source review shows a mixed but more mature landscape than these early READMEs suggest. Shared PyTorch FiLM forward surrogate, compact model, per-step dynamics model, CVAE, deterministic regressor, timestep inverse-dynamics model, and masked proposals all have substantive model/training code and unit tests. Legacy `option2_nn_surrogate/surrogate.py` and `option3_inverse_nn/inverse_cvae.py` remain stubs, but they are not the current shared implementations.

The experimental record nevertheless does not justify broad acceleration claims. The NM-00 inventory marks two conventional checkpoint paths absent; it quarantines unsupported mean-baseline training claims where no curves or checkpoint hash are available. NM-04’s trainer tests do not imply a native teacher corpus. NM-05’s baseline tests do not imply successful training. NM-06 says synthetic fixtures and no native training success. NM-07 is explicitly diagnostic because earlier ablation numbers were hard-coded, not measured. NM-08’s inference contracts use synthetic fixtures. NM-09’s checkpoint matrix is diagnostic and has no native ODE replay qualification on reduced planar mechanisms. NM-10’s historical speed receipt calls its costs and baseline estimates unmeasured. The corresponding receipts are linked from `docs/plans/neural_motion_matching/evidence`.

There is an internal inconsistency: the later model-card turnover states measured speedups and lists several models as promoted, while the artifact and validation receipts retain diagnostic, synthetic-only or unmeasured status. A textbook must cite both and decline to repeat promoted speed values until source-hashed model weights and run logs reconcile the statuses. The lesson is not to discard neural approaches; it is to make artifact lineage a hard scientific interface. A model-card verdict must be derived from immutable checkpoint, train data/split, command, runtime, candidate, replay and benchmark receipts, rather than static prose or an estimate.

The neural ablation findings still guide method design qualitatively. A forward surrogate can offer smooth gradients, but optimization can exploit inaccurate directions and propose non-physical states; trust regions and comparison of surrogate to native directional derivatives are necessary. A hybrid that uses the learned model to initialize an exact/native polish step is safer. A physics-structured residual can preserve an analytical rigid-body prior and bound the learned correction. Diffusion or repeated sampling is useful when multimodality matters, but may not meet interactive latency. These are design conclusions recorded in a diagnostic receipt; they are hypotheses and safeguards, not quantified performance findings.

5. From Historical Failures to Adaptive Model Constraints

The most promising adaptive-dimension idea must respond to the confounds identified above. A trajectory residual alone cannot decide whether to alter segment length, marker attachment, camera offset, initial pose, inertia, actuator gain or torque basis. Free joint optimization is especially dangerous: extra model dimensions can absorb a wrong coordinate map or unmodeled marker motion and appear to “improve” a fit. Conversely, a fixed model can hit a structural error floor even after optimizer bounds are no longer active. Run 102’s head/back cluster residual provides an example: its handoff attributes a residual floor to a topology without an independent neck coordinate. The consequence is not “add any degree of freedom”; it is to formulate a topology-change hypothesis and test it against held-out head markers, orientation, dynamic replay, contact and model-complexity cost.

A robust adaptive model should learn only low-dimensional, physically interpretable shape factors. Let the base segment lengths be l_0 ; learn $l = l_0 \odot \exp(Bz)$ where z is a small latent and columns of B encode permitted proportional changes, such as shared upper-limb scale or leg-to-trunk ratio. Put anthropometric priors on z , enforce positive lengths and mass/inertia consistency, and preserve topology. Marker attachments should be separate parameters with tighter priors. Camera extrinsics should be calibrated independently when possible. Control coefficients should use a normalized time basis and physically interpretable bounds. This variable partition makes compensation visible.

Fit in stages: calibration and marker validity; geometry and initial state; control fit with model dimensions fixed; only then a restricted joint update if sensitivity analysis shows geometry is observable. Compute a scaled Jacobian of held-out marker/orientation residuals with respect to each parameter block. Use singular values and profile fits to find confounded directions. If a parameter changes little in the data likelihood or trades almost perfectly against torque coefficients, label it unobserved and keep its prior. Confidence should reflect empirical parameter coverage and calibration, not subjective probability that a reconstruction is a real golfer.

Validation should use more than random frame splits. Adjacent frames leak nearly identical poses. Hold out complete capture trials, view angles, movement phases or marker groups. For a single capture, hold out contiguous time windows and report extrapolation separately. Evaluate both the fit target and measurements excluded from fitting. A dimension model that lowers fit-frame RMS but worsens a held-out segment or fails contact is rejected. On small data, prioritize synthetic truth-recovery for implementation correctness, then native

simulator-generated trials, then real-capture evaluation with honest uncertainty.

6. An Evidence-Preserving Research Sequence

The historical record suggests this order of work. First freeze the source capture, event interpretation, frame transformations, units, masks and marker attachments. Then establish a geometric residual floor using a qualified model and held-out targets. Confirm model topology, initial state and native actuator/control path. Next, run direct native fits with normalized bases and repeatable restarts; save every failed candidate. If an optimizer saturates, examine coordinate conditioning and sensitivity before widening bounds. Introduce surrogate/inverse networks only after a versioned native dataset and group-based split exist. Require calibrated OOD rejection and native forward polish. Finally, cold replay the selected candidate from the original initial state with solver settings stricter than the fit. Evaluate each phase, marker group, orientation, contact, torque, joint range, continuity and energy consistency, and publish failures and unavailable quantities.

This sequence converts historical “best fit” stories into reproducible evidence. A report can then distinguish four outputs: (1) kinematic visualization candidate; (2) dynamic tracking solution with feedback or transcription constraints; (3) independent forward replay that passes declared numerical gates; and (4) physically qualified model fit with contact and force evidence. Keep the status attached to the artifact, not only the prose. If a later geometry, control basis, marker map, model file or runtime changes, invalidate the old qualification and rerun the relevant gates.

The historical fits show a field that has made real progress: native execution and cross-engine parity, model-coordinate corrections, an expanding cost and optimizer toolkit, carefully preserved negative experiments, and receipts that separate accepted from rejected gates. They also show why the central research question remains open. A low trajectory error is only one measurement of a coupled system. The next method should improve identifiability and evidence quality before it tries to reduce that number further.

Measurement and Preprocessing for Dynamic Matching

A forward dynamics fit is only as meaningful as the map from the world to the model. Before optimizing controls or adapting dimensions, the analyst must define what was measured, where it was measured, which coordinate frames produced it, and how uncertainty enters. Many apparent model failures arise before the equations of motion are evaluated: marker offsets, camera calibration, time synchronization, force-plate assignment, filtering, or a mismatch between the model's generalized coordinates and the measured observables. The preparation stage should therefore be treated as part of the estimation model rather than as invisible clerical work.

State, Parameters, and Observations

Let the generalized configuration be $q \in Q$, velocity $v \in T_q Q$, internal actuator states z , and static or slowly varying model parameters p . A state is $x = (q, v, z)$. The dynamics can be written in first-order form as $\dot{x} = f(x, u, p, w)$,

where u is control and w collects external loads, contact forces, and possible disturbances. A measurement at time t_k is $y_k = h_k(x(t_k), p, c) + \epsilon_k$,

where h_k is a sensor-specific observation function, c represents calibration and nuisance parameters, and ϵ_k is measurement error. The distinction between f and h matters. A camera measures image coordinates or reconstructed marker positions, not joint torques; an IMU measures specific force and angular velocity, not generalized coordinates; a force plate measures a wrench in its own frame, not the contact force of whichever foot the analyst hopes it measured.

An appropriate state estimator compares predicted observations h_k with sensor outputs. It should not compare a marker measurement directly with a joint angle unless a valid observation model maps the configuration to that angle. For a 3D marker i attached to segment s , a simple model is $y_{ik} = R_s(q_k)r_i(p) + t_s(q_k) + \epsilon_{ik}$; more realistic models can add soft tissue displacement, marker placement offsets, camera projection, or visibility-dependent covariance. Frame definitions must be explicit: world, laboratory, segment, sensor, and model frames; handedness; axis direction; origin; angular units; and time origin.

Noise Models and Weighting

If errors are approximately Gaussian with covariance Σ_k , the negative log-likelihood contribution is, up to a constant, $r_k^T \Sigma_k^{-1} r_k$, where $r_k = h_k(x, p, c) - y_k$. Thus weighting is not cosmetic. It encodes expected measurement variability. A 2 mm marker uncertainty, a 0.02 rad orientation uncertainty, and a 30 N force uncertainty should not receive equal raw numeric weights.

Whitening residuals with a Cholesky factor $L_k L_k^T = \Sigma_k$ gives $L_k^{-1} r_k$, a dimensionless residual whose squared norm has a likelihood interpretation.

Real motion data contain outliers and correlated errors. A marker may be occluded, mislabeled, or displaced by skin motion. Filtering creates temporal correlation, and markers on one rigid segment share calibration and soft-tissue errors. A robust loss such as Huber or Student-t can reduce outlier influence, but it does not repair a systematic coordinate-frame error. If residual structure persists over contiguous intervals or moves together across a segment, model the shared nuisance source instead of downweighting every point. A practical model may include a per-trial marker offset, a time offset, or a low-dimensional soft-tissue deformation term with a prior.

The time base must be modeled as carefully as the spatial frame. If a sensor stream has a time offset δ , then its prediction is $h(x(t_k + \delta))$. A small offset in a high-acceleration movement can look like a wrong inertial parameter or actuator delay. Synchronization can be estimated from hardware trigger signals or jointly through a bounded delay parameter, but the latter can be confounded with muscle activation dynamics. Calibration, synchronization, and physiological delay should therefore have distinct priors and should not be collapsed into one unconstrained phase shift.

Filtering and Derivatives

Inverse dynamics commonly requires velocity and acceleration, so data are smoothed and differentiated. Differentiation is ill-conditioned: a finite difference divides measurement noise by the sample interval, and a second derivative amplifies it further. Smoothing reduces variance but introduces bias and phase delay. Filtering choices must therefore be documented by filter family, cutoff, order, edge treatment, sample frequency, and whether zero-phase filtering was used. Applying zero-phase filtering offline avoids phase lag but is noncausal and can create endpoint artifacts; a causal filter preserves real-time behavior but delays the signal.

Smoothing splines and state-space estimators offer a more coherent alternative to “filter then differentiate.” One can estimate a smooth trajectory by minimizing $\sum_k \|\hat{q}(t_k) - q_k\|_W^2 + \lambda \int \|\hat{q}^{(m)}(t)\|^2 dt$,

where the first term fits observations and the second penalizes roughness. The smoothing strength λ expresses a bias-variance choice. Better still, estimate latent states directly from a dynamic model and sensor likelihood, avoiding a separate differentiation pipeline. But this transfers risk to the model: a poor forward model can bias the state estimate. The right comparison is not “filtered truth versus model truth”; it is a sensitivity analysis across defensible noise and process models.

For rotations, ordinary componentwise filtering of Euler angles can fail near singularities and can average orientations incorrectly. Use a representation with valid geometry—rotation matrices projected onto $SO(3)$, unit quaternions with sign continuity, or local Lie algebra coordinates—and define residuals with the group logarithm. If orientations R_{obs} and R_{pred} belong to $SO(3)$, an intrinsic residual is $r_R = \text{Log}(R_{pred}^T R_{obs}) \in \mathbb{R}^3$. This respects the geometry and avoids treating angular components as three unrelated scalars.

Contact and External Loads

Ground reaction measurements are often the strongest constraints on dynamics and among the easiest to mishandle. A force plate reports a wrench about a plate-fixed origin. To compare it with a model wrench about another origin, transform both force and moment consistently. If r shifts the moment reference, then $M_{new} = M_{old} + r \times F$ (with sign depending on the direction

of the shift). A correct vertical force with an incorrect center of pressure can still imply the wrong net moment. Verify the plate coordinate transform, calibration, sign, thresholding, and which foot is in contact at each frame. When a foot contacts multiple plates or both feet share a plate, force assignment is an inference problem, not a trivial label.

A forward simulation needs a contact model: prescribed measured wrench, compliant contact, complementarity constraints, or a hybrid contact schedule. Each choice imposes a different scientific assumption. Prescribing the measured wrench can make kinematic tracking easier but does not test the contact model. A compliant penalty model is smooth and convenient but may add stiffness and permit penetration. Complementarity can represent unilateral contact and friction more directly but yields difficult nonsmooth or nonlinear programs. If the contact schedule is known from measurements, it can be fixed for estimation while contact mechanics are tested separately. If it is unknown, phase or contact variables must be inferred with explicit constraints.

External loads should be measured and represented as uncertain quantities. A convenient Gaussian load residual is $r_F = F_{model} - F_{meas}$; yet force-plate errors may be biased by calibration, drift, or center-of-pressure uncertainty. For a measured wrench w , use a covariance over all six components or separate force and moment covariance blocks with units accounted for. Avoid allowing a large unit imbalance to make translational forces dominate rotational moments simply because newton-meters have numerically smaller values.

Model Coordinate Compatibility

Kinematic fitting assumes that the model can express the observed motion. A model with the wrong joint topology, missing degree of freedom, or incompatible coordinate convention cannot match every observation no matter how clever the optimizer is. Before fitting, map each measurement to a model quantity and verify range, sign, neutral pose, and anatomical meaning. Joint angles derived from marker trajectories may not coincide with model coordinates because of joint definitions, skin motion, or functional-axis estimation. Therefore marker-space tracking and generalized-coordinate tracking answer different questions. Marker-space residuals preserve the observation geometry; coordinate tracking can be easier to interpret but embeds an upstream kinematic reconstruction.

The model itself brings priors: segment dimensions, inertial properties, joint centers, muscle paths, tendon slack lengths, and actuator limits. Scaling all lengths while leaving inertias unchanged is inconsistent. A uniform geometric scale factor s implies mass scales roughly with s^3 under constant density and inertia roughly with s^5 ; real subject-specific distributions require more informed scaling. When “learning dimensions,” parameters should have physical units, plausible bounds, and dependency rules. A dimension parameter is not independent if it changes muscle moment arms, marker locations, contact geometry, and segment inertias simultaneously.

A Pre-Fit Diagnostic Sequence

A reproducible preprocessing report should include: sample counts and missingness; sensor calibration and frame transforms; synchronization offsets; filtering and derivative method; marker residual distributions by segment; force/moment residual plots; contact labels; range and velocity checks; and the model-to-observation mapping. Plot raw observations and model projections together before running a global optimizer. Check whether a rigid-body segment can fit its own markers. Verify that the measured support wrench has a plausible magnitude and center of pressure. Compare finite-difference and spline derivatives at several cutoffs. These checks often reveal errors that an optimizer would otherwise absorb into implausible parameters.

The baseline should retain raw data and transformations in a machine-readable provenance record. A derived signal should identify its parent signal, units, filter, time base, and coordinate frame. This is crucial for scientific repeatability and for separating model adaptation from preprocessing changes. Do not overwrite raw data or treat an undocumented filter setting as part of the model.

Worked Derivation: A Marker Residual With Calibration Uncertainty

Suppose a single segment pose is (R, t) and a marker has known segment coordinate r . The predicted lab-frame location is $m = Rr + t$. The observed marker is $y = m + \epsilon$, with $\epsilon \sim \mathcal{N}(0, \Sigma_y)$. If the marker placement is uncertain by $\delta r \sim \mathcal{N}(0, \Sigma_r)$, then linearizing gives $\delta m \approx R\delta r$; hence the effective observation covariance is $\Sigma_e f f^T = \Sigma_y + R\Sigma_r R^T$.

For a parameter-dependent covariance, the full Gaussian contribution is $\frac{1}{2}r^T \Sigma(\theta)^{-1}r + \frac{1}{2} \log \det \Sigma(\theta)$, up to constants. The determinant term cannot be discarded when learning uncertainty or when rotating anisotropic placement covariance changes the effective covariance. Otherwise increasing uncertainty can reduce the quadratic residual without explaining the observations. The preceding quadratic forms assume fixed covariance or describe only the residual part of the likelihood.

The quadratic contribution to the objective becomes $(m - y)^T \Sigma_e f f^{-1} (m - y)$. This derivation shows why treating all markers as exact rigid points can over-weight a marker with uncertain placement. If several markers share a segment-level skin displacement b , the model $y_i = Rr_i + t + b + \epsilon_i$ induces correlated residuals. Estimating one regularized b is more principled than independently inflating each marker covariance, although it introduces an additional latent variable and prior.

When This Chapter's Assumptions Fail

Gaussian errors are not universal; marker visibility and force-plate clipping create non-Gaussian failures. Rigid segment assumptions fail under soft tissue motion and flexible structures. A fixed contact schedule fails for unexpected contact. Constant dimensions fail across growth or deforming anatomy. A time-invariant camera calibration fails if the rig moves. The practical response is to expose the assumption and test its consequences, not to claim that a single optimizer removes these limitations.

Exercises With Solutions

Exercise 1: Weighting Two Sensors

A marker residual is 4 mm with standard deviation 2 mm. An IMU orientation residual is 0.03 rad with standard deviation 0.01 rad. Which contributes more under independent Gaussian noise?

Solution. The standardized residuals are 2 and 3, respectively. Their squared contributions are 4 and 9; the orientation residual is more surprising under its stated model. Raw numbers alone would mislead because the units differ.

Exercise 2: Timing Error

A marker follows $y(t) = A \sin(\omega t)$. A sensor offset is δ . Show the first-order position error.

Solution. $y(t + \delta) \approx y(t) + \delta A \omega \cos(\omega t)$. Thus the error grows with frequency and amplitude and is largest near zero crossings. A time error can therefore mimic an amplitude or phase-dependent dynamics defect.

Exercise 3: Force-Plate Moment Shift

A plate measures force F and moment M about origin O . Derive the moment about $O + r$.

Solution. The moment about a new point equals the old moment minus the displacement from the old point crossed with the force if r points from O to the new point: $M_{O+r} = M_O - r \times F$. The sign is easiest to verify from the force line of action. State the vector direction in any implementation or report.

References

Rigid-body dynamics conventions and efficient mechanics algorithms are covered in (Featherstone 2008). Direct biomechanical source links collected for the larger review include OpenSim (Delp et al., 2007; DOI 10.1109/TBME.2007.901024), OpenSim locomotion review (live URL; full citation metadata pending verification), and the 2024 systematic review of simulation-to-measurement differences (live URL; full citation metadata pending verification). The direct source retrieval record is in the accompanying literature ledger.

Inverse Kinematics and Inverse Dynamics as Baselines

Inverse methods are essential in a forward-dynamics matching project because they summarize the observations in mechanically interpretable quantities and provide initial guesses. Their role is often overstated. Inverse kinematics estimates configurations that explain measured positions or orientations. Inverse dynamics then asks what net generalized loads would produce the observed motion under a specified model and external loading. The combined pipeline is useful, but it does not demonstrate that a muscle-actuated forward simulation can generate the motion, and it does not uniquely identify muscle forces or neural control.

Inverse Kinematics

For markers $y_i(t_k)$ and model marker locations $m_i(q_k, p)$, a weighted least-squares IK estimate solves $\hat{q}_k = \arg \min_q \sum_i (m_i(q, p) - y_i)^T W_i (m_i(q, p) - y_i) + R(q)$.

Here p contains fixed body dimensions and marker attachment locations; $R(q)$ may encode coordinate bounds, joint couplings, or a temporal prior. At each time sample, the model configuration is selected to minimize marker discrepancy. The problem is generally nonlinear because the forward kinematic map is nonlinear. It may be solved by Gauss-Newton, Levenberg-Marquardt, or another local method. When each time point is solved independently, the result can be noisy or jump between local minima; temporal regularization or a trajectory-level formulation improves continuity but changes the estimator and should be reported.

The local geometry is described by the Jacobian $J_m = \partial m / \partial q$. Near a candidate q , the residual has approximation $r(q + \delta q) \approx r(q) + J_m \delta q$. A Gauss-Newton step solves $(J_m^T W J_m + \Lambda) \delta q = -J_m^T W r$, where Λ may stabilize weak directions. A small singular value of $W^{1/2} J_m$ indicates that the markers poorly observe a coordinate direction. Such a coordinate can vary substantially without changing marker fit. Regularization makes a numerical answer possible; it does not add information to the data.

IK can be formulated in marker space or in a previously estimated joint-coordinate space. Marker-space IK retains direct connection to camera observations but depends on marker attachments and body geometry. Coordinate tracking avoids repeated marker fitting but inherits the assumptions and errors from the upstream reconstruction. A good workflow retains both: use IK for interpretable motion summaries and marker-space residuals in the final tracking estimator when raw markers are available.

Inverse Dynamics

For a rigid multibody system with coordinates q , a common equation is $M(q, p)\ddot{q} + C(q, \dot{q}, p) + g(q, p) = \tau + J_c(q)^T \lambda + Q_{ext}$. M is the mass matrix, C includes velocity-dependent terms, g gravity, τ generalized actuator force, $J_c^T \lambda$ constraint or contact forces, and Q_{ext} external applied loads. If q , $\dot{q}$, $\ddot{q}$, model parameters, and external loads are known, inverse dynamics computes the required net generalized force τ_{req} by rearranging the equation. In practice, the system can include unmodeled forces and residual actuators, so reported net loads are conditional on the chosen model and force inputs.

This is a computationally powerful decomposition. It can reveal whether a model and force-plate record are dynamically compatible, help estimate net joint moments, and seed controls. But the inverse map from net joint moment to individual muscle force is underdetermined. With n muscles and d generalized coordinates, the relation $\tau = R(q)f_m$ maps many muscle force vectors to the same joint moment whenever the muscle set is redundant. Even if a unique nonnegative solution exists in a special case, the result is sensitive to moment arms, force limits, co-contraction, passive structures, and activation state. An optimizer selecting the minimum squared activation chooses a solution according to a criterion; it does not discover a directly observed muscle force.

Why Differentiation Is Dangerous

Suppose measured position is $q_k = q(t_k) + \epsilon_k$ with independent noise variance σ^2 . A central second difference is approximately $(q_{k+1} - 2q_k + q_{k-1})/\Delta t^2$. Its noise variance is $(1 + 4 + 1)\sigma^2/\Delta t^4 = 6\sigma^2/\Delta t^4$. Halving the sample interval increases this variance by a factor of sixteen under this simple white-noise model. Higher sampling rate does not automatically make acceleration estimates better; it can expose higher-frequency sensor noise unless the estimation procedure accounts for it.

Filtering before differentiation controls variance but biases the acceleration and may introduce phase distortion. Conversely, a very flexible smoothing spline reproduces noise. Inverse dynamics should therefore report its dependence on smoothing parameters, derivative method, endpoint handling, and external-force synchronization. A trajectory-level optimization can avoid differentiating noisy measured positions directly by treating velocities and accelerations as latent states constrained by dynamics, but that trades derivative noise for model dependence.

Residual Forces and Dynamic Inconsistency

In musculoskeletal workflows, residual actuators often represent unmodeled force needed to reconcile kinematics, external loads, and model dynamics. These are not physiological muscle forces. Large residuals can reflect incorrect body mass or center of mass, marker-based kinematics inconsistent with force data, bad force-plate assignment, missing degrees of freedom, contact errors, or time mismatch. Residual thresholds are useful warnings, but a low aggregate threshold cannot tell which assumption is correct. Residual magnitude, direction, duration, and mechanical work should all be inspected. A brief large impulse can matter more than a modest bias sustained over a full stride.

OpenSim's Residual Reduction Algorithm (RRA) addresses part of this issue by solving a torque-driven tracking problem that can adjust kinematics and mass properties while reducing residual forces and moments. The result is a dynamically improved target, not a proof that adjusted dimensions are anatomically correct. A downstream muscle-driven simulation

should report both the original data fit and the adjusted trajectory fit, together with the adjustments to the model. Otherwise, a preprocessing modification may be mistaken for successful prediction.

Muscle Redundancy and Optimization Criteria

A static optimization commonly selects muscle activations a at each sample to minimize a cost such as $\sum_i a_i^p$, subject to $R(q)f(a, l, v) = \tau_{req}$ and activation bounds. The exponent p , muscle strength scaling, passive forces, and constraints alter recruitment. With $p = 2$, the optimizer tends to distribute effort; with other norms, it may favor sparse or different patterns. If the optimization is solved independently at each time sample, temporal smoothness and activation dynamics are omitted unless explicitly added. Thus this is a convention for resolving redundancy, not a data-driven identification of control.

Forward optimal control improves upon per-frame allocation by adding activation dynamics and the equations of motion over time. Yet the solution still depends on its objective: activation minimization, metabolic energy, EMG fit, joint-load penalties, or task performance can produce distinct controls and trajectories. The correct interpretation is conditional: “the control history that minimizes this declared objective under these model and data assumptions.”

Weighted Least Squares, Constraints, and Uncertainty

IK is often described as a least-squares problem, but the choice of residual space determines what the result means. Marker-position residuals can be weighted by camera reconstruction covariance, while coordinate residuals can be weighted by confidence in a joint-angle estimate. These are not interchangeable objectives. If markers are correlated because they share a segment rigid-body fit or camera calibration, a block covariance is more appropriate than independent scalar weights. Robust losses can reduce the effect of marker swaps, but persistent directional residuals usually indicate a calibration, attachment, or model problem and should not be hidden with a robust penalty.

Joint limits and couplings may be encoded as hard constraints or soft priors. A hard bound prevents anatomically impossible configurations but can cause estimates to pile up at the boundary; a soft penalty permits small violations while expressing uncertainty. Report active constraints and distinguish a measurement-driven coordinate from one determined mostly by its prior. In local uncertainty analysis, a prior precision contributes to the approximate information matrix, so a small posterior variance does not necessarily imply the markers observed that coordinate well. Compare the data-only information with the prior-augmented information.

Using Inverse Methods Well

A disciplined sequence is to use IK and ID as diagnostics, not ground truth. First fit a scaled model to markers and inspect residuals by marker and segment. Then compute filtered derivatives and net joint loads, documenting sensitivity to smoothing. Check external-force consistency and residual-actuator demand. Use those results to initialize controls or to formulate observation terms. Finally solve a forward constrained problem and compare its independently generated observations to the raw measurements. If the forward result diverges, distinguish whether the cause is poor optimization, bad initialization, model misspecification, or incompatible data.

A useful diagnostic identity is the local covariance of IK coordinates. Under a locally linear model with Gaussian marker noise, $\text{Cov}(\hat{q}) \approx (J_m^T W J_m)^{-1}$ when the inverse exists. Near

singular configurations, this covariance becomes large. A coordinate may appear precise only because a strong prior constrains it. Reporting this uncertainty helps identify which parts of a model can reasonably be adapted from the available markers.

Worked Derivation: Why Joint Moments Do Not Identify Muscle Forces

Consider a single hinge with two muscles having moment arms $r_1 = 0.03$ m and $r_2 = -0.02$ m. The measured net flexion moment is $\tau = 10$ N m. The moment equation is $0.03f_1 - 0.02f_2 = 10$, with $f_1, f_2 \geq 0$. One solution is $(f_1, f_2) = (333.3, 0)$ N; another is $(400, 100)$ N, since $12 - 2 = 10$ N m. A co-contraction of both muscles changes individual force substantially while preserving the same net moment. A cost function can select one point on this line—for example minimizing squared normalized activations—but no net moment measurement distinguishes those solutions.

With m muscles and d moment equations, the linearized null space has dimension at least $m - \text{rank}(R)$ before inequality and muscle-force constraints. Bounds may shrink the feasible set but usually do not make the true force identifiable. EMG, tendon-force measurements, activation dynamics, physiological priors, and multiple movement conditions add information, though each has uncertainty.

Failure Modes to Record

- Joint angles are differentiated twice without a convergence or smoothing sensitivity analysis.
- A force plate is assigned to the wrong foot or expressed about the wrong origin.
- A residual actuator is interpreted as muscle force.
- An IK fit is judged only by its average error, hiding a few large outlier markers.
- A static optimizer's muscle recruitment is presented as uniquely inferred.
- Coordinate limits or priors are mistaken for measurement information.
- One preprocessing choice is fixed silently and no alternative is tested.

Exercises With Solutions

Exercise 1: Rank and Redundancy

A model has 18 muscles acting around 6 generalized coordinates. What does the count alone imply about unique muscle-force recovery?

Solution. The count suggests at least 12-dimensional null space if the moment-arm matrix has full row rank, before inequalities and muscle dynamics. Exact rank and feasible bounds matter, but the count immediately rules out uniqueness from six net moments alone.

Exercise 2: Acceleration Noise

A central second difference is used at $\Delta t = 0.01$ s. If position noise standard deviation is 1 mm, estimate the standard deviation of acceleration noise under independent samples.

Solution. The variance is $6\sigma^2/\Delta t^4$; the standard deviation is $\sqrt{6}(0.001)/(0.01)^2 \approx 24.5$ m/s². This large value explains why raw finite differences are unsuitable for typical marker noise.

Exercise 3: What Does Low Residual Prove?

RRA reduces residual moments by adjusting torso center of mass. What can be concluded?

Solution. The adjusted model and kinematics are more consistent with the observed loads under that formulation. It does not prove the original center of mass was wrong by exactly the fitted amount, because marker error, force errors, omitted dynamics, and other parameters can create similar residuals.

References

The locally available bibliography includes Featherstone, *Rigid Body Dynamics Algorithms* (Featherstone 2008) . Relevant live-linked records collected for the full review: Thelen, Anderson, and Delp (2003), CMC; Thelen and Anderson (2006), gait forward simulations; OpenSim official RRA documentation; and the 2024 systematic review. Source-access and metadata status are recorded in the parent research ledger.

Shooting Methods for Forward Model Matching

Shooting methods make a forward simulation the core of the estimator. The optimization chooses an initial state, control trajectory, and possibly model parameters; the equations of motion then generate states by integration. The predicted measurements are compared with observed kinematics and other sensor channels. This preserves a direct causal chain from controls and parameters to motion, but it also exposes the optimization to the sensitivity and instability of that chain.

Single Shooting Formulation

Let observations occur at times $t_0, \dots, t_N$, controls be parameterized by coefficients θ , and model parameters be p . Starting from x_0 , the numerical flow map integrates $\dot{x} = f(x, u(t; \theta), p)$, $x(t_0) = x_0$,

to produce $x_k = \Phi_{t_k, t_0}(x_0, \theta, p)$. A tracking objective is $J(x_0, \theta, p) = \sum_k \rho(L_k^{-1}[h(x_k, p) - y_k]) + R(\theta, p, x_0)$,

possibly with terminal, force, EMG, contact, or task terms. Here $L_k L_k^T = \Sigma_k$ whitens observation residuals. Bounds constrain controls, states, and parameters. The problem can be solved by a gradient-based nonlinear optimizer, derivative-free search, or stochastic method; derivatives of the objective with respect to parameters can be computed through sensitivities or automatic differentiation of the integrator.

The essential feature is that states are not independent decision variables. They are determined by integration. Therefore, if the simulated system is unstable or highly sensitive, a small change in control or initial condition can create a large final-state error. Conversely, a long-horizon objective may give a weak gradient to a parameter whose effect is rapidly damped or hidden by feedback-like controls. This is the classical credit-assignment difficulty of trajectory optimization.

Control Parameterization

The control history must be finite-dimensional. Common parameterizations include piecewise constants, piecewise-linear values, splines, basis functions, or a feedback policy. Coarse control knots regularize the problem and reduce dimension, but may be unable to represent fast activation changes or impacts. Fine knots increase flexibility and can fit measurement noise, while worsening conditioning and creating unrealistic rapid switching. Activation dynamics provide a physical low-pass filter: muscle excitation may be the control and activation a state. For a first-order model, $\dot{a} = (u - a)/\tau(u, a)$ with distinct activation/deactivation time

constants, the optimizer cannot instantaneously jump the muscle force. This distinction is important: optimizing activation directly may allow controls that no excitation can realize.

A useful basis has interpretable scale and smoothness. If $u(t) = \sum_j B_j(t)\theta_j$, then a regularizer such as $\int \|\dot{u}(t)\|^2 dt$ becomes a quadratic form in θ . But smoothness is a modeling assumption; impacts and rapid control transitions may be real. One should compare the fit and control spectrum across knot densities and regularization strengths. Mesh refinement should improve fit until measurement noise or model error limits meaningful resolution.

Sensitivities and Gradients

For a parameter θ_j , define sensitivity $S_j(t) = \partial x(t)/\partial \theta_j$. Differentiating the ODE gives the variational equation $\dot{S}_j = f_x S_j + f_{\theta_j}$,

with an initial condition determined by whether x_0 depends on θ_j . The measurement derivative is $\partial h/\partial \theta_j = h_x S_j + h_{\theta_j}$. This gives the gradient of a residual-based objective. For many controls and parameters, forward sensitivities become expensive because each parameter requires a sensitivity state. Adjoint methods solve a backward costate equation; they are most attractive when there are many decision variables but relatively few scalar objectives. Automatic differentiation can generate derivatives through the discretized solver but requires careful treatment of solver branches, event handling, and contact discontinuities.

Gradient accuracy should be checked against finite differences on small representative cases. A seemingly sophisticated differentiable solver can produce incorrect gradients if tolerances are loose, event times shift, or a branch changes. Before fitting high-dimensional geometry, test the gradients for each parameter group and test the integration error independently.

Adjoint Gradient Derivation

Consider a smooth ODE with a parameterized control, $\dot{x} = f(x, u(t; \theta), p)$, and objective

$$J = \Phi(x(T), p) + \int_0^T L(x(t), u(t; \theta), p, t) dt.$$

For a perturbation in θ , the state variation δx satisfies $\delta \dot{x} = f_x \delta x + f_\theta \delta \theta$, where f_θ includes the chain rule through the control basis and any direct parameter dependence. The first variation of the objective is

$$\delta J = \Phi_x \delta x(T) + \Phi_\theta \delta \theta + \int_0^T [L_x \delta x + L_\theta \delta \theta] dt.$$

Introduce a costate $\lambda(t)$ and add zero in the form $\int_0^T \lambda^T (f_x \delta x + f_\theta \delta \theta - \delta \dot{x}) dt$. Integrating the $-\lambda^T \delta \dot{x}$ term by parts produces the boundary contribution $-\lambda(T)^T \delta x(T) + \lambda(0)^T \delta x(0)$ and the integral $\int_0^T \dot{\lambda}^T \delta x dt$. Choose the terminal condition $\lambda(T) = \nabla_x \Phi(x(T), p)$ and the backward differential equation

$$-\dot{\lambda} = f_x^T \lambda + \nabla_x L.$$

The coefficient of the arbitrary state variation inside the integral then vanishes. For a fixed initial state, the remaining parameter gradient is

$$\nabla_\theta J = \nabla_\theta \Phi + \int_0^T [\nabla_\theta L + f_\theta^T \lambda] dt.$$

If the initial state depends on θ , add $(\partial x_0/\partial \theta)^T \lambda(0)$; any terminal direct dependence is already in $\nabla_\theta \Phi$. For control parameters, $f_\theta = f_u \partial u/\partial \theta$ and similarly $L_\theta = L_u \partial u/\partial \theta$, summed over

the control channels. This method computes a gradient with one backward costate solve per scalar objective, independent of the number of parameter components in the leading solve cost. It is attractive for a large control vector and a small number of scalar objectives. If there are many separate objectives or outputs, forward sensitivities can be more efficient because each sensitivity pass yields derivatives for all outputs.

The derivation assumes differentiable dynamics and fixed event structure. Hybrid event times and contact switches add jump conditions to the costate and parameter sensitivity. Ignoring those terms yields incorrect gradients near impacts. A practical implementation should compare adjoint directional derivatives with finite differences on smooth test problems before using them in hybrid biomechanics.

Multiple Shooting

Multiple shooting divides time into intervals. At each interval boundary t_j , the state s_j becomes an explicit optimization variable. The dynamics are integrated locally from s_j using controls over that interval, and continuity constraints enforce $c_j = \Phi_{t_{j+1}, t_j}(s_j, \theta_j, p) - s_{j+1} = 0$.

The objective still compares states and predicted measurements to data. Compared with single shooting, multiple shooting introduces more variables and equality constraints, but it can improve conditioning because the optimizer can adjust intermediate states directly during early iterations. It can also initialize each segment from a measured or inverse-kinematic state estimate. A defect penalty may ease optimization, but final solutions must satisfy the continuity constraints tightly; otherwise the trajectory is a sequence of inconsistent segments, not one forward solution.

Shooting intervals should be short enough that local integration sensitivity is manageable, but not so short that the nonlinear program becomes dominated by unnecessary boundary variables. For a gait cycle, intervals may align with phases or contact events; for smooth movement, a uniform mesh may suffice. A measured trajectory supplies a strong initial state sequence, while control initialization can use inverse dynamics or static optimization. Continuity constraints then adjust the guess toward dynamic consistency.

Contact and Hybrid Events

Impacts create discontinuous velocity changes. A smooth ODE integrator must locate events and apply a reset map $x^+ = \Delta(x^-, p)$. Event times depend on states and parameters, which can make derivatives nonsmooth. One approach fixes contact times from data and solves a sequence of smooth phases with reset constraints. Another estimates event times or contact modes jointly, which is more general but increases nonconvexity. Penalty contact models smooth the transition but may introduce stiffness, requiring small integration steps and making sensitivities difficult.

Hybrid dynamics require explicit accounting for mode-dependent constraints. During stance, a foot may be constrained or compliant; during flight, contact force should be zero. Treating stance as a bilateral joint can generate attractive forces, while a penalty model can permit penetration. The event model should match the scientific use: if the task is to infer muscle control given measured contact timing, fixing the contact schedule is reasonable; if the model is meant to predict contact timing, fixing it would leak the answer.

Strengths and Limits

Single shooting is conceptually simple and guarantees dynamic consistency up to integration error. It has few decision variables and works well for stable systems, short horizons, and good initial guesses. It is often ideal for final validation even when another method finds the solution. Its weaknesses become severe for long, unstable, stiff, or contact-rich systems. Initial state error propagates; gradients can vanish or explode; a bad trajectory may provide no useful local direction.

Multiple shooting reduces long-range sensitivity and makes a data-informed initial trajectory available as optimization variables. It is often more robust for biomechanics trajectories with many observations. Its costs are larger nonlinear programs, continuity constraints, and the possibility of “cheating” during intermediate iterations by allowing defects. Tight convergence and independent reintegration are required. Neither method resolves model identifiability. If body dimensions and controls both influence the same observed motion, optimization can trade one against the other. Priors and multiple trials are still needed.

Regularization and Identifiability

A tracking estimator without control or parameter penalties can fit noise. A generic regularizer is $R = \lambda_u \int \|u(t)\|^2 dt + \lambda_s \int \|\dot{u}(t)\|^2 dt + (p - p_0)^T \Lambda_p (p - p_0)$,

where p_0 is a prior mean and Λ_p is a prior precision. The scales should correspond to prior uncertainty or a principled task objective. A free-running trajectory with low effort but poor fit is not useful; an exact data fit requiring actuator saturation and implausible dimensions is also suspect. Report each objective component separately and examine tradeoff curves rather than presenting only a weighted sum.

The regularized local precision approximation is $H = J_r^T W J_r + \Lambda$, where J_r is the residual Jacobian with respect to all decision variables. Strongly correlated columns reveal compensations: for example, larger muscle strength paired with lower activation can preserve force; a longer segment paired with altered joint-center position can preserve marker locations. Singular vectors identify combinations that data do not resolve. Adapt only parameters whose effects are distinguishable under the observed motions.

Worked Derivation: Why Shooting Can Be Ill-Conditioned

Consider the scalar unstable system $\dot{x} = ax + u$, with fixed $x(0) = 0$ and constant control u . Its terminal state is $x(T) = u(\exp(aT) - 1)/a$ for $a \neq 0$. The derivative with respect to control is $(\exp(aT) - 1)/a$, which grows exponentially when aT is large and positive. Thus a small control change creates a large terminal change, making a long single-shooting horizon sensitive. For stable $a < 0$ and large T , the terminal derivative approaches $-1/a$; early control effects disappear, so fitting early observations from a terminal objective becomes difficult. Multiple shooting inserts intermediate states and local continuity constraints, reducing the long sensitivity chain.

Practical Algorithm

1. Construct a fixed-parameter model and validate its forward integrator on short simulations.
2. Initialize $x(t_k)$ from IK or a state estimator; align clocks and contact phases.

3. Initialize controls using inverse-dynamics or a simple controller, then smooth and bound them.
4. Fit controls with geometry fixed, using multiple shooting if long-horizon conditioning is poor.
5. Add a small, interpretable parameter block with priors; inspect parameter-control correlation.
6. Increase control resolution only when residual structure supports it.
7. Tighten continuity and integrator tolerances, then perform an independent open-loop rollout.
8. Report state and observation residuals, control magnitude and smoothness, parameter values, defects, contact violations, and rollout error.

Exercises With Solutions

Exercise 1: Piecewise Constant Control Count

A 2-second trial has control knots every 0.1 seconds. How many piecewise-constant intervals are there, and what is the main tradeoff in doubling the knot rate?

Solution. There are 20 intervals. Doubling the knot rate approximately doubles control variables per input channel and increases flexibility and potential overfit; it may represent faster control changes but worsens conditioning and optimization cost.

Exercise 2: A Defect-Tolerant Solution

A multiple-shooting optimizer achieves tiny observation error but continuity defects of 0.2 meters at several interval boundaries. Is this a successful fit?

Solution. No. The state sequence is dynamically discontinuous. It cannot be generated by one forward integration. Continuity constraints are part of feasibility and must be driven below a physically meaningful tolerance, then checked by independent reintegration.

Exercise 3: Control-Parameter Confounding

A joint torque is $\tau = spu$, where s is muscle strength, p moment arm, and u activation. With only joint angle observations, can all three be uniquely identified?

Solution. Generally not. The product is what drives the motion, so scaling strength up and activation down or changing moment arm can preserve torque. Independent anatomical/EMG/force information or informative movements are required.

References

The direct-method comparison by Porsa, Lin, and Pandy reports direct shooting and collocation on a large musculoskeletal model: DOI 10.1007/s10439-015-1538-6, indexed in PubMed/repository records. A broad treatment of practical optimal control is available in (Betts 2010), alongside multibody dynamics context in (Featherstone 2008).

Direct Collocation for Dynamic Tracking and Model Fitting

Direct collocation turns a continuous-time optimal-control problem into a sparse nonlinear program by representing states and controls at a finite set of mesh points and enforcing approximate equations of motion as algebraic constraints. It is a natural approach when the goal is to fit a forward model to observed kinematics because the trajectory, control, model parameters, contact variables, and measurement residuals can be optimized together. The method is powerful, but it does not eliminate numerical modeling choices. Results depend on the transcription, mesh, scaling, solver, and independent feasibility checks.

Continuous Problem

Let $\dot{x} = f(x, u, p)$ over $t \in [0, T]$. A tracking objective may be written $J = \int_0^T [r_y(x, p, t)^T W_y r_y(x, p, t) + l_u(u) + l_p(p)] dt + \Phi(x(T), p)$,

subject to path constraints $g(x, u, p) \leq 0$, endpoint constraints, and perhaps a phase-dependent contact model. Here $r_y = h(x, p) - y$ is a measurement residual. If parameters are static, they are shared throughout the trajectory. If they are slowly varying, a state equation or temporal prior should describe that evolution explicitly; independently fitting each time point's dimensions makes no physical sense.

The optimal control problem is not just “fit a curve.” It asks for a curve that is dynamically feasible and has controls and parameters consistent with the model and priors. A trajectory can fit markers perfectly while violating force balance; a dynamically feasible trajectory can differ from noisy observed markers; a model can fit with implausible body dimensions unless priors and bounds prevent it. The objective and constraints make these preferences explicit.

Direct Collocation Transcription

Choose nodes $t_0, \dots, t_N$ and interval widths $h_k = t_{k+1} - t_k$. Treat x_k and typically u_k as decision variables. A simple trapezoidal defect is $d_k = x_{k+1} - x_k - (h_k/2)[f(x_k, u_k, p) + f(x_{k+1}, u_{k+1}, p)] = 0$.

The nonlinear program minimizes a quadrature approximation to J subject to all defects, bounds, path constraints, and boundary conditions. Higher-order schemes use midpoint states, polynomial interpolation, or collocation points inside each interval. Hermite-Simpson, for example, adds an interpolated midpoint and enforces consistency at it, improving accuracy for smooth dynamics without requiring an extremely fine mesh.

The key computational benefit is sparsity. Each defect couples only neighboring mesh variables, yielding a sparse Jacobian and Hessian structure. Large biomechanical problems may have many thousands of state, control, and parameter variables, but each local dynamics constraint depends on only a small subset. Sparse interior-point or sequential quadratic programming solvers can exploit that pattern. Automatic differentiation can provide derivatives, but model code must be differentiable and carefully scaled.

Collocation states are independent decision variables during optimization. Therefore they need not equal the result of integrating the controls from the initial state until the defect constraints converge. This freedom makes the NLP easier to initialize than single shooting, especially for unstable long-horizon problems. It also creates a vital obligation: after convergence, independently integrate the model using the optimized controls and parameters. If the trajectory differs materially, the mesh may be too coarse, defects too loose, controls interpolated inconsistently, or the contact/event treatment wrong.

Objective Construction

A useful objective separates data terms from regularization: $J_{data} = \sum_{k,s} \rho(L_{k,s}^{-1}(h_s(x_k, p) - y_{k,s}))$, $J_{reg} = \lambda_u \sum_k \|u_k\|^2 + \lambda_s \sum_k \|u_{k+1} - u_k\|^2 / h_k + (p - p_0)^T \Lambda_p (p - p_0)$.

The data term should use sensor-specific covariances or carefully justified weights. Robust losses can protect against outliers. Control effort, smoothness, parameter priors, and residual-actuator penalties should remain separately reported. An arbitrary weight can make a preferred outcome look optimal while hiding a large error elsewhere. Weights can be selected from measurement uncertainty, prior uncertainty, or an explicitly stated scientific tradeoff; sensitivity plots should show whether conclusions depend on them.

For marker tracking, compare predicted marker locations with measured marker positions. For coordinate tracking, compare generalized coordinates and speeds. For force tracking, compare external wrenches and centers of pressure. For EMG-informed fitting, compare processed EMG with excitation or activation only after defining the measurement-to-model transformation and delay. A multimodal objective is more informative only when its terms are calibrated and their correlations are considered.

Equality and Inequality Constraints

For a constrained mechanical system, one may impose position constraints $\phi(q, p) = 0$, velocity constraints $J(q, p)v = 0$, and acceleration-level equations. A redundant formulation can be numerically problematic. If the model already exposes a constrained dynamics solve, enforcing its differential-algebraic system consistently is preferable to independently adding inconsistent constraints. Alternatively, use reduced coordinates for simple constraints. Coordinate singularities, redundant constraints, and Baumgarte stabilization can introduce drift or artificial dynamics. Constraint residuals must be monitored at all levels.

Muscle states include activation bounds, fiber length bounds, tendon force relations, and force-length-velocity constraints. Contact adds unilateral normal force, nonpenetration, friction cones, and complementarity or a smooth approximation. Exact complementarity is nonconvex and difficult. Smoothing can improve derivatives but creates a parameter controlling approximation error. A compliant penalty law is simple but stiff. The chosen contact representation should be included in the model comparison rather than hidden in implementation details.

Mesh Refinement and Numerical Feasibility

An optimal solution to the discretized NLP is not automatically optimal or feasible for the continuous-time model. A coarse mesh may miss rapid transitions, impacts, or high-frequency muscle states. Mesh refinement should be based on defect error, state curvature, contact transitions, and reintegration mismatch. An h-refinement increases node density; p-refinement increases polynomial degree. Adaptive schemes place nodes where the solution changes rapidly. Report mesh spacing and demonstrate convergence of key outputs, not merely solver convergence.

Solver tolerances should be stated for optimality, constraint feasibility, and integration. Variables and constraints must be scaled to comparable numeric ranges. A meter-scale position, 1000 N muscle force, 0.01 activation, and $1e-6$ dynamic defect can otherwise produce poor conditioning. Nondimensionalization or explicit scaling can significantly improve behavior. Initial guesses matter: use data-derived states, an RRA/CMC solution, inverse-dynamics controls, or a simpler model. Continuation can gradually increase model complexity or tracking weight.

Collocation, Multiple Shooting, and Single Shooting

Single shooting has a compact decision vector but can be sensitive over long horizons. Multiple shooting adds segment states and continuity constraints. Collocation includes states at mesh points and local algebraic defect constraints, usually creating a larger but sparse NLP. The approaches can perform similarly on some predictive biomechanics tasks; practical performance depends on dynamics formulation, solver, mesh, and initialization. A fair comparison holds objective, bounds, mesh resolution, derivative quality, and stopping criteria as similar as possible. For systems with stiff muscle dynamics or contact, implicit formulations can improve stability and efficiency.

Collocation excels when the full trajectory is unknown and should be optimized, including motion prediction, tracking, and parameter estimation. A fixed observed motion can also be solved by inverse methods, but that suppresses the question of how parameters affect motion. Jointly optimizing state trajectory, controls, and model parameters exposes those interactions and lets the data pull the simulated motion while equations constrain it.

OpenSim Moco

OpenSim Moco provides a software framework for musculoskeletal optimal control using direct collocation. Moco supports motion tracking, prediction, parameter optimization, model fitting, EMG-constrained problems, and user-defined costs and constraints. Its MocoTrack workflow is designed for trajectory tracking while retaining optimization over the motion and controls. This is conceptually useful for forward matching: users can compare tracking loss, muscle effort, and model parameter changes in one OCP. Moco is a toolkit, not a guarantee of a scientifically well-posed objective. Users still need correct measurements, weights, priors, model scaling, mesh settings, and validation. The project's indexed documentation notes advanced users must understand optimal control, numerical methods, and local minima. The original Moco GitHub repository is archived and states development moved to OpenSim core, so implementation references should pin a current release and API.

Worked Derivation: Trapezoidal Defect Error

For scalar dynamics $\dot{x} = f(x)$, exact integration over a short interval is $x_{k+1} = x_k + hf(x_k) + (h^2/2)f_x(x_k)f(x_k) + O(h^3)$. Trapezoidal integration predicts $x_k + (h/2)[f(x_k) + f(x_{k+1})]$. Expanding $f(x_{k+1}) \approx f(x_k) + hf_x f + O(h^2)$ gives the same first and second order terms, leaving a local error of order h^3 for smooth dynamics. Over $N = T/h$ intervals, accumulated global error is typically order h^2 . This textbook order does not apply cleanly across impacts, discontinuous controls, nonsmooth contact, or stiffness; those require phase-aware refinement or different formulations.

Worked Example: Zero Endpoint Defect Can Miss the Interior Path

Consider $\dot{x} = \sin(\omega t)$ on one period $[0, 2\pi/\omega]$ with $x(0) = 0$. The exact solution is $x(t) = (1 - \cos(\omega t))/\omega$. It returns to zero at the final node, while reaching the interior maximum $2/\omega$ at the midpoint. A transcription or diagnostic that checks only endpoint values can therefore report a zero net change while completely missing the excursion. The trapezoidal formula on the whole period also gives zero increment because both endpoint derivatives vanish, so its interval defect is zero when both endpoint states are set to zero. Yet the within-interval state error is $2/\omega$ at midperiod.

Refine the interval by adding the midpoint $t = \pi/\omega$. The observed or collocated midpoint state now exposes the excursion; the two half-period increments are each $2/\omega$, and the node trajectory can represent the maximum. In real fitting, observations may not be available exactly at the most informative point, so evaluate dense-output defects and compare against independently integrated states at interior check points. If the model has characteristic frequency ω , an interval much longer than $1/\omega$ cannot resolve its curvature even if endpoint defects are small. Adaptive refinement should place nodes where dense integration error, state curvature, force changes, or contact transitions exceed a declared tolerance.

This example distinguishes algebraic feasibility of a coarse transcription from accuracy of the represented continuous trajectory. Refinement should be assessed on held-out interior checks as well as the objective: halve the maximum step, reintegrate, and verify that marker residuals, forces, peak states, and key parameter estimates converge. A model whose reported fit changes materially under modest mesh refinement is not numerically settled.

Failure Modes and Reporting

- Tiny mesh defects are reported but the solution is never reintegrated.
- A node-sampled control is applied with a different interpolation during rollout.
- Contact transitions occur between nodes and are missed.
- A coarse mesh smooths away activation or impact dynamics.
- Objective weights are tuned until the desired answer appears, with no sensitivity analysis.
- Parameters hit bounds, yet bound activity is omitted from interpretation.
- Local solver convergence is reported as evidence of global optimality.
- Constraints hold at nodes but not between nodes.

A defensible result reports the transcription scheme, mesh and refinement, solver and tolerances, variable scaling, initial guess, objective decomposition, active bounds, defect norms, path-constraint violations, and independent rollout metrics. If the problem has multiple initializations, report variation across them. If model parameters are adapted, report confidence intervals or profiles where possible.

Exercises With Solutions

Exercise 1: Defect Scaling

A trapezoidal transcription uses interval width $h = 0.02$ s. What happens to its asymptotic global error if a smooth solution's mesh is halved?

Solution. For a second-order method, global error scales approximately as h^2 . Halving h should reduce asymptotic error by about four, provided the problem is smooth and solver/integration errors are smaller.

Exercise 2: Why Reintegration?

A collocation fit has marker RMSE 3 mm and dynamic defects near solver tolerance. Why can rollout still be wrong?

Solution. The continuous integration error, control interpolation, off-grid constraints, contact event handling, and accumulated defect effects may differ from the transcription. Independent integration tests the actual forward model's ability to generate the fitted trajectory.

Exercise 3: Objective Weights

A 1 cm marker residual and 10 N force residual are added as raw squares. Why is the resulting sum uninterpretable?

Solution. The terms have different units and scales, and the numeric sum changes under unit conversion. Whiten by measurement covariance or define an explicit dimensionless tradeoff, then report each term separately.

References

Primary resources retrieved for this review: Dembia et al., OpenSim Moco, DOI 10.1371/journal.pcbi.1008493; Moco official overview; MocoTrack API documentation; Porsa et al., direct methods in OpenSim, DOI 10.1007/s10439-015-1538-6; Dorschky et al., direct-collocation tracking of 3D locomotion, DOI 10.1016/j.jbiomech.2017.04.038. The Moco and DOI pages were retrieved or indexed in the parent ledger; metadata for the locomotion paper's full author/volume citation remains pending.

Control Parameterizations and Continuation Methods

The Control Representation Is a Modeling Decision

A forward dynamics model maps initial state, controls, parameters, and numerical solver settings to a trajectory. Matching therefore depends strongly on how control is represented. Let the sampled dynamics be $x_{k+1} = \Phi_k(x_k, u_k, \theta)$, with observations $y_k = H(x_k) + \epsilon_k$. An optimizer may choose every control sample, a lower-dimensional basis, a feedback policy, or a structured actuator waveform. Each choice changes the feasible trajectories and the dimension and conditioning of the search. A good parameterization encodes known regularity and actuator structure without forcing the answer toward an unsupported prior.

Piecewise-constant controls are simple: one u_j applies over a time interval. Piecewise-linear controls reduce discontinuities, but require a precise endpoint and interpolation rule. Cubic splines provide smooth derivatives, although unconstrained spline coefficients can overshoot actuator limits between knots. B-splines with nonnegative basis functions make convex bounds easier to enforce. Fourier bases compactly represent periodic motion but impose periodicity and can ring near abrupt events. Wavelets or local radial basis functions give localized flexibility at the cost of more coefficients.

Control knot placement should follow physics and data. Uniform time knots can waste variables in slow phases and undersample rapid transitions. Event-aligned knots may help at impact or release, but event time itself may be uncertain. Adaptive knot refinement should be driven by a declared residual or defect criterion and logged as part of the optimization. Comparing a high-knot model to a low-knot model without matching regularization or effective degrees of freedom is not a fair comparison.

Regularization and Its Interpretation

A common objective adds effort, magnitude, or smoothness penalties:

$$J(u, \theta) = J_{obs} + \lambda_0 \int_0^T \|u(t)\|^2 dt + \lambda_1 \int_0^T \|\dot{u}(t)\|^2 dt + \lambda_2 \int_0^T \|\ddot{u}(t)\|^2 dt.$$

The magnitude penalty discourages large controls. The first derivative penalty discourages rapid changes; the second derivative penalty discourages jerk-like variation. These terms are not interchangeable and have different physical units. Their weights should be scaled from actuator limits, sensor noise, or explicit priors. Report the normalized contributions at the fitted solution; a tiny residual with an enormous control penalty can indicate scaling failure, while a negligible penalty means regularization is inactive.

Smoothness penalties can misrepresent actual control systems. Muscle activation has activation dynamics and bounds; a motor can have rate limits; neural feedback may have latency; passive tissues contribute force without commanded excitation. If the model already contains these actuator dynamics, penalizing the command derivative may double-count smoothness. If physiological control is not observed, an effort penalty is a modeling assumption, not recovered neural truth.

Control and model parameters are often confounded. Greater torque can compensate for larger inertia; gain can trade against damping; an unmodeled passive force can be represented by a biased control. Analyze local identifiability with the joint sensitivity matrix over controls and physical parameters. Fixing selected parameters may be necessary, but the fixed values and resulting conditional interpretation must be explicit. Better excitation or independent force measurements are often more useful than a more powerful optimizer.

Direct Transcription and Shooting

Single shooting chooses controls and propagates from one initial state. It has relatively few explicit state variables but long products of transition Jacobians. This can cause exploding or vanishing gradients and makes early control changes affect the full remaining trajectory. Multiple shooting introduces state variables at segment boundaries and constrains them with defects:

$$d_j = x_{j+1} - \Phi_j(x_j, u_{j:j+1}, \theta) = 0.$$

It enlarges the optimization but improves conditioning, enables parallel segment simulation, and makes initialization more flexible. Defect penalties alone do not enforce exact dynamics; final candidates need defect tolerances or constrained solves.

Direct collocation approximates the state within each interval using polynomial basis functions and enforces the dynamics at selected collocation points. It transforms trajectory matching into a sparse nonlinear program. Collocation is especially useful when controls, states, and path constraints must be optimized together. Its solution is a discretized trajectory; independent integration of the resulting controls is essential because collocation defects and between-node interpolation can still yield replay error.

Differential dynamic programming and iterative LQR use local models of dynamics and cost to improve a nominal control sequence. They can be efficient for smooth systems and provide feedback gains around the trajectory. Their performance depends on derivatives, regularization, and local basin. Crocodyl's FDDP approach is an example present in upstream matching paths, while other engines expose different optimization interfaces. Treat them as algorithm families with specific dynamics and discretization, not a universal engine-independent solver.

Continuation and Homotopy

A difficult fit can be solved through a family of easier problems. Let $J(z; \alpha)$ depend on continuation parameter $\alpha \in [0, 1]$. Start at $\alpha = 0$ where the problem is simplified, then gradually move toward the target at $\alpha = 1$, using each solution to initialize the next. Possible schedules include gradually adding high-frequency controls, reducing smoothing, tightening constraint tolerances, increasing contact sharpness, increasing observation weights, or moving from a surrogate to the final engine.

For example, replace a nonsmooth hinge $\max(0, -\phi)$ with softplus $s_\kappa(\phi) = \kappa^{-1} \log(1 + e^{-\kappa\phi})$. Small κ gives a smoother transition; increasing κ approaches the hinge. Yet the fitted solution can follow a branch that becomes infeasible or unstable as κ grows. At every stage, track objective components, constraint residuals, event schedule, and parameter movement. If a

continuation step causes a large jump, reduce the increment, try a different path, or report branch sensitivity.

Continuation is not proof of global optimality. Multiple branches may exist, especially in closed-chain kinematics and hybrid contact. Use multiple initializations, record unsuccessful branches, and compare candidates under the final unsmoothed model. Never report the easier surrogate’s objective as the final physical fit. A schedule can also bias the solution; compare plausible schedules on held-out predictions when continuation materially determines the outcome.

Derivative-Free Optimization

Derivative-free methods are useful when gradients are unavailable, unreliable at hybrid transitions, or too costly to validate. Nelder-Mead, Powell search, pattern search, covariance matrix adaptation, Bayesian optimization, and evolutionary methods have different assumptions and scaling. They can handle black-box objectives but suffer rapidly as dimension grows. A free control at hundreds of time knots is generally a poor candidate for naive derivative-free search.

Reduce dimension first through control bases, block coordinate updates, or physically informed parameter groups. Use derivative-free search for a small set of difficult event or model-choice variables, then gradient-based local refinement within a smooth mode. Bayesian optimization is suitable for expensive low-dimensional tuning, not usually for a high-dimensional control sequence. Covariance adaptation may handle correlated low-dimensional parameters but its covariance estimate becomes costly as dimension grows. Report objective calls and wall time, not just generations.

Hybrid optimization can combine methods: use derivative-free initialization for event timing, then optimize continuous controls with verified gradients; use deterministic grid sweeps for discrete schedules; or compare candidate dimensions with information criteria before continuous refinement. Ensure every candidate is evaluated by the same authoritative objective and final replay. A derivative-free method is not automatically safer: a discontinuous failure penalty can create arbitrary optima, and stochastic search may exploit simulation artifacts.

Proposed Adaptive Control-Dimension Strategy

An adaptive basis can begin with a small number of control modes and add local basis functions where the trajectory defect or held-out residual warrants more freedom. Let $u(t) = \sum_{i=1}^m c_i \psi_i(t)$. A proposed refinement indicator combines dynamics defect, observation sensitivity, and coefficient uncertainty; basis expansion is allowed only when added modes improve held-out predictive likelihood or a predeclared penalized objective. Prune modes with negligible posterior contribution only after checking that they are not required in a brief high-acceleration phase.

The model should learn control dimension separately from anatomical generalized-coordinate dimension. Otherwise it can confuse richer actuation with richer body structure. A hierarchical prior could place sparsity on control coefficients while a separate discrete prior governs structural dimensions. Cross-validation must hold out entire trials or participants when the claim concerns generalization. This strategy is proposed for future work; it is not an existing upstream feature.

Upstream Evidence and Caveats

Upstream contains Drake fitting, Pinocchio/Crocoddyl FDDP, and MuJoCo knot optimization implementations under `src/engines/physics_engines`. Their parameterizations and discrete dynamics differ. Before calling one “best,” compare actual control variables, interpolation, effort terms, bounds, contact handling, and final evaluator. The immutable source links for this chapter should point to the audited repository revision `d7c7d07add0d7407a5a3376d5fe4a97aee9112a0`; a revision link documents code identity, while evidence receipts establish what a particular run did.

Worked Illustrative Example

Consider an illustrative unit-mass oscillator $\ddot{q} + q = u$ over a fixed interval. A single constant control may fit the final position but miss an observed mid-trajectory reversal. Adding a second knot permits a sign change and may reduce residual, while an $\int \dot{u}^2 dt$ penalty resists that change. A cubic basis may fit smoothly but overshoot force bounds between knots. This simple example demonstrates that parameterization controls the model’s expressive power; it is not empirical evidence about a human swing.

Exercises and Solutions

Exercise 1. When can multiple shooting improve an optimization without changing the physical model? **Solution.** It changes the decision variables and numerical conditioning while retaining the same continuous segment dynamics and continuity constraints. If defects remain nonzero in the final result, however, the candidate does not satisfy that model.

Exercise 2. Why can a smoother control produce a worse forward match? **Solution.** The measured trajectory may include rapid but real changes, or actuator dynamics may already account for smoothness. Over-regularization biases the model and worsens prediction.

Exercise 3. What is required after a homotopy fit? **Solution.** Evaluate using the target objective and final dynamics, verify constraints and events, run independent forward replay, and report sensitivity to continuation schedule and initialization.

Further Reading

The AffineDrift bibliography includes (Jacobson and Mayne 1970) for differential dynamic programming and (Olson 2026c) for local model residuals. These references provide context; implementation-specific claims require the pinned source and candidate receipt.

Choosing a Representation in Practice

The best representation depends on which uncertainty dominates. If measurement timing is reliable but control is unknown, a compact basis with smoothness prior may be appropriate. If the trajectory contains impacts, fixed uniform knots can place no decision point near the event; optimize event times or use a hybrid segmentation. If actuator commands are bounded, encode bounds in the parameterization or enforce them at all quadrature points, not just knot endpoints. If control observations exist, treat them as data with calibration uncertainty rather than a prior of infinite strength.

Assess representation bias by fitting simulated data generated from a richer known control and measuring recovery as knot density changes. Inspect residual autocorrelation: systematic high-frequency residual structure suggests missing flexibility, while isolated noise fitting suggests overparameterization. Plot control uncertainty bands, not only one optimized waveform. A parameterization can yield a stable prediction while individual coefficients remain nonidentifiable, especially when adjacent spline bases overlap strongly.

When increasing basis dimension, warm-start old coefficients and initialize new coefficients at zero so the prior trajectory is preserved. Recompute derivatives and verify the expanded model reproduces the old rollout before optimization. This nested property makes refinement diagnostically interpretable. If the baseline changes at expansion time, compare objective changes only after correcting the embedding or documenting the intentional model change.

Constraints, Contact, and Hybrid Events

The Mathematical Structure

Rigid-body forward matching is not a smooth ODE everywhere. A model may move freely, enter sustained contact, slide, stick, release, or undergo an impulsive collision. Each mode has its own continuous equations and admissible state set. Treating those changes as incidental details of a simulator hides a major source of model mismatch.

For holonomic constraints $g(q) = 0$, the velocity constraint is $J(q)v = 0$, where $J = \partial g / \partial q$. Acceleration consistency gives $J\dot{v} + \dot{J}v = 0$. The constrained acceleration and reaction force solve the saddle system

$$\begin{bmatrix} M & -J^T \\ J & 0 \end{bmatrix} \begin{bmatrix} \dot{v} \\ \lambda \end{bmatrix} = \begin{bmatrix} \tau - h \\ -\dot{J}v \end{bmatrix}.$$

Sign conventions vary; the invariant requirement is that the computed acceleration satisfy dynamics and constraint equations to tolerance. Redundant constraints make the KKT matrix singular unless dependencies are removed or a principled regularization is used. A penalty formulation instead adds approximately $-kg - d\dot{g}$ forces. It is easier to integrate but introduces stiffness, penetration, and a parameter-dependent approximation to ideal constraints.

Unilateral contact is described by gap $\phi(q) \geq 0$ and normal force $\lambda_n \geq 0$ with complementarity $\phi\lambda_n = 0$. Coulomb friction imposes a tangential force bounded by $\mu\lambda_n$, with sticking/sliding mode selection. Complementarity is nonsmooth. Solvers use complementarity problems, convex cones, regularized forces, or smoothed approximations. Each choice defines a different forward model and derivative. A soft-contact law may be useful for gradients and stable fitting, but fitted stiffness and damping partly encode discretization and are not automatically physical material constants.

Impact is a reset map. If pre-impact velocity is v^- , impulse p changes it to $v^+ = v^- + M^{-1}J^T p$. A Newton restitution rule sets post-impact normal velocity from pre-impact normal velocity; perfectly inelastic impact uses zero normal separation speed. The impact solution must respect nonnegative impulse and nonpenetration. The state jumps even though position is usually continuous. A single smooth integrator step across impact smears the event and can yield incorrect energy loss and timing.

Event Time Changes the Gradient

Let an event occur when $g(x, t) = 0$. A small perturbation δx changes the event time. Linearization gives

$$\delta t_e = -\frac{g_x \delta x^-}{g_t + g_x f^-},$$

provided the denominator is nonzero (transversal crossing). After a time-independent reset $x^+ = R(x^-)$, perturbations transported to a common post-event time are mapped by the saltation matrix

$$\Xi = R_x + \frac{(f^+ - R_x f^-)g_x}{g_t + g_x f^-}.$$

This expression shows why differentiating only the reset map misses event-time sensitivity. Near grazing contact the denominator approaches zero and sensitivity can become very large; this is an intrinsic conditioning issue, not necessarily an AD bug. With multiple simultaneous contacts, event ordering and reset composition require explicit policy.

The event map is often only piecewise differentiable. If perturbations change which event occurs first, the classical derivative may not exist. An optimizer can exploit mode switching, producing a lower training residual but an implausible or unstable trajectory. Report mode sequence and event times alongside the scalar fit score. For matching, contact timing may be part of the observed evidence, so event residuals should be first-class measurements rather than hidden implementation choices.

Formulations for Fitting

There are four useful formulations, with different tradeoffs.

1. **Penalty or compliant contact.** Replace constraints with smooth forces. Standard ODE solvers and AD can be applied, but stiffness limits step size and smooth approximations can bias impact. Include contact law, smoothing width, integrator, and tolerance in the model identity.
2. **Complementarity or constrained time stepping.** Solve contact impulses/forces jointly at each step. This captures nonpenetration robustly but active-set changes make derivatives piecewise. Use a solver that exposes the actual discrete residual and, if differentiating, define a derivative for the selected active set.
3. **Multiple shooting with mode schedule.** Optimize states and controls at segment boundaries while enforcing each segment's dynamics. Fix or enumerate contact modes during a local solve, then validate alternative schedules. This can improve conditioning and localize mismatch.
4. **Hybrid trajectory optimization.** Optimize continuous phases, event times, and reset conditions explicitly. It is the most faithful representation when event timing matters, but introduces additional variables and transversality constraints.

A fifth option is to fit a smooth surrogate first and use it only as an initializer for the physical hybrid model. The final candidate must be re-evaluated with the declared target law; surrogate objective success is not physical validation.

Contact-Aware Objective and Diagnostics

A defensible objective separates kinematic fit from event and physical penalties:

$$J = J_{\text{markers}} + w_q J_{\text{orientation}} + w_t J_{\text{event-time}} + w_c J_{\text{contact}} + w_u J_{\text{effort}} + w_p J_{\text{prior}}.$$

For measured contact onset t_i , $J_{\text{event-time}} = \sum_i ((\hat{t}_i - t_i)/\sigma_{t_i})^2$. Contact consistency can penalize penetration, negative normal force, slip during inferred sticking, and energy creation at impact. Do not score an unobserved contact state as if it were measured: distinguish measured, inferred, and assumed mode labels. If force plates or instrumented surfaces exist, use their uncertainty model; if not, report contact as a model prediction.

Useful diagnostics include maximum and RMS constraint violation, penetration depth, complementarity product, friction cone violation, impulse magnitude, event ordering, energy jump, and solver convergence. Report these per time interval. A low marker RMSE can hide substantial penetration or energy injection. Conversely, a contact law that is too stiff can worsen marker fit by making the numerical system unresolvable at the selected step size.

Repository Evidence

The upstream evidence includes replay notes at `evidence/matched/driver_full_mujoco_replay/README.md`, which state that a regularized Coulomb contact law is reused while stock MuJoCo contacts stay disabled in that replay. The README also records that integrator refinement changed marker positions substantially, a concrete warning that solver and contact details alter the result. The corresponding Crocoddyl replay README records the same shared contact-law approach for that candidate. The repository’s MuJoCo contact implementation and tests are under `src/engines/physics_engines/mujoco/python/motion_matching` and `tests/unit/motion_matching/test_contact_law.py` (the evidence README lists the qualification test command). These facts establish a particular replay configuration, not universal equivalence between engines or experimentally measured contact truth.

The matching inventory spans Drake, MuJoCo, Pinocchio/Crocoddyl, OpenSim, MyoSuite and Simscape evidence. They do not share a single mathematical contact formulation. A parity comparison therefore needs a contact-law statement, not just an engine name. For the adaptive-dimension model proposed by the project, a newly activated latent coordinate should not be allowed to absorb contact-model error without a model comparison that separates those explanations.

Proposed Design for Adaptive Dimension Under Contact

Treat model dimension d as a discrete model index and continuous parameters θ_d as within-model variables. A proposed objective is

$$J_d = J_{data,d} + \lambda_d d + \lambda_{event} J_{event,d} + \lambda_{phys} J_{phys,d},$$

where the complexity penalty discourages adding dimensions merely to chase noise. Candidate dimensions should be compared on held-out trials, with the same observation masks, event law, and compute budget. Evidence for a new dimension should include improved predictive log likelihood or a predeclared information criterion, stable parameter estimates under resampling, and a mechanistic interpretation. Selection uncertainty should be retained rather than presenting one dimension as certain.

For hybrid systems, dimension proposals should be localized by phase: an extra generalized coordinate may be warranted for free motion but redundant during a constrained phase. Any dimension change across a phase boundary must have a defined state embedding and velocity mapping that preserve physical quantities as appropriate. This is a research proposal; upstream code currently offers multiple fit configurations, not a general Bayesian or adaptive-dimension model selector.

Worked Impact Example

In an illustrative unit-mass example, a mass approaches a rigid wall at $v^- = -2$ m/s. With restitution $e = 0.5$, the reset gives $v^+ = -ev^- = 1$ m/s (positive away from the wall under this sign convention). Kinetic energy changes from 2 J to 0.5 J; 1.5 J is dissipated. A method that

simply clips penetration without a velocity reset can report zero gap while retaining inward velocity, causing immediate re-penetration. This illustrates why position agreement alone cannot qualify contact handling.

Exercises and Solutions

Exercise 1. Why can a penalty stiffness that improves penetration worsen the fit? **Solution.** Increasing stiffness reduces static penetration but increases the fastest natural frequency approximately as $\sqrt{k/m}$. A fixed time step then under-resolves the contact transient, creating phase and force errors. Damping and numerical dissipation also change the trajectory.

Exercise 2. Which evidence is needed to claim the fitted contact coefficient is a physical friction coefficient? **Solution.** The force law and units, independent calibration or measured tangential/normal forces, identifiability analysis, sensitivity to timestep and regularization, and uncertainty intervals. Matching kinematics alone generally confounds friction with control, geometry, and actuator effects.

Exercise 3. What does a small denominator in the saltation formula mean? **Solution.** The trajectory crosses the event surface nearly tangentially. Event time becomes highly sensitive to state perturbations; uncertainty and numerical error are amplified. The model may require finer event resolution or a probabilistic treatment.

Further Reading

The AffineDrift bibliography includes (Olson 2026b) and (Goebel et al. 2012) for hybrid flow and reset concepts. See also the maintained contact and replay evidence paths above. Distinguish mathematical references from empirical validation of any particular candidate.

Numerical Choices and Conservation Checks

Constraint stabilization controls drift in $g(q)$. Baumgarte stabilization modifies acceleration constraints with terms proportional to α and β , damping position and velocity error; excessive gains make the system stiff, while weak gains permit drift. Projection methods instead correct the state onto the constraint manifold after a step, but the projection itself changes the discrete map and must be differentiated if used in gradient fitting. Position projection without a compatible velocity projection can produce a state that is geometrically valid but dynamically inconsistent.

For ideal bilateral constraints, verify residuals ($|g(q)|$), ($\|Jv\|$), and $\|J\dot{v} + \dot{J}v\|$. For conservative free motion, monitor total energy and momentum. Energy drift should decrease predictably with step refinement for the selected integrator. With damping and impact, the expected energy balance includes dissipated work and impulse losses; “energy conservation” is then the wrong criterion. For contact, monitor the maximum penetration and complementarity residual, but also evaluate sensitivity to contact regularization. A model that meets a tolerance only at one hand-selected timestep is not numerically robust.

Inter-engine comparison should be staged. First compare kinematics from the same generalized coordinates. Then compare unconstrained accelerations under identical force vectors. Next compare constrained accelerations in a simple fixture with analytically known contact. Only after those agree should a complex swing replay be compared. A discrepancy can come from joint-axis definitions, gravity sign, rotor armature, friction, actuator transmission, contact activation, integrator, or observation maps. Logging each stage makes the investigation falsifiable.

When a contact mode is inferred from video or marker data, encode uncertainty in the event label. One approach is to compare candidate schedules and marginalize their likelihoods; another is to use a soft probability for contact onset while retaining a hard mechanics model within each hypothesis. A smooth probability does not make the physical state itself smooth, so report the approximation and validate the selected schedule with independent replay. For uncertain onset $t_c \sim p(t_c)$, event likelihood can be evaluated across plausible onset times instead of fixing one timestamp and overstating precision.

A contact benchmark should vary approach speed, incidence angle, friction, and timestep independently. One easy normal collision is not enough to validate a frictional law. Include separation with zero force, low-speed grazing, stick-slip transition, and simultaneous contact cases; state clearly which are supported and which remain outside the qualified domain.

Differentiable Simulation and Gradient Checking

Why Gradients Matter in Forward Matching

Forward matching asks for parameters and controls whose simulated motion explains measured kinematics. Let the generalized configuration be $q \in \mathcal{Q}$, velocity $v \in T_q\mathcal{Q}$, control u , and unknown physical or controller parameters θ . A continuous model has

$$\dot{q} = N(q)v, \quad M(q; \theta)\dot{v} + h(q, v; \theta) = B(q; \theta)u + J(q)^T\lambda,$$

where N maps generalized velocity to configuration derivative, M is inertia, h collects gravity, Coriolis and passive forces, and contact multipliers λ enforce active constraints. A forward solver maps the initial state and controls to predicted observations, $\hat{y}_k = H(q_k, v_k; \theta)$. A weighted fit objective might be

$$J(\theta, u) = \frac{1}{2} \sum_k r_k^T W_k r_k + \frac{1}{2} \sum_k \|u_k\|_R^2 + \frac{1}{2} \|L(\theta - \theta_0)\|^2, \quad r_k = \hat{y}_k - y_k.$$

The gradient is valuable because the decision dimension grows with time knots, actuator channels, body parameters, and latent model dimensions. But the computed gradient is only meaningful if it differentiates the same discrete map that is evaluated and qualified.

The distinction between continuous and discrete derivatives is foundational. If $x_{k+1} = \Phi_h(x_k, u_k, \theta)$ is the actual numerical step, optimize the composition of these steps. Differentiating the differential equation and then approximating its sensitivity with a different step can yield a gradient that does not point downhill for the implemented objective. For explicit Euler, $x_{k+1} = x_k + hf(x_k, u_k, \theta)$; its exact discrete sensitivity is $A_k = I + hf_x$, $B_k = hf_u$, and $E_k = hf_\theta$. Runge-Kutta and adaptive solvers have more complex computational graphs, including stage evaluations and potentially step-size decisions.

Three Derivative Routes

Forward sensitivity propagates $S_k = \partial x_k / \partial \theta$:

$$S_{k+1} = \Phi_x S_k + \Phi_\theta.$$

It is attractive when there are few parameters and many outputs. Reverse mode propagates a costate backward. For a discrete objective $\ell_N(x_N) + \sum_{k=0}^{N-1} \ell_k(x_k, u_k, \theta)$,

$$\bar{x}_N = \nabla_x \ell_N, \quad \bar{x}_k = \nabla_x \ell_k + \Phi_x^T \bar{x}_{k+1}, \quad \nabla_\theta J = \sum_k (\nabla_\theta \ell_k + \Phi_\theta^T \bar{x}_{k+1}).$$

This is usually preferable when the parameter/control vector is large and the scalar loss has many residual components. Automatic differentiation (AD) implements forward or reverse chain rules through code; AD is not synonymous with differentiable physics. Dynamic array mutation, foreign simulator calls, contact branches, hidden state, and adaptive control flow can block or distort derivatives.

A third route is adjoint sensitivity for continuous dynamics. It can reduce memory cost by integrating a costate backward, but requires accurate reconstruction of the forward trajectory and correct treatment of events. It is not automatically equivalent to reverse differentiation through a particular solver. Checkpointing trades memory for recomputation. In long swing trajectories, checkpoint boundaries should be deterministic and stored in provenance.

Matching Observation Geometry

The observation residual must respect its geometry. For marker positions, $r = p_{\text{model}} - p_{\text{data}}$ is a Euclidean residual after both systems use the same frame, unit, and marker definition. For rotations, subtracting Euler angles can create branch discontinuities. Use a group residual such as $r_R = \text{Log}(R_{\text{data}}^T R_{\text{model}}) \in \mathbb{R}^3$, with a declared convention and a safeguard near the logarithm cut locus. For joint angles, wrap only genuinely periodic coordinates; do not wrap bounded hinge angles across physically distinct configurations. A robust loss (Huber, Cauchy) changes the derivative and should be stated explicitly. Missing observations should be masked in both residual and normalization; dividing by a fixed total count while observations disappear silently changes effective weights.

Weights carry units. If position noise has standard deviation σ_p , a statistically interpretable term is $r_p^T \Sigma_p^{-1} r_p$, not an arbitrary sum of millimetres and radians. Whitening improves optimizer scaling and exposes which sensors dominate. Parameter scaling also matters: optimize dimensionless transformed variables, for example log mass or log positive stiffness, then apply chain rule back to physical parameters. Bound transforms must not make near-bound derivatives vanish unintentionally.

Gradient Verification as an Evidence Ladder

A plausible loss decrease does not establish correct derivatives. Begin with analytic toy systems where the exact solution and gradient are known. For a scalar function f , compare AD derivative g with centered differences

$$g_{FD}(\epsilon) = \frac{f(\theta + \epsilon d) - f(\theta - \epsilon d)}{2\epsilon}, \quad e = \frac{|g^T d - g_{FD}|}{\max(1, |g^T d|, |g_{FD}|)}.$$

Sweep epsilon logarithmically: truncation error decreases as $O(\epsilon^2)$, while floating-point cancellation grows as epsilon shrinks. A single epsilon can accidentally look good. Use randomized normalized directions d , coordinate checks for selected high-impact parameters, and report absolute plus relative discrepancy.

Test every layer separately: observation residual; one integrator step; rollout; total objective; parameter transforms; and any custom contact derivative. Compare AD against finite differences at smooth interior states. Then test a full small trajectory, followed by the production-sized case. Validate gradients at several states and parameter regimes, especially near bounds and near—but not across—events. Finite differences across an impact compare different event sequences and cannot validate a smooth local derivative. For an event, perturbations must be categorized by whether they preserve event order; otherwise use a hybrid derivative with event-time sensitivity or report nonsmoothness.

A useful directional test checks the first-order Taylor remainder:

$$\rho(\epsilon) = \frac{|J(\theta + \epsilon d) - J(\theta) - \epsilon \nabla J^T d|}{\epsilon}.$$

For a correct derivative in a smooth region, this tends toward zero linearly until roundoff dominates. A second-order test divides by ϵ^2 when Hessian-vector products are claimed. Gradient checks must use identical initial conditions, interpolation, integrator tolerances, contact law, and random seeds. Otherwise disagreement measures a changed experiment rather than derivative error.

Upstream Implementation Evidence and Limits

The repository contains a Drake AutoDiff fitting path in `src/engines/physics_engines/drake/python/motion_m` and a test named in its module docstring, `tests/test_drake_fit_swing_autodiff.py`, guarding an AutoDiff issue. MuJoCo includes analytical Jacobian utilities at `src/engines/physics_engines/mujoco/python/m`. Pinocchio/Crocoddyl uses action-model derivatives and FDDP in `src/engines/physics_engines/pinocchio/pyth` and `full_body_fit.py`. These are distinct derivative routes, not evidence that every matching route has interchangeable or independently certified gradients. The Crocoddyl action uses implicit Euler; Drake and MJX paths have their own discrete semantics. Any comparison must align objectives, controls, masks, contact, and time grids before interpreting score differences.

A proposed improvement is a shared derivative qualification suite whose backend-neutral fixture records the exact step map, observation map, and parameter vector, then runs directional checks per engine. It should emit a machine-readable receipt with tolerances, seed, smooth/event classification, normed discrepancies, and environment identity. The suite must not certify event-crossing gradients using smooth finite differences. This is a proposal, not a current repository capability.

Worked Illustrative Scalar Example

Let $x_{k+1} = ax_k + bu_k$, $x_0 = 1$, with one observation $y_2 = 2$, $u_0 = u_1 = 1$, and loss $J = \frac{1}{2}(x_2 - y_2)^2$. Since $x_2 = a^2 + ab + b$,

$$\frac{\partial J}{\partial a} = (x_2 - 2)(2a + b), \quad \frac{\partial J}{\partial b} = (x_2 - 2)(a + 1).$$

At $a = 0.5, b = 0.4, x_2 = 1.05$, so the gradient is $(-1.61, -1.725)$. This tiny example exercises rollout and reverse accumulation. An implementation reporting another derivative has a defect before any biomechanics complexity is introduced.

Exercises and Solutions

Exercise 1. Explain why differentiating continuous dynamics with a continuous adjoint may disagree with centered differences of a fixed-step numerical rollout. **Solution.** The finite differences estimate the derivative of the discrete map Φ_h^N ; the continuous adjoint estimates sensitivity of the ODE flow. Their difference is integration and discretization error, and can be amplified by unstable or stiff dynamics. Convergence of both as $h \rightarrow 0$ is informative, but does not make them identical at finite h .

Exercise 2. A marker residual has standard deviation 4 mm, while a joint-angle residual has standard deviation 0.02 rad. Give normalized squared terms. **Solution.** Use $r_p^2/(4 \text{ mm})^2$ and

$r_q^2/(0.02 \text{ rad})^2$, assuming independent scalar errors. Correlations require the full covariance inverse.

Exercise 3. Why is a gradient check through impact ambiguous? **Solution.** Perturbations can shift event time, change impact ordering, or switch contact modes. The loss can be nonsmooth or piecewise smooth; a centered difference may straddle branches and estimate neither one-sided derivative. Check mode-preserving directions separately and use a hybrid event derivative when justified.

Further Reading

Use the repository bibliography entries (Olson 2026c) for residual-aware tangent approximations, (Olson 2026b) for event sensitivities, and standard AD or optimal-control references linked from the main bibliography. Local implementation paths above identify evidence; the proposed qualification harness remains future work.

Conditioning, Identifiability, and Optimizer Use

A derivative can be correct yet unhelpful. If two columns of the sensitivity matrix $S = \partial \hat{y} / \partial \theta$ are nearly collinear, the observations respond similarly to two parameters. The local information matrix $F = S^T W S$ then has a large condition number, so small measurement or numerical errors cause large parameter changes. Singular-value decomposition $S = U \Sigma V^T$ reveals which combinations are observable: right singular vectors associated with small singular values are weakly informed directions. Report singular values at the fitted solution and repeat after changing the motion window or observation channels. A low residual alone says nothing about uniqueness.

The optimizer sees gradients in the chosen parameterization. If θ is transformed from unconstrained z , then $\nabla_z J = (\partial \theta / \partial z)^T \nabla_\theta J$. Log transforms are natural for positive quantities but can make the physical gradient scale strongly depend on the current value. Normalize control and parameter coordinates so one unit in the optimizer has a roughly meaningful effect. A trust-region method can be safer than a large quasi-Newton step when the local linearization is poor. Gradient clipping may stabilize a run but changes the search direction; log clipping events and compare with an unmodified diagnostic gradient.

For trajectory optimization, adjoint gradients can be dominated by long-horizon instability. If a discrete dynamics Jacobian has expanding directions, backward costates can grow exponentially. Multiple shooting limits propagation length by introducing intermediate state variables and continuity defects. This costs variables but often improves conditioning and makes failed segments local. A practical diagnostic plots the norm of state sensitivity and adjoint by time, then checks whether high gradients coincide with contact transitions, poorly observed phases, or unstable modes. Do not respond automatically by increasing regularization: first determine whether the issue is physical identifiability, numerical integration, or a derivative defect.

Gradient Reliability Reporting Template

For a published differentiable fit, provide: objective definition and units; derivative method and software version; exact discrete integrator and tolerance; custom derivative rules; gradient check directions and seeds; epsilon sweep; maximum and median normalized directional error; tested parameter/state regions; event exclusion or hybrid treatment; and whether checks ran on the same compiled path used in optimization. If a component is nondifferentiable, report the reason.

tiable, document the optimizer’s treatment (subgradient, smoothing, active-set derivative, or derivative-free step). A statement such as “uses autodiff” omits all of these facts.

A compact reliability table can classify each term as smooth/verified, smooth/unverified, piecewise smooth/event-sensitive, or derivative-free. Include data preprocessing in the graph: if the objective resamples or filters input signals based on parameters, those operations contribute derivatives too. If preprocessing is fixed, state that it is outside the optimization. This prevents the false impression that every part of the scientific pipeline is differentiated.

A Practical Check Sequence

A reviewable sequence is: freeze a tiny deterministic fixture; validate forward values; verify local derivatives; check rollout derivatives; run a small optimizer; then evaluate a separate longer trajectory. Preserve every fixture and tolerance as data. This sequence locates defects cheaply: if the scalar transform fails, there is no reason to investigate contact or optimizer convergence yet. Once the pipeline passes, keep a representative gradient report attached to the run receipt so later library upgrades can be compared against an explicit baseline.

For every claim about gradient quality, state the tested domain explicitly; a check at one nominal pose does not cover the full parameter box or a different contact branch. Keep failed checks, explain root cause, and rerun after correction.

Muscle-Driven Matching of Forward Dynamics Models

Muscle-driven matching asks which control history, model parameters, and sometimes state trajectory allow a musculoskeletal model to reproduce measured movement while obeying its dynamics. It is a stronger test than prescribing the measured kinematics into the model: the motion must emerge from actuators and external forces. It is also a more underdetermined problem. Many muscle recruitment patterns can explain similar motion, measurements do not directly reveal neural commands, and model properties such as tendon slack length or maximum isometric force can trade off with activation.

From Excitation to Motion

A simplified Hill-type muscle-tendon model transforms neural excitation e to activation a , activation and muscle-tendon state to force F_m , and muscle force through moment arms to joint torque. A first-order activation equation is often represented as $\dot{a} = (e - a)/\tau$, with separate time constants for activation and deactivation. Muscle force depends on active force-length and force-velocity factors, passive fiber force, tendon force-length behavior, and maximum isometric force. The generalized force contribution is $\tau_m(q) = R(q)F_m$, where sign and moment-arm conventions matter. Forward dynamics then evolves the coordinates and muscle states.

This chain creates multiple sources of lag and uncertainty. An excitation waveform can lead activation; fiber and tendon dynamics can delay force; joint motion reflects inertia and external loads. Fitting excitation directly to kinematic data without these states may attribute all lag to control timing. Conversely, highly flexible tendon parameters may absorb a bad time offset. Muscle models differ in whether tendon is rigid or elastic, whether equilibrium is solved algebraically, and how pennation and compliant tissues are represented. These choices alter the dynamics and optimization conditioning.

Computed Muscle Control

Computed Muscle Control was designed as a practical method to generate dynamic simulations that track desired kinematics. In the original formulation, static optimization determines muscle forces or excitations, while feedforward and feedback components drive the simulated trajectory toward a desired path. This approach efficiently produced muscle-driven simulations from experimental data and remains historically important. Thelen and colleagues demonstrated it in pedaling; subsequent work generated forward walking simulations by combining residual elimination with CMC.

CMC's advantages are speed and accessibility. The controller provides a correction when

simulation state differs from the desired path, and static optimization resolves instantaneous muscle redundancy according to a declared cost. It is effective for building an initial muscle-driven simulation or generating controls for later forward rollout. Its limitations follow from the architecture. Static optimization treats each step largely locally, so it may not represent full-horizon anticipatory tradeoffs. The feedback controller can mask model mismatch: large corrective excitation may keep tracking error small even if the model's passive dynamics or dimensions are wrong. A final open-loop simulation is therefore essential. Report controller gains, desired trajectory source, residual actuators, muscle objective, and achieved excitation limits.

Forward Optimal Control

A trajectory-level muscle matching problem can optimize excitations and states across the entire time horizon: $\min_{x,e,p} J_{track}(x,p) + \lambda_e \int \|e\|^2 dt + \lambda_s \int \|\dot{e}\|^2 dt + R_p(p)$,

subject to activation dynamics, muscle-tendon force equilibrium, multibody dynamics, contact, and physiological bounds. Dynamic optimization accounts for history dependence and can trade effort at one time against motion fidelity later. Direct collocation or multiple shooting typically transcribes this problem. The objective can be modified to include metabolic cost, EMG residuals, joint-load penalties, or task success, but each criterion encodes a hypothesis about motor coordination.

A precise statement of the inference is crucial. If only kinematic markers are observed, muscle control is inferred through the model and the chosen regularizer. If ground reaction forces and moments are added, whole-body force balance constrains the solution. If EMG is included, excitation or activation can be tethered to processed electromyograms with muscle-specific mapping and time delay. If tendon forces or ultrasound fascicle lengths are observed, additional internal states become more identifiable. No one sensor should be treated as exact physiological truth.

Tracking Variants

Prescribed kinematics with muscle optimization uses IK trajectories as fixed inputs and estimates muscle forces or activations. It is useful for load estimation conditional on motion but does not test whether the forward model produces that motion.

Feedback controller tracking drives a forward simulation toward target kinematics online. It is computationally efficient and tolerates mismatch, but controller action can compensate for inaccurate parameters.

Optimal-control tracking allows motion to deviate from data according to noise-weighted loss while enforcing the dynamics. It handles uncertainty and model mismatch more transparently; its computational and modeling burden is higher.

EMG-informed tracking constrains controls with muscle-specific electrical activity. EMG amplitude is affected by electrode placement, crosstalk, normalization, muscle fiber conduction, and electromechanical delay. The relationship between EMG and force is not a universal calibration curve. A sound estimator uses uncertainty, subject-specific calibration when available, and reports sensitivity to EMG weighting.

External-force tracking fits measured ground reaction forces and center of pressure in addition to motion. This can substantially improve dynamic consistency but depends on correct force transformation and foot assignment. When the same force record both drives contact and appears in the objective, distinguish whether it is prescribed input or an independently predicted output; otherwise the reported fit may be tautological.

Muscle Model Calibration and Scale

The maximum isometric force and optimal fiber length are commonly scaled from generic muscle parameters using subject size or physiological cross-sectional area. Such scaling is only a starting prior: two people with the same segment dimensions can differ in strength, architecture, training, and pathology. If the observed movement never approaches high force, maximum strength may be practically unidentifiable even with excellent kinematic tracking. A fit may simply select stronger muscles and lower activation, or weaker muscles and higher activation, with nearly identical generalized force. Activation bounds can truncate this tradeoff but cannot establish which strength value is true.

Parameterize calibration so that each quantity has a clear role. A global force scale may be better supported than independent scales for every muscle; a small set of muscle-group multipliers can provide a regularized intermediate model. Preserve muscle-specific architecture where there is direct evidence, and avoid simultaneously freeing force scale, excitation scale, EMG gain, and tendon slack length without independent observations. Compare fitted parameters across multiple tasks and test whether the same calibration predicts movement outside the training range.

Residual Actuators

Residual actuators are generalized-force terms introduced to stabilize or reconcile a simulation. They can be useful numerical aids and diagnostics but should not silently bear the movement. A muscle-driven result is more convincing when residual magnitudes and work are small relative to physiological actuator contributions, and when conclusions are robust to residual penalty strength. Since force magnitude alone ignores duration, compare residual impulse and work: $I_{res} = \int \tau_{res}(t) dt$, $W_{res} = \int \tau_{res}(t)^T \dot{q}(t) dt$. A large residual that performs substantial positive or negative work can change the inferred muscle demands even if its peak is short.

Residual penalties should have physically meaningful scales. Penalizing force alone can make an optimizer choose a large high-frequency correction with low time-integrated force. Penalizing residual work alone can permit large force near zero speed. Reporting peak, RMS, impulse, work, and direction over time offers a more complete picture.

Adaptive Dimensions and Co-Estimation

Geometry and muscle parameters influence the tracking model through different pathways. Segment dimensions alter marker projection, joint centers, lever arms, inertia, and sometimes contact geometry. Muscle maximum force scales torque at a given activation. Tendon slack length shifts operating length and force. When these parameters are all free with control, a single motion trial often cannot distinguish them. The problem is not solved by adding more optimizer iterations.

A promising design adapts dimensions in a low-dimensional, interpretable layer. Start with segment scale factors, functional joint-axis parameters, and perhaps a small number of inertial adjustments, each with anatomical priors. Derive dependent quantities consistently: changing femur length must update marker locations and muscle paths where applicable, not just the rendered skeleton. Positive quantities such as lengths and strength can be represented logarithmically; inertia matrices should remain positive definite; rotations should be updated on $SO(3)$ rather than by unconstrained Euler increments. Include multiple trials and speeds so a single parameter set must explain distinct accelerations and postures.

A hierarchical model can share population parameters while fitting subject-specific deviations. For example, $p_i \sim \mathcal{N}(\mu_p, \Sigma_p)$ for subject i , with μ_p and covariance learned across a cohort. This prevents each subject fit from drifting arbitrarily and quantifies population variability. Adaptation should be staged: fit sensor calibration and body scale, then dynamic inertial parameters, then selected muscle parameters, and only then allow a richer shape model if residuals persist. Jointly fitting everything from the start obscures which evidence supports each change.

Practical Fit Protocol

1. Validate a passive/torque-driven forward simulation before adding muscle complexity.
2. Fix geometry and fit controls using a simple actuator set; verify markers, forces, and rollout.
3. Add muscle states and physiologic bounds; initialize with static optimization or CMC.
4. Optimize full-horizon tracking with explicit sensor covariances and effort/EMG priors.
5. Inspect muscle excitations, activations, fiber/tendon states, saturation, and residual actuators.
6. Adapt only well-motivated model parameters with priors; analyze correlations with control variables.
7. Validate on a held-out stride or task and test open-loop rollout away from the exact training data.

Results should identify the objective and all weights, model version, muscle formulation, activation dynamics, contact model, parameter bounds, residual actuators, solver and mesh, and the actual measured channels used. A muscle force estimate should be framed as conditional on those choices.

Worked Derivation: EMG Delay and Activation

Consider $\dot{a} = (e - a)/\tau$ with constant excitation $e = e_0$ from an initially inactive muscle $a(0) = 0$. The solution is $a(t) = e_0(1 - \exp(-t/\tau))$. If measured EMG rises at time zero but force is delayed, one plausible explanation is activation dynamics; another is an electromechanical delay δ , tendon compliance, or timing misalignment. For a ramp-like signal, a delay and a larger τ can produce similar trajectories over a short observation window. Joint kinematics alone may not distinguish them. Multi-channel EMG, tendon/fascicle measurements, repeated activation patterns, or calibrated priors can improve identification. This example explains why simultaneously freeing time offset, activation time constant, and tendon slack length is poorly conditioned.

Exercises With Solutions

Exercise 1: Feedback Masking

A CMC simulation tracks joint angles with small error but uses large residual pelvis torques and saturated muscle activations. Is the model validated?

Solution. No. The controller achieves tracking through artificial and saturated actuation. The forward model's ability to explain the motion physiologically is weak. Report those corrections, reduce reliance on residual actuators, examine model/data mismatch, and perform an open-loop rollout.

Exercise 2: Effort Norm

Why does minimizing $\sum_i a_i^2$ not uniquely identify biological recruitment?

Solution. It selects the feasible activation vector with smallest squared activation under the assumed muscle model and constraints. Alternative objectives, strength parameters, or EMG evidence can select other solutions. The objective is a prior on coordination, not a direct measurement.

Exercise 3: More Sensors

What does a measured ground reaction wrench add beyond joint-angle tracking?

Solution. It constrains external force and moment balance, reducing some tradeoffs between muscle action and unobserved net loading. It does not by itself resolve muscle redundancy or prove internal forces, and it depends on correct frame transformation and contact assignment.

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Filtering, Smoothing, and Data Assimilation for Dynamics Matching

Why State Estimation Matters

A forward dynamics model is fitted to kinematic records, but the observations are not the state itself. Marker positions are noisy functions of configuration; velocity and acceleration are usually derived; force inputs may be incomplete; and motion capture can contain occlusion, calibration drift, and timing errors. Treating a differentiated trajectory as exact data transfers sensor noise directly into the estimated dynamics. State estimation offers a principled way to represent uncertainty and infer hidden trajectories jointly with model parameters.

Filtering estimates the current state using observations available up to the present time. Smoothing estimates past states using observations from both before and after the time of interest. Offline model matching usually has access to an entire trial, so smoothing is often more appropriate for latent trajectory reconstruction. Filtering remains important for on-line adaptation, causal prediction, and deployment. The two tasks should not be conflated: a smoother can use future measurements and therefore cannot be used as an online filter without changing the information set.

State-Space Formulation

Write the continuous dynamics and observation process as

$$\dot{x}(t) = f(x(t), u(t), \theta) + w(t), \quad y_k = h(x(t_k), \theta) + v_k,$$

where x contains configuration, velocity, and optional latent states; u is an input; w represents process discrepancy; and v_k is measurement noise. For mechanics, f may arise from a constrained forward model, while h maps the physical state to marker or sensor coordinates. A discrete-time integrator gives a transition model

$$x_{k+1} = F_{\Delta t}(x_k, u_k, \theta) + w_k.$$

This representation makes assumptions visible. Process noise does not mean that physical motion is intrinsically random in every application. It can stand for unmodeled forces, integration error, uncertain inputs, or unresolved dynamics. If it is too flexible, it can absorb precisely the structural mismatch the research seeks to identify. Its covariance must therefore be interpreted and tested, not merely tuned until the filter follows every observation.

Extended and Unscented Kalman Filters

The extended Kalman filter (EKF) linearizes the nonlinear transition and observation functions around the current mean. If the predicted state is $\hat{x}_k^-$ with covariance P_k^- , the transition Jacobian $A_k = \partial F / \partial x$ propagates uncertainty approximately as

$$P_{k+1}^- = A_k P_k^- A_k^T + Q_k.$$

At an observation, the Jacobian $H_k = \partial h / \partial x$ produces the innovation and Kalman gain. The EKF is computationally efficient when derivatives are available and uncertainty remains local. It can fail when the posterior is strongly nonlinear, multimodal, or near constraints that make Gaussian linearization misleading. Angle wrapping and contact events require special handling.

The unscented Kalman filter (UKF) propagates a set of deterministically chosen sigma points through nonlinear functions, then reconstructs mean and covariance. It avoids explicit Jacobians and can capture some curvature more faithfully than first-order linearization. It still represents uncertainty with a single Gaussian and can behave poorly under multimodality or hard mode switches. Both methods need physically sensible covariance scaling and consistent treatment of constrained coordinates.

For kinematic fitting, the EKF/UKF can jointly track q and $\dot{q}$ from noisy marker positions without direct numerical second differentiation. However, if the model parameter vector is augmented into the state, parameter learning can be slow or confounded with process noise. A parameter that changes gradually should not be treated as arbitrary high-variance motion state. Alternatives include batch smoothing, dual estimation, or hierarchical Bayesian parameter inference.

Ensemble and Particle Methods

An ensemble Kalman filter represents uncertainty with a collection of state samples and approximates covariance from their spread. It is useful for large systems where covariance matrices are too expensive, and can assimilate observations without analytic derivatives. The standard ensemble update remains approximately Gaussian and may underrepresent multimodality. Finite ensemble size introduces sampling error; localization and inflation techniques can help in large spatial systems but require domain-specific justification for multibody models.

Particle filters represent a posterior with weighted state samples. They can capture non-Gaussian and multimodal uncertainty, which is valuable when marker configurations correspond to multiple plausible poses or contact modes. Their main challenge is degeneracy: in high dimensions, most weights can collapse onto a few particles. Resampling can reduce effective sample size and lose diversity. Particle smoothers additionally need to connect plausible state histories; naive backward simulation can be expensive and suffer path degeneracy over long sequences.

Ensemble and particle methods are not automatic remedies for unknown inputs. If torque is uncertain, include it as a modeled latent input with a physically meaningful process model or use a separate force measurement. Assigning unconstrained process noise to acceleration can produce a state estimate that fits kinematics while leaving the forward dynamics untested.

Smoothing and Joint Trajectory Estimation

A classical Rauch–Tung–Striebel (RTS) smoother runs a forward Kalman filter and a backward pass to refine the state history using future observations. For nonlinear dynamics, extended

or unscented smoothing variants approximate the same logic. Batch nonlinear least squares and multiple shooting provide alternatives: optimize a complete state trajectory and parameters while penalizing observation residuals and dynamics defects. Direct collocation similarly enforces discretized dynamics at mesh points.

Consider a one-dimensional point mass with noisy position samples $y_k = x_k + v_k$ and a model $\ddot{x} = a(u; \theta)$. Differentiating observed positions twice yields a noisy acceleration estimate. A smoother instead estimates a sequence $(x_k, \dot{x}_k)$ that balances the position observations against integrated acceleration dynamics. The posterior covariance shows where velocity and acceleration are poorly determined, especially near trial boundaries or long gaps. This uncertainty should propagate to parameter estimates. A smooth curve alone is not evidence that the assumed force law is correct: compare innovations, residual structure, and forward predictions.

In batch estimation, one objective is

$$\mathcal{L}(x_{0:T}, \theta) = \sum_k r_{y,k}^T R_k^{-1} r_{y,k} + \sum_k r_{d,k}^T Q_k^{-1} r_{d,k} + \mathcal{R}(\theta),$$

where $r_y = y - h(x)$ is the observation residual, $r_d = x_{k+1} - F(x_k, u_k; \theta)$ is the dynamics defect, and $\mathcal{R}$ encodes parameter constraints or priors. This decomposition is useful because it shows whether a fit trusts the measurements, the dynamics, or the regularizer. Very large Q weakens the dynamics constraint; very small Q can force an incorrect model to explain noisy observations. Sensitivity analyses over plausible covariance choices are essential.

When R_k or Q_k are estimated, the Gaussian objective also includes their log determinants (and any hyperparameter priors). The displayed weighted least-squares expression assumes fixed covariances; omitting normalization while learning them rewards noise inflation.

Process Noise Versus Model Error

Process noise can make inference robust to small disturbances, but it can also mask missing dimensions. Suppose the true system has a hidden activation state; a position-only model may track it by increasing acceleration process noise. The state estimate appears adequate, but the inferred model has not explained the memory. Compare residual autocorrelation and dependence on prior inputs. Structured residuals suggest model error; approximately white residuals are more consistent with the assumed noise model, though whiteness alone is not proof of correctness.

A useful model comparison fits the same observation noise across candidates and uses process noise tied to a common physical interpretation. If each candidate receives an independently tuned, unconstrained process covariance, the noisiest model may win by absorbing mismatch. Report process-noise units, covariance structure, estimation method, and sensitivity. For contact, Gaussian process noise is usually a poor surrogate for an impact reset or unilateral constraint.

Unknown Inputs and Data Assimilation

Unknown torque or external force can be modeled as an augmented state, a stochastic input process, a sparse event sequence, or a bounded nuisance variable. Each choice imposes assumptions. A random-walk torque prior favors slowly varying inputs; a white-noise prior favors rapid variation; a sparse prior favors few impulses. The choice determines which physical effects can be distinguished from passive dynamics. Where possible, measure force or actuator commands directly, synchronize clocks, and calibrate scale.

In human movement, an input model may represent muscle activation, joint torque, or a controller. These are not interchangeable. A state estimator cannot turn an unspecified control policy into ground truth. At minimum, separate measured inputs, inferred inputs, and external disturbances in the data record, and show how inferred input uncertainty affects model conclusions.

Recommended Workflow

Begin with sensor calibration and a measurement-noise model. Fit a kinematic smoother without dynamics to understand what the sensors support. Then fit the constrained dynamics jointly with trajectories, using either a batch smoother or a clearly specified filter. Inspect innovations by sensor and time, posterior uncertainty near boundaries, constraint residuals, and sensitivity to process noise. Compare differentiated-data and integrated-trajectory baselines only as diagnostic alternatives. Finally, evaluate forward rollouts on complete held-out trials and measured inputs.

For online adaptation, use a causal filter and update model parameters more slowly than state estimates unless evidence supports faster change. Monitor innovation statistics and out-of-domain conditions. For offline research, use smoothing to exploit all observations but report that state estimates are retrospective. Keep the preprocessing and smoother parameters fixed within validation folds to prevent leakage from test trials.

Filtering on A Constrained Configuration Space

Joint angles live on circles and rigid-body orientations live on rotation groups; ordinary Euclidean Gaussian updates can average across a wrap boundary or leave the constraint manifold. For an angle, represent uncertainty with wrapped residuals or embed the orientation as $(\cos q, \sin q)$. This is one-to-one on the circle (angles modulo 2π), although it cannot distinguish different winding counts for an unwrapped angle; Gaussian noise in the two embedded coordinates also induces a nonuniform angular error model. For rotations, use a local tangent perturbation and retract to the group. If the mechanism has holonomic constraints, project estimated states and covariances onto the tangent space or formulate coordinates that satisfy the constraints by construction. Projection changes covariance and should be included in uncertainty propagation.

Contact adds discrete modes. A single Gaussian filter that averages stance and flight hypotheses can represent an impossible intermediate state. An interacting multiple-model filter maintains mode-specific estimates and mixes them with transition probabilities; a particle method can carry explicit mode labels; a hybrid smoother can optimize mode sequence and continuous states jointly. These approaches add complexity and depend on mode transition assumptions. Force sensors, foot switches, or event timing can substantially improve mode estimation. Do not report a precise continuous state if uncertainty is actually split across incompatible contact hypotheses.

Choosing A Method for the Data Regime

Use a deterministic smoother or batch least squares when the observation map is smooth, noise is modest, and a good initial trajectory is available. Use EKF or UKF for online tracking with relatively unimodal uncertainty and moderate state size. Ensemble methods are attractive when the state is large and covariance factorization is expensive. Particle methods are indicated when the posterior is strongly multimodal or contains discrete alternatives, provided dimensionality is manageable. For offline parameter learning, a smoother and outer

parameter optimizer is often simpler to debug than an elaborate joint filter, though it may be computationally heavier.

These are starting points, not a universal ranking. Benchmark candidates on synthetic trajectories with known truth and the same sensor timing, missingness, event frequency, and parameter scale as the actual study. Measure state error, uncertainty coverage, parameter recovery, runtime, and robustness to initial conditions. A method that wins only because it receives a better tuned noise covariance has not established a fair comparison.

Limitations

Kalman methods rely on Gaussian approximations; particle methods can be computationally prohibitive; smoothing can be overconfident under a wrong model; process-noise tuning is often subjective; and joint state-parameter estimation can be weakly identifiable. A good state estimate is not a validated dynamics model. Model comparison must still assess held-out prediction, physical constraints, and identifiability.

Exercises

1. Explain why a smoother can estimate a past state more accurately than a filter, and why that same smoother cannot be used as a causal online estimator.
2. A model fits marker trajectories after increasing acceleration process noise by 100 times. What diagnostics determine whether this represents plausible disturbances or missing dynamics?
3. Compare EKF, UKF, and particle filtering for a pose estimate with two plausible configurations separated by an occlusion gap.

Solution Sketches

1. The smoother conditions on future observations as well as past data, reducing uncertainty retrospectively. An online filter at time t cannot use future samples that have not arrived.
2. Examine units and magnitude against known force variability, innovation autocorrelation, inferred accelerations, sensitivity to the covariance, constraint violations, and held-out forward predictions. Test candidate hidden-state/contact models; do not accept noise inflation solely because it lowers position error.
3. EKF/UKF Gaussian approximations may average two modes into an implausible pose. A particle filter can represent both hypotheses, at higher computational cost and with degeneracy risk. Add an informative sensor or motion prior if feasible.

Uncertainty, Identifiability, and Validation

A forward model can match measured kinematics for the wrong reasons. A flexible control trajectory can compensate for inaccurate mass properties; a geometric parameter can compensate for a camera-frame error; a compliant contact coefficient can compensate for force-plate timing; and a muscle-strength parameter can trade off against activation. A research-grade matching method must therefore answer two separate questions: does the model reproduce held-out observations under forward simulation, and are the inferred dimensions and internal forces sufficiently identifiable to support the intended scientific claim?

Structural and Practical Identifiability

A parameter is structurally identifiable if ideal, noise-free, sufficiently rich observations uniquely determine it under the model. Practical identifiability asks whether finite, noisy observations constrain it tightly enough in the experiment. These differ. A parameter may be identifiable in theory but practically uncertain because the movement barely excites its effects. A parameter may appear numerically stable only because bounds or priors constrain it.

Let residual vector $r(\theta)$ contain whitened marker, force, EMG, and other measurement residuals, with decision vector θ including states, controls, and model parameters. Near a fitted point θ^* , linearize $r(\theta^* + \delta) \approx r(\theta^*) + J\delta$, where $J = \partial r / \partial \theta$. The regularized Gauss-Newton precision approximation is $H \approx J^T J + \Lambda$, where Λ represents prior precision or regularization. Singular value decomposition $J = U\Sigma V^T$ reveals weak directions. Small singular values correspond to parameter combinations that produce little observable change. Columns with high correlation reveal compensations. These diagnostics should be computed after meaningful nondimensionalization; otherwise units distort numerical comparisons.

The full Hessian can also include second-order residual curvature and objective terms, but the Jacobian analysis is often easier to interpret. Profile likelihoods provide a more nonlinear view: fix one parameter over a grid, re-optimize other variables, and observe how the objective changes. A flat profile indicates weak identification. Bayesian posterior sampling or Laplace approximations can quantify uncertainty, though multimodality and model misspecification complicate interpretation.

Why One Trajectory Is Often Insufficient

A single movement typically explores a narrow region of configuration and velocity space. Muscle moment arms change with posture, but if the posture barely varies, different anatomical geometries can be indistinguishable. Segment inertias matter most during acceleration;

steady or slow motions provide little information about them. Contact parameters are hard to identify if normal force and foot penetration barely vary. Activation dynamics are difficult to estimate without varied excitation timing. Therefore, a more complex estimator cannot recover information that the experiment did not excite.

Multiple tasks help because shared parameters must explain different motions. Walking at several speeds changes acceleration and contact loading; reaching with varied directions changes joint leverage; repeated trials expose sensor noise and stochastic control variation. A hierarchical fit can share global or subject-level anatomy while assigning trial-specific controls and initial states. The data likelihood then distinguishes persistent model properties from trial-specific control differences.

Optimal experiment design formalizes this idea. One may choose a movement to maximize a criterion of the expected information matrix, such as log-determinant (D-optimality), minimum eigenvalue (E-optimality), or minimizing the trace of inverse information (A-optimality). These criteria focus on different goals: reducing parameter-volume uncertainty, improving the weakest direction, or average variance. In practice, safe and repeatable motions with informative variation are preferable to mathematically optimal but infeasible tasks. The selected criterion and parameter scaling must be stated.

Priors, Regularization, and Model Capacity

Priors are not an admission of failure; they encode available anatomical and physical knowledge. Segment dimensions, mass distributions, joint ranges, and muscle strengths have plausible populations. A Gaussian prior can be written $p \sim \mathcal{N}(p_0, \Sigma_p)$, giving penalty $(p - p_0)^T \Sigma_p^{-1} (p - p_0)$. Positivity or manifold structure may require a transformed distribution. Muscle force and segment length are positive; inertia matrices must be positive definite and obey triangle inequalities; rotations belong to $\text{SO}(3)$. A penalty in raw Cartesian parameter space can violate these constraints, so parameterize with log lengths, Cholesky factors, or manifold coordinates.

Model complexity should increase only when residual structure and cross-validation justify it. An adaptive geometry model could begin with global scale, then individual segment lengths, then joint-center shifts, then shape or muscle-path corrections. Each expansion should answer a predefined question and be tested on held-out data. A richer model will nearly always lower training residual; only transfer to new movement, speed, or session demonstrates useful adaptation. Penalization, hierarchical shrinkage, or Bayesian model comparison can discourage unnecessary dimensions.

A key issue is the distinction between measurement nuisance and physical parameters. Camera translation, marker offsets, soft tissue motion, model dimensions, and joint center locations can all move predicted markers. If only marker residuals are available, these variables can be collinear. Use calibration objects, functional motions, anatomical priors, and multiple sensor types to separate them. Do not allow a subject's femur length to change just because camera calibration is uncertain.

Uncertainty Propagation

Once a parameter uncertainty distribution is available, propagate it through the fitted forward model. For each posterior or profile sample, simulate muscle forces, joint loads, predicted motion, and contact quantities. Report intervals for the scientific outputs, not only point estimates of geometry. Sensitivity derivatives $\partial y / \partial p$ can approximate output variance via $\text{Cov}(y) \approx J_p \text{Cov}(p) J_p^T$, but this first-order approximation fails for nonlinearities, contact

switches, and multimodal distributions. Sampling or ensemble methods are safer for those cases.

Residuals should also be decomposed by source. A total marker RMSE can hide systematic offsets on one segment; a total force error can hide poor center-of-pressure prediction; a single dynamic-defect norm can hide a severe constraint violation. Report residual time series, distributions, and stratification by sensor, segment, movement phase, and contact state. Check autocorrelation: temporally structured residuals suggest omitted dynamics, delays, or model inadequacy rather than independent sensor noise.

Validation Design

Validation has several levels:

1. **Numerical verification:** confirm equations, units, sign conventions, derivatives, integrator behavior, and constraint enforcement on analytic or simple cases.
2. **Training-fit diagnostics:** compare predicted and measured channels used in estimation; inspect each residual and active bound.
3. **Dynamic feasibility:** independently integrate the fitted controls and parameters; measure rollout error and contact/constraint violations.
4. **Held-out prediction:** use a held-out trial, stride, speed, task phase, or sensor stream. Avoid random frame-level splits because adjacent frames leak nearly identical information.
5. **Parameter transfer:** test whether inferred body dimensions and system parameters remain stable across trials and predict a distinct task.
6. **External validation:** where available, compare internal quantities with independent measurements such as tendon force, ultrasound, instrumented implants, or validated EMG-driven estimates, respecting their own limitations.

Baselines should include a transparent inverse-kinematics/inverse-dynamics workflow and a recognized forward-tracking method such as CMC or MocoTrack when applicable. Baseline comparisons require the same data, model, objective channels, and evaluation metrics. A method should not win by using more sensors or a different target without disclosure. Report compute time and optimization failures, not only best-case accuracy.

Validation Metrics

Use separate metrics for observation fit and physical consistency. For marker positions, report per-marker RMSE, median, high quantile, and maximum, with robust/outlier handling disclosed. For orientations, use geodesic rotation distance. For forces and moments, report componentwise errors and center of pressure. For state, compare joint angles and speeds. For dynamics, report defect norms, generalized-force balance residuals, holonomic constraint error, and contact complementarity or penetration. For controls, report excitation/activation bounds, smoothness, saturation time, and muscle-force plausibility. For residual actuators, include peak, RMS, impulse, and work. For inferred parameters, report prior-to-posterior changes, uncertainty, covariance/correlation, and bound activity.

Free rollout is particularly important. A collocation solution can be dynamically feasible at its mesh nodes yet fail when controls are integrated between nodes. Starting from the fitted initial state, propagate controls without feedback corrections and compare predicted observations. A separate perturbation test can offset the initial state or parameters slightly and determine whether the model is stable and robust. If the task is inherently feedback controlled, distinguish validating the plant model from validating the controller.

Model Misspecification and Residual Structure

An estimator assumes some model class is adequate. If the true system contains unmodeled compliance or variable contact, the optimizer may force geometry or controls to compensate. Residuals can guide model expansion: a consistent phase lag may suggest missing activation/tendon dynamics; angle-dependent marker bias may suggest geometry or soft tissue; force error localized at touchdown may suggest contact; cross-axis error may suggest joint-axis misalignment. These interpretations are hypotheses. A residual pattern can have multiple causes, so change one model family at a time and test on held-out data.

A statistically low residual is not enough if the errors are autocorrelated or systematically biased. Conversely, a residual above nominal measurement noise may arise from unavoidable model discrepancy. A useful model can still predict the target quantity if its uncertainty is calibrated and its limitations are clear. Claims should match validation: successful marker tracking supports kinematic reconstruction; accurate held-out force prediction supports dynamic transfer; validated tendon force supports a stronger internal-load claim.

Worked Example: Two Parameters That Trade Off

Suppose measured angular acceleration is approximately $\ddot{q} = rF_{max}a/I$, with moment arm r , maximum force F_{max} , activation a , and inertia I . A single observed acceleration constrains only the ratio $rF_{max}a/I$. If activation is also optimized freely, increasing F_{max} and decreasing a preserves the same acceleration. Decreasing inertia can be offset by reducing force. The log sensitivity of acceleration is $d \log(\ddot{q}) = d \log(r) + d \log(F_{max}) + d \log(a) - d \log(I)$.

A single scalar observation identifies one linear combination of four log parameters. Varying posture changes r ; varying acceleration and load can inform I ; EMG constrains a ; independent strength assessment constrains F_{max} . This simple derivation explains why diversified measurements and excitation are more important than optimizer sophistication.

Exercises With Solutions

Exercise 1: Training and Test Leakage

Why is a random 80/20 split of individual frames weak for validating gait tracking?

Solution. Neighboring frames are highly correlated and share the same trial, calibration, subject, and cycle dynamics. The test frames are nearly interpolations of training frames. Hold out a full stride or trial, and ideally test a distinct speed or task.

Exercise 2: Weak Singular Vector

An SVD of the scaled residual Jacobian shows a weak vector mixing femur length and camera translation. What is the next useful experiment or measurement?

Solution. Add independent camera calibration or an anatomical/functional measurement of femur length; alternatively use movements that expose the segment endpoints from varied orientations. More optimization iterations cannot separate parameters whose effects are observationally identical.

Exercise 3: Criterion Choice

When might E-optimal experimental design be preferable to maximizing the determinant of information?

Solution. If one poorly observed parameter direction is especially important, maximizing the smallest eigenvalue can improve the weakest direction. D-optimality reduces total uncertainty volume but may leave one critical direction weak.

References

Primary retrieved sources informing this chapter include Dembia et al. (2020), OpenSim Moco, whose indexed text explicitly cautions about local minima and numerical methods; the 2024 systematic review of methods for reducing motion-to-simulation mismatch, which classifies parameter adjustment, filtering, EMG-informed tracking, controllers, and trajectory optimization; and Porsa et al. (2016), direct methods in OpenSim. The existing local bibliography includes (Olson 2026c) for the repository’s residual-aware perspective and (Featherstone 2008) for multibody dynamics context; these local works should be positioned as AffineDrift contributions rather than independent validation evidence.

Bayesian Parameter and Model Inference

From A Best Fit to an Uncertainty Statement

Optimization returns a parameter vector that minimizes an objective. It does not by itself show how strongly the observations support that value, whether multiple solutions exist, or whether alternative structures predict equally well. Bayesian inference makes assumptions about parameters and data explicit through a prior $p(\theta \mid m)$, a likelihood $p(y \mid \theta, m)$, and a posterior

$$p(\theta \mid y, m) = \frac{p(y \mid \theta, m)p(\theta \mid m)}{p(y \mid m)}.$$

Here m denotes a model structure, and the denominator is the marginal likelihood or evidence. The posterior describes parameter uncertainty conditional on the model, prior, observation process, and data. It does not erase structural uncertainty or compensate for an incorrect likelihood.

For forward dynamics matching, y may be a full kinematic sequence and the likelihood may integrate the equations from initial conditions. A common Gaussian observation model is

$$y_k \mid x_k, \theta, m \sim \mathcal{N}(h_m(x_k, \theta), R_k),$$

with a transition model that includes process discrepancy. If R_k and process covariance are fixed, posterior concentration can be interpretable. If both are freely adjusted, a model may explain residuals by inflating noise rather than improving its dynamics. Noise parameters therefore need priors, calibration, or sensitivity analysis.

Priors Encode Real Knowledge and Regularization

Priors can encode positivity of mass, smoothness of force laws, plausible parameter ranges, or group-level shrinkage for candidate terms. A log transform enforces positive scalar parameters; a Cholesky factorization can enforce a positive-definite inertia matrix; hierarchical priors can shrink related coefficients together. Positive-definiteness alone does not guarantee a physically realizable rigid-body inertia: mass must be positive and the principal moments must satisfy the inertia triangle inequalities (equivalently, the associated second-moment matrix must be positive semidefinite). These are not neutral numerical choices. They affect posterior uncertainty and model evidence, especially when the data are weak.

A useful prior should be stated in physical units and tested for sensitivity. Uniform priors over a parameter can become highly informative after nonlinear reparameterization. Broad priors may assign substantial mass to physically impossible regions unless constraints are built in. When a posterior is concentrated far from prior boundaries, modest prior changes may have little effect; near boundaries, conclusions can be prior-driven. Report the prior scale and transformed parameterization.

Laplace Approximation

The Laplace approximation replaces the posterior near a mode $\hat{\theta}$ with a Gaussian. For negative log posterior $J(\theta)$ and Hessian $H = \nabla^2 J(\hat{\theta})$, the local approximation is

$$p(\theta \mid y, m) \approx \mathcal{N}(\hat{\theta}, H^{-1}).$$

The approximation is attractive because it is fast and provides approximate uncertainty and evidence. Around a well-identified, unimodal, interior mode with a smooth likelihood, it can work well. It can fail when the posterior is skewed, multimodal, bounded, strongly correlated, or nearly singular. A Hessian inverse at one optimum cannot reveal a distant second mode. For dimension selection, Laplace evidence can be especially sensitive to whether the mode lies near parameter boundaries and whether the local Gaussian approximation is credible.

Evidence approximations include both fit quality and an Occam factor related to posterior volume relative to prior volume. This is often described as automatic complexity control, but the penalty is conditional on prior scale. A candidate with many parameters can be penalized because only a tiny prior region fits, yet changing the prior width changes that volume ratio. MacKay's work on Bayesian interpolation develops evidence-based regularization ideas; the direct DOI URL is <https://doi.org/10.1162/neco.1992.4.3.415>, pending link validation.

Profile Likelihood

Profile likelihood is a frequentist alternative for diagnosing practical identifiability. For parameter of interest θ_i , fix it at value a and re-optimize nuisance parameters:

$$\ell_{prof}(a) = \max_{\theta_{-i}} \ell(\theta_i = a, \theta_{-i}).$$

A sharp profile indicates the data constrain the parameter locally; a flat or one-sided profile indicates weak identification. Profiles reveal nonlinear ridges and boundary effects that a quadratic approximation misses. They are computationally more expensive because the model must be refit over a grid, and their confidence interpretation relies on regularity assumptions. Correlated parameters should be profiled jointly when a single-parameter profile hides a ridge.

Raue and colleagues discuss profile likelihood for practical identifiability in nonlinear dynamical models. A canonical reference link is <https://doi.org/10.1093/bioinformatics/btp358>; live validation is pending. Profile likelihood is not Bayesian posterior uncertainty: it does not incorporate a prior, and its confidence set is based on likelihood-ratio behavior. Using both profiles and posterior summaries can be informative because they expose different dependencies.

Markov Chain Monte Carlo

MCMC generates dependent samples whose stationary distribution is the posterior. Hamiltonian Monte Carlo and its adaptive variants can explore continuous, correlated parameter

spaces more efficiently than random-walk proposals when gradients are available. Differential evolution, slice sampling, and ensemble methods can be useful in other settings. No sampler is universally reliable. Chains can remain trapped in one mode, mix slowly along weakly identified ridges, or produce apparently stable summaries while missing alternative structures.

Use multiple dispersed chains, inspect trace plots and effective sample sizes, compare rank-normalized convergence diagnostics, and verify posterior predictive behavior. These diagnostics assess aspects of convergence but cannot prove that all modes have been found. Synthetic parameter-recovery experiments are valuable: generate data from known settings, fit the model, and check interval coverage and recovery rates across a range of noise and excitation. For high-dimensional latent models, exact posterior sampling may be infeasible; variational approximations are faster but often understate uncertainty and should be calibrated against smaller problems.

Unknown states add another layer. Jointly sampling an entire trajectory and parameters can create strong correlations and high dimensionality. State-space smoothers, marginalized likelihoods, or particle MCMC can help, but each adds computational and modeling choices. Approximate state estimates plugged into a parameter likelihood can understate uncertainty because the uncertainty in the trajectory is ignored.

Comparing Candidate Models

For model structure m , the marginal likelihood is

$$p(y | m) = \int p(y | \theta, m) p(\theta | m) d\theta.$$

Posterior model odds combine evidence with prior model probabilities. The evidence averages fit over parameter space, so a highly flexible model is not rewarded merely for having one excellent parameter setting. But the result depends on the prior distribution and on the completeness of the candidate set. If every model omits the correct contact mode, the highest-probability candidate is merely the least inadequate among those considered.

When model dimensions differ, reversible-jump MCMC can move among parameter spaces of varying dimensions. Green's foundational article is "Reversible jump Markov chain Monte Carlo computation and Bayesian model determination," with DOI <https://doi.org/10.1093/biomet/82.4.711>; live validation is pending. The dimension-matching transformation and proposal probabilities must be correct, and chain mixing across structures must be demonstrated. In many applied studies, a small predeclared set of candidates plus robust held-out validation is more practical than an unrestricted transdimensional sampler.

Worked Example: Two Damping Models

Compare m_1 , which uses a single damping coefficient b in two regimes, with m_2 , which uses separate coefficients b_1, b_2 . Suppose the data include only regime 1, and the prior for b_1 under m_2 matches the prior for b under m_1 . Let η denote any other shared parameters. Because b_2 does not appear in the likelihood, the evidence factors as

$$p(y | m_2) = \int p(y | b_1, \eta, m_2) p(b_1, \eta | m_2) \underbrace{\left[\int p(b_2 | b_1, \eta, m_2) db_2 \right]}_1 db_1 d\eta.$$

The bracketed integral is one. Under matched priors for shared parameters and identical regime-1 likelihoods, this gives exactly the same evidence as m_1 ; the unobserved b_2 incurs no automatic Occam penalty. Its posterior remains its prior conditional on shared parameters, so the data say nothing about it. If regime-2 observations are present but weak, their likelihood depends slightly on b_2 ; then evidence can reflect the prior-to-posterior volume change, and a profile may be broad. A new informative trial in regime 2 can identify the coefficient. Do not equate a parameter absent from the likelihood with one that has a weak but nonzero likelihood effect.

A robust report therefore includes posterior distributions, prior-to-posterior comparison, profile curves for critical parameters, evidence sensitivity, and held-out predictions. If two model structures produce practically indistinguishable predictions on all feasible protocols, report that equivalence rather than presenting a posterior probability as a universal truth.

Calibrating Uncertainty

Uncertainty intervals should be checked for coverage on synthetic and, where available, repeated experimental data. Posterior predictive checks compare simulated trajectories with features not used merely to construct the likelihood: event timing, peak velocity, cross-joint coordination, residual autocorrelation, constraint error, and response to perturbations. A model can have well-behaved parameter chains but poor predictive calibration if the likelihood omits temporal correlation or contact events.

For adaptive dimension learning, separate uncertainty into at least three levels: parameter uncertainty within a candidate model; model uncertainty across candidate structures; and data or measurement uncertainty. A single standard error does not capture all three. If the candidate set changes during ongoing adaptation, preserve the history of proposals and evidence so that model selection remains auditable.

Parameter Correlation and Reparameterization

Weakly identified models often produce long posterior ridges. Sampling in the original coordinates can be inefficient when two parameters trade off, such as mass scale and torque scale or passive damping and feedback gain. Reparameterization can align the sampler with identifiable combinations: estimate the ratio and overall scale separately, or use a centered/non-centered hierarchical form depending on data strength. This can improve computation, but it does not add information. Always translate the result back to physical quantities and state which combinations the experiment actually constrains.

For multibody systems, physically valid parameterizations matter. Sampling unconstrained entries of an inertia matrix can generate nonphysical bodies; a structured parameterization must enforce positive mass and a realizable central inertia (principal moments satisfying triangle inequalities, equivalently a positive-semidefinite second-moment matrix), as well as any required positive definiteness. Parameterizing mass, center of mass, and a valid second-moment matrix can help. Constraints can create truncated or curved posterior geometry. A Laplace approximation should account for the transformed coordinates and Jacobian when it approximates density in physical space. Reporting an uncertainty interval in transformed log-mass space as if it were symmetric in mass is incorrect.

Evidence Computation and Model Priors

Marginal likelihood can be estimated by bridge sampling, nested sampling, thermodynamic integration, or Laplace approximation. Each method has failure modes. Importance sam-

pling can have unstable weights when proposal tails are too light; nested sampling depends on accurately exploring constrained likelihood contours; thermodynamic integration requires adequate sampling across inverse temperatures; Laplace depends on local Gaussian geometry. Evidence estimates should include numerical uncertainty and convergence diagnostics. Comparing tiny differences without Monte Carlo error is false precision.

Prior model probabilities matter as well. If one candidate adds a coordinate that is considered a priori unlikely, the posterior odds may favor a simpler structure before data are seen. This can be a reasonable scientific preference, but it should be reported separately from likelihood support. Sensitivity analysis should vary both parameter priors and model priors over defensible ranges. Evidence is especially fragile when candidate parameterizations contain redundant dimensions or singular limits, such as a larger model reducing to a smaller one when a coefficient is zero.

Posterior Predictive Checks for Dynamics

Draw parameter and state samples from the posterior, simulate forward under measured and novel inputs, and compare trajectory summaries with observations. Checks should preserve temporal structure: compare autocorrelation, cross-correlation between joints, peak and phase timing, impulse response, and event transitions. Pointwise marker coverage alone can look good even when the model has the wrong phase dynamics. For long-horizon prediction, report the distribution of trajectory error and constraint violations across posterior draws, not just the mean trajectory. A mean over trajectories with different contact timings may be physically meaningless.

Posterior predictive success is conditional on the chosen model class. It can reveal model inadequacy when replicated data fail to resemble observed data, but agreement does not prove the model's internal mechanism. Combine predictive checks with synthetic recovery, intervention response, parameter profiles, and independent physical measurements. In adaptive model selection, store these checks per model version so changes in fit are traceable.

Limitations

Bayesian inference is conditional on the prior, likelihood, and model set. Laplace approximations can miss skewness and multiple modes; MCMC can fail to mix; profile likelihood can be costly and is not a posterior; and approximate variational methods can understate uncertainty. Evidence comparisons are sensitive to prior volume. Predictive validation remains necessary even when posterior calculations appear numerically stable.

Exercises

1. Explain why a narrow posterior can still be misleading if process noise is fixed to an unrealistically small value.
2. A Laplace approximation reports a tiny standard error, while a profile likelihood is flat on one side. What may be happening?
3. Model A has lower training loss than Model B, but lower marginal likelihood. Give a plausible explanation and a necessary sensitivity check.

Solution Sketches

1. The model is forced to attribute all deviations to parameters, producing overconfident estimates under a misspecified noise model. Revisit the measurement/process covariance and examine posterior predictive residuals.
2. The posterior may be asymmetric, boundary-limited, or ridge-shaped; the local Hessian captures only curvature near the mode. Inspect multi-start fits, joint profiles, and posterior samples.
3. A's larger parameter space may fit well only in a small region of its prior, producing a stronger complexity penalty. Check prior-scale sensitivity, candidate completeness, convergence of the evidence estimator, and held-out predictive performance.

Identifiability Before Dimension Selection

Why This Chapter Comes First

A forward model predicts acceleration or future motion from state and input. Matching it to measured kinematics looks like an optimization problem, but optimization is downstream of a more basic question: do the available observations distinguish the candidate mechanisms at all? If two parameter settings or two model structures produce the same measured trajectories under every input in the experiment, no optimizer can select the physically correct one from those data. It can only choose according to initialization, regularization, or prior assumptions.

This distinction is central when a model is allowed to learn its dimensions. A smaller model can predict observed motion just as well as a larger one because the experiment never excited the extra coordinate. A latent state can mimic an omitted force, a time delay, or a contact mode. The resulting fit may be numerically excellent and structurally wrong. The correct output in such a case is an equivalence class or an explicit statement that the data do not identify the requested quantity.

Structural and Practical Identifiability

Structural identifiability asks whether ideal, noise-free, sufficiently rich observations uniquely determine unknown parameters or structure under a specified model and input class. Practical identifiability asks whether a finite, noisy experiment determines them with useful precision. Structural failure cannot be fixed by collecting more samples of the same uninformative motion. Practical weakness may improve with more trials, better sensors, or better excitation.

For a smooth state-space model,

$$\dot{x} = f(x, u, \theta), \quad y = h(x, \theta),$$

where x is the state, u is the known input, y is the observation, and θ is a parameter vector, one may study local sensitivity of the predicted output to parameters. For sampled outputs $y_k(\theta)$, define the sensitivity matrix $S_{ki} = \partial y_k / \partial \theta_i$. If a column is zero, that parameter has no local effect on the data. If two columns are nearly proportional, their effects are confounded. The Fisher information approximation is

$$\mathcal{I}(\theta) = S^T R^{-1} S,$$

where R is observation-noise covariance. Small eigenvalues indicate combinations of parameter changes that barely alter the observations. This is a local diagnostic, not proof of global uniqueness, but it is a useful way to expose weak directions.

In nonlinear systems, observability concerns reconstruction of hidden state from outputs and inputs; parameter identifiability concerns uniqueness of model parameters. The questions interact. If a coordinate is unobservable, parameters acting only through that coordinate may also be unidentifiable. Conversely, state reconstruction can be possible while inertial parameters remain confounded. Hermann and Krener developed a differential-geometric observability framework for nonlinear systems. A canonical DOI URL is <https://doi.org/10.1109/TAC.1977.1101601>; live validation is pending. Structural identifiability depends on the observation map, equations, and inputs, so any review source added for this claim should be verified before publication.

A Mechanical Example: Two Pendulum Parameters

Consider a planar pendulum with known length l , angle q , mass m , gravity g , viscous coefficient b , and applied torque u :

$$ml^2\ddot{q} + b\dot{q} + mgl \sin q = u.$$

Dividing by ml^2 gives

$$\ddot{q} = \frac{u}{ml^2} - \frac{b}{ml^2}\dot{q} - \frac{g}{l} \sin q.$$

If $u = 0$, l is known, and g is fixed, observations of a sufficiently varied decay may reveal the ratio b/m , but not necessarily m and b separately: multiplying both by a common positive constant leaves the normalized equation unchanged. With measured, calibrated torque input, the acceleration response to u can identify the inertia scale and break that symmetry. If the input torque is itself unknown up to a scale, the ambiguity returns. The issue is not an optimizer defect; it is an invariance of the observation model.

This illustrates why kinematic-only model matching requires care. Kinematics provide positions, and perhaps velocities and accelerations after estimation. They do not provide the forces that generated those accelerations. Unknown control, passive dynamics, external forces, and contact can substitute for one another. A model fitted only to one unperturbed swing can explain the path while having incorrect inertial or actuator parameters.

Observation Maps and Coordinate Ambiguity

Let the physical configuration be $q \in \mathbb{R}^n$, while the instrument reports markers $y = h(q, \alpha) + \epsilon$, with calibration parameters α and noise ϵ . Marker count is not coordinate count. Markers may be redundant, occluded, or insensitive to a rotational degree of freedom. A latent state z adds another distinction: a learned model may represent the same input-output behavior in several coordinate systems related by an invertible transformation. The numerical dimension of that representation does not establish anatomical meaning.

Constraints also change what “dimension” means. A rigid body with six pose coordinates may have fewer independent coordinates after a joint constraint. A mechanism can contain coordinates that are fixed in the experiment but available in the anatomy. Thus report separately: raw model coordinates, independent constrained coordinates, observed outputs, latent state size, active library terms, and unknown parameter count.

A Practical Diagnostic Sequence

First specify the observation and input model. Identify which quantities are measured, calibrated, inferred, or unknown. Second, simulate the candidate model under the exact experimental protocol and compute local sensitivities. Third, inspect singular values and parameter correlations, using several parameter points because nonlinear sensitivities can vary. Fourth, profile important parameters: fix one parameter over a grid and re-optimize the others. Flat profiles expose weak identification that a local Hessian can obscure. Fifth, test recovery on synthetic data generated from known parameters, including realistic noise, sampling, filtering, and missing observations. Finally, deliberately construct observationally equivalent candidate structures. A selection procedure should abstain on those cases.

The experiment should be designed to excite uncertain modes. In the pendulum example, known torque variation separates inertial scale from damping. For a multi-joint system, independent perturbations and varied initial conditions can expose coupling that a stereotyped trajectory hides. Contact requires trials with distinct contact regimes if mode behavior is to be identified. Excitation must remain within a safe and valid domain; arbitrary persistently exciting inputs are not automatically appropriate for human movement.

From Diagnostics to Model Decisions

A useful model report does not merely print an optimizer covariance. It includes parameter sensitivity singular values, profiles or posterior correlations for influential parameters, recovery performance in controlled simulations, and a plain description of unresolved ambiguities. For discrete structures, compare the predictions of plausible alternatives on trajectories and conditions not used in fitting. If two structures remain indistinguishable, preserve both as candidates or select the simpler one only as a declared engineering convention.

The forward model should also separate measurement residual from dynamics residual. A large marker residual may indicate a bad pose estimate or observation map; a small marker residual with a large rollout error indicates dynamics mismatch. A model that reproduces fitted samples but drifts badly over a forward rollout is not a successful forward model. Long-horizon prediction, constraint violation, energy behavior where appropriate, and event timing should be reported alongside observation fit.

Limitations

Sensitivity analysis is local, and rank can change with the trajectory. Fisher information assumes a useful noise model and does not account automatically for optimizer multimodality. Profile likelihood is more robust to some nonlinearities but can be expensive. Bayesian posterior concentration can reflect prior assumptions when data are weak. Observability tests for idealized models may be difficult to apply to high-dimensional constrained systems. These tools are diagnostics, not certificates that a complex model is correct.

Diagnostic Example: Sensitivity Rank in A Two-Coordinate Model

Suppose an observation vector stacks sampled marker positions, and a candidate model has parameters $\theta = (I_1, I_2, k)$. A finite-difference approximation to the sensitivity column for parameter θ_i is $[y(\theta + \delta e_i) - y(\theta - \delta e_i)] / (2\delta)$. After whitening by measurement covariance, form a

singular value decomposition $S = U\Sigma V^T$. The right singular vector associated with the smallest singular value describes a parameter combination that barely changes predicted measurements. For example, it may increase I_1 while decreasing I_2 and adjusting k . This is more informative than reporting that each individual standard error is “large,” because the ambiguity is a direction through parameter space.

The calculation must respect constraints. Finite-difference perturbations that create negative mass or an invalid inertia matrix are not valid probes. Use transformed parameters, such as logarithms for positive scalars, and interpret sensitivities in scaled coordinates. Repeat at several fitted solutions and initial conditions. If the weak direction rotates substantially, a single local covariance is especially misleading.

A useful practical distinction is between a parameter that is unidentifiable but irrelevant to the requested prediction and one that affects the deployment task. If all candidate values predict the same output over the target domain, the parameter need not be estimated for that purpose. If they diverge under deployment inputs, the unresolved parameter is a material risk and the data collection protocol should change.

Exercises

1. For the pendulum equation above, show that if torque is unknown and modeled as $u(t) = c\tilde{u}(t)$, the acceleration depends on c/m . Which additional measurement or perturbation could identify m and c separately?
2. Suppose two coordinate sensitivity columns are nearly collinear. Explain why collecting ten times as many samples from the same trajectory may not resolve the ambiguity. Suggest a different experiment.
3. A latent model with dimension 8 predicts held-out marker positions slightly better than a model with dimension 5. List at least three checks needed before claiming that the physical system has eight degrees of freedom.

Solution Sketches

1. Substitution yields $\ddot{q} = (c/(ml^2))\tilde{u} - \dots$; only the ratio is identified without an external scale. Calibrated torque, known mass, or an independent force measurement can break the ambiguity.
2. Repeated samples increase precision along already informative directions but preserve nearly dependent sensitivity columns. Apply a perturbation that affects the two coordinates differently, alter initial conditions, or add a sensor with a different observation map.
3. Check whether latent dimension is stable across seeds and priors; demonstrate identifiable mapping from latent variables to coordinates; compare on independent trajectories and inputs; verify dynamics and constraints; rule out delay, contact, or omitted actuation as the source of extra state.

Learning Physical Parameters and Anatomical Dimension

Distinguish Parameters From Structure

A model can contain unknown numerical parameters without uncertainty about its structure. For example, a specified two-link mechanism may have uncertain segment inertias, damping, and actuator gains. Structural adaptation asks whether to add or remove a coordinate, actuator, joint coupling, muscle group, or contact mode. These are different decisions and should not be collapsed into a single parameter vector.

For generalized coordinates q , a constrained mechanical system may be written

$$M(q; \theta) \ddot{q} + C(q, \dot{q}; \theta) + g(q; \theta) = B(q; \theta)u + J(q)^T \lambda + \tau_{ext}.$$

Here M is the mass matrix, C collects velocity-dependent terms, g gravity, Bu actuation, and $J^T \lambda$ constraint forces. The coordinate dimension belongs to a candidate topology. The vector θ contains numerical quantities within that topology. Contact forces λ may be governed by complementarity or a compliant law rather than being arbitrary free parameters.

If measured kinematics are $q(t)$, one can compute an inverse-dynamics residual for chosen parameters and assumed inputs. But if the inputs and external loads are unknown, they can absorb errors in M, C, g . A parameter optimizer that is permitted to invent arbitrary torques may perfectly reproduce the motion while saying little about the true forward dynamics. The model must state what generates input, whether input is observed, and what regularization or prior constrains unobserved forces.

Physical Parameter Estimation

Rigid-body inverse dynamics is often linear in a suitable set of inertial parameters, even though the full forward equation is nonlinear in state. For a known trajectory and known external loads, one can write

$$\tau(t) = Y(q, \dot{q}, \ddot{q})\pi,$$

where Y is a regressor and π collects inertial parameters. Stacking observations gives $\mathbf{r} = \mathbf{Y}\pi$. A least-squares estimate exists when the relevant columns of $\mathbf{Y}$ are sufficiently independent. If they are dependent, only combinations of parameters are identifiable. Physical constraints such as positive mass and physically realizable inertia further restrict the feasible set.

For forward matching, derivative noise matters because $\ddot{q}$ magnifies measurement noise. Rather than differentiate each sample twice and treat acceleration as exact, fit a trajectory

smoother with an explicit observation-noise model and use its posterior derivatives, or optimize an integrated trajectory against the measurements. Direct collocation provides another route: state and control values at mesh points are optimization variables, with dynamics enforced through discretized defects. The smoothing or collocation choices must be validated because an overly flexible trajectory can hide dynamics error.

Use scaled parameters and physically meaningful bounds. Parameter scaling is especially important when masses, lengths, damping, and neural residual weights differ by orders of magnitude. When possible, estimate base inertial parameters, not a redundant list of link quantities. Preserve positive-definiteness of the generalized mass matrix $M(q)$ for independent coordinates, but do not treat that condition as sufficient for valid link inertias. Each rigid body also requires positive mass and physically realizable principal moments satisfying the triangle inequalities, equivalently a positive-semidefinite second-moment matrix; parameterize these constraints directly. Standard multibody algorithms and movement-modeling frameworks are described by Featherstone and the OpenSim authors (Featherstone 2008; Delp et al. 2007).

What Counts as Anatomical Dimension?

An anatomical degree of freedom is grounded in physical structure and joint organization. A behavioral realization dimension is the number of independent state variables needed to reproduce observed input-output behavior. A statistical latent dimension is a representation size chosen by a learning objective. These may coincide in a carefully specified system with complete measurements and sufficient excitation, but they need not.

Consider a human arm model where the shoulder is represented by three rotational coordinates. In a particular reaching task, the subject may keep one rotation nearly fixed. The recorded path then occupies an approximately two-dimensional surface. A dimension-reduction algorithm can correctly summarize that task with two coordinates. It has not shown the shoulder anatomically has two degrees of freedom. Conversely, two muscle activations can produce nearly indistinguishable net joint torques in the observed task. Kinematics may reveal neither that there are two actuators nor how their forces split.

Anatomical claims require an explicit bridge from the data to structure: a specified skeletal model, imaging or physical examination, sensors that identify the relevant coordinate, or controlled perturbations that distinguish candidate structures. A model constrained by known anatomy can still learn effective parameters or a task-specific reduced coordinate system, but label those quantities precisely.

Candidate-Set Construction

A robust dimension-learning study begins with finite, interpretable candidates. Examples include a locked versus free joint; one versus two actuator groups; a fixed versus state-dependent transmission; a rigid versus compliant contact hypothesis; or alternative observation mappings. Candidate structures should be mechanically valid and sufficiently different to produce testable predictions. Unbounded search over arbitrary coordinate subsets is difficult to interpret because each candidate changes parameterization and potentially sensor mapping.

For each candidate, specify initial conditions, input map, external-force model, parameter priors or bounds, and observation model. Make candidate models comparable in their ability to explain nuisance effects. If one model includes marker offsets and another does not, their fit difference cannot be attributed cleanly to physical dimension. Similarly, if only the larger model receives a neural residual, complexity comparison is not controlled.

A staged strategy is easier to diagnose: first fit known topology and physical parameters; then test a sparse, structured residual; then evaluate a short list of structural alternatives; finally test whether a latent state is needed for predictive memory. Some interactions remain—for example, changing coordinate dimension changes parameter identifiability—so selected joint comparisons should follow staged screening.

Worked Example: A Hidden Second Coordinate

Suppose a planar double pendulum has coordinates q_1, q_2 , but a camera observes only the endpoint position $y = h(q_1, q_2)$. During a restricted motion, the endpoint remains close to a curve. A one-coordinate model may reproduce the observed endpoint trajectory with a learned curve parameterization. Yet under a torque perturbation applied at the second joint, the two-coordinate system may move off that curve while the one-coordinate candidate cannot. The distinction is invisible in the original data and visible under a targeted input.

This example suggests the right question is not “Which dimension has the lower training loss?” but “Under what input and observation conditions do these hypotheses make different predictions, and did the experiment include those conditions?” If no feasible input separates the candidates with current sensors, the distinction is not empirically resolved. A selected low-dimensional model may still be useful for prediction within the restricted domain, but the report should not call it anatomical discovery.

An Adaptive Model Proposal

A research candidate for AffineDrift is a mechanics-first hierarchy with explicit model gates. Let m index candidate topology and contact structure, z represent optional latent memory, and r_ϕ be a bounded residual force:

$$M_{m,\theta}(q)\ddot{q} + C_{m,\theta}(q, \dot{q}) + g_{m,\theta}(q) = B_{m,\theta}(q)u + J_m(q)^T \lambda + r_\phi(q, \dot{q}, u, z).$$

The candidate model would first verify $M \succ 0$, coordinate constraints, and valid contact semantics. A sparse group prior could turn on or off interpretable force families. If held-out residual structure remains and identifiability analysis supports memory, a latent state could be added. Model comparison would include a complexity cost and a reject/abstain outcome. This is a proposed architecture for research; it is not represented as implemented in the repository.

The hierarchy should maintain separate scores: measurement fit, forward rollout, physical feasibility, information adequacy, and complexity. A single weighted scalar can conceal trade-offs. A model with better fit but frequent constraint violation should not silently win over a feasible model. For each structural choice, log the evidence, priors, candidate inventory, and observed regimes.

Limitations and Failure Modes

Physical constraints do not guarantee correct physics. An incorrect actuator map can be mechanically elegant. A force residual can still absorb unmodeled anatomy. Known-looking coordinate labels may not align with sensor coordinates. Parameter bounds can make an unidentifiable parameter appear precise at a boundary. Model selection penalties can prefer an overly simple model outside the training domain. Human movement also changes with learning, fatigue, pain, or strategy; a structural change detector may confuse these effects with anatomy.

The solution is not a universal dimension estimator. It is a protocol that states the estimand, checks identifiability, compares physically plausible alternatives, and limits claims to tested inputs and observations.

Constraints, Coordinates, and Effective Degrees of Freedom

A coordinate count should be stated with its constraint convention. If generalized coordinates satisfy holonomic constraints $c(q)=0$, then local independent configuration dimension is $n - \text{rank}(\partial c / \partial q)$, provided the constraint Jacobian has constant rank near the motion. The corresponding velocity constraints are obtained by differentiating $c(q)=0$. Redundant coordinates can be computationally convenient, but they are not independent degrees of freedom. In addition, a model may include compliant deformation coordinates that are not rigid-joint degrees of freedom. This is why a textbook or software model's coordinate count cannot be compared casually to a learned latent size.

For an adaptive system, each proposed dimension change should have a semantic type. “Unlock coordinate q_3 ” changes configuration topology. “Add one actuator group” changes the input map. “Add one latent memory state” changes realization dimension. “Add one SINDy term” changes equation support. These operations require different tests, and an automated selector should log exactly which one it performed. The selection objective should penalize each according to scientific and computational consequences rather than treating every scalar parameter equally.

Effective coordinates can be learned from a task trajectory using manifold methods, but the manifold is conditional on the sampled movement distribution. A trajectory may occupy a low-dimensional set because of a controller, task instruction, or participant strategy. To estimate capacity rather than habitual use, perturbations should probe directions outside the observed manifold. A controlled reduced model can therefore be valuable even when it is not an anatomy model; the naming and evaluation target determine whether the inference is correct.

Exercises

1. Give an example where a task-specific low-dimensional representation is valid for prediction but invalid as an anatomical claim.
2. Explain why allowing arbitrary unknown torque at every time step weakens inference about inertia.
3. Design one perturbation and one sensor addition that could distinguish a locked joint candidate from a weakly moving free joint.

Solution Sketches

1. Reaching trajectories may lie on a low-dimensional coordination manifold while the shoulder and elbow retain additional anatomical motion capacity. The reduction describes the task manifold, not the skeleton.
2. Arbitrary torque can compensate for almost any acceleration error, making the mapping from inertial parameters to observed trajectory non-unique. Restrict torque through measured input or a justified control model.
3. Apply a safe perturbation around the suspected joint while measuring a marker cluster or joint encoder that directly observes the coordinate. Repeat across initial states and quantify whether predictions separate relative to sensor noise.

Sparse Discovery and Model Order Selection

Discovery Is Conditional on A Library

Sparse identification methods ask whether observed dynamics can be expressed using a small number of candidate functions. In the basic SINDy formulation, measurements of state $x(t)$ and derivative $\dot{x}(t)$ are assembled into

$$\dot{X} = \Theta(X)\Xi,$$

where $\Theta(X)$ is a library of candidate features and Ξ contains coefficients. A sparse regression seeks a coefficient matrix with few active entries while explaining the derivative data. The method’s “model order” is conditional on the chosen state, candidate library, scaling, derivative estimate, and threshold or prior. It does not automatically infer the anatomical dimension of the system.

Brunton, Proctor, and Kutz introduced sparse identification of nonlinear dynamical systems as a way to discover governing equations from data. This foundational method is summarized by Brunton, Proctor, and Kutz (Brunton et al. 2016). A canonical DOI URL is <https://doi.org/10.1073/pnas.1517384113>; live validation is pending because network access was paused by repository instructions. SINDy is a useful equation-discovery baseline, but its assumptions should be made visible in any application.

A Mechanical Formulation

For a second-order system, one can define a library over configuration, velocity, and known inputs:

$$\ddot{q} = \Theta(q, \dot{q}, u)\Xi.$$

A generic polynomial library may include constants, coordinate terms, velocity terms, products, and control interactions. For mechanics, a more defensible formulation separates known structure from unknown forces. For example, use a known mass matrix and gravity model, then represent only uncertain passive or actuator forces in a structured library. Alternatively, express inverse dynamics as $\tau = Y(q, \dot{q}, \ddot{q})\pi$, then select among candidate force terms. The latter relies on known or measured net torque and suitable trajectory derivatives.

Mechanical constraints can be built into the library: terms may be grouped by joint, actuator, symmetry pair, or physical force family. Coefficients for related coordinates can be tied. Positive inertia should be parameterized rather than encouraged by a penalty after the fact.

Conservation or dissipativity constraints may be useful in appropriate regimes, but should not be imposed where active control or contact legitimately changes energy.

Why Support Recovery Is Difficult

Sparse regression works best when active and inactive features are distinguishable in the data. If two columns of $\Theta(X)$ are highly correlated, many coefficient sets produce similar derivatives. For example, over a small angle range, $\sin q \approx q$; a linear restoring term and a gravitational term can be hard to distinguish. Similarly, a passive damping term and a feedback controller proportional to velocity can have identical observational effects if control is unmeasured. Thresholding one coefficient to zero may simply assign shared explanatory power to the other.

Derivative estimation adds a second difficulty. Finite differences amplify noise, especially for second derivatives. Smoothing reduces variance but can attenuate sharp events. Weak-form methods integrate against test functions to avoid pointwise differentiation, though they introduce choices about test functions, window size, and boundary effects. In every case the derivative or integral estimator is part of the model-selection pipeline and should be included in synthetic recovery tests.

Scaling is essential. If one feature has numerical values near 10^3 and another near 10^{-3} , coefficient penalties can favor one solely due to units. Normalize features, retain the transform for interpretation, and use physically meaningful coefficient units. Tune sparsity on held-out trajectories or conditions, not on random time points whose neighboring samples leak nearly identical states into training and validation.

Sparse Penalties and Bayesian Alternatives

A common objective combines squared residual with an ℓ_1 penalty:

$$\min_{\Xi} \|\dot{X} - \Theta\Xi\|_2^2 + \lambda \|\Xi\|_1.$$

Sequential thresholded least squares alternates regression and coefficient removal. Group lasso can select whole feature families. Bayesian shrinkage, including ARD-style group precisions, can produce posterior uncertainty and relevance rankings. None removes the need for a meaningful feature library or informative experiment. Correlated features can lead to unstable support across data subsets; report selection frequency or posterior uncertainty rather than a single binary support.

For discrete candidate models, marginal likelihood or reversible-jump sampling can compare different numbers of groups or states, but the result depends on priors and model definitions. A practical workflow uses sparse methods to screen a broad library, then compares a limited set of physical hypotheses in greater detail. The selected structure should survive prior and penalty sensitivity analyses.

Worked Example: Damping or Feedback?

Suppose recorded motion obeys

$$I\ddot{q} + b\dot{q} = u, \quad u = -k\dot{q}.$$

Then the observed equation is $I\ddot{q} + (b + k)\dot{q} = 0$. If only $q(t)$ is measured and u is unobserved, a regression library containing $\dot{q}$ can estimate an effective decay coefficient proportional to $b + k$, but cannot separate physical damping b from feedback gain k . A sparse algorithm may select a damping term and report a compact model. That term is predictive under the same control policy, but its physiological interpretation is unsupported.

If the input is measured, variation in u can separate the terms, provided it is sufficiently rich and not exactly collinear with velocity. A perturbation protocol can also help: apply known torque disturbances or alter feedback conditions. The discovery method does not create information absent from its inputs.

Selecting Model Order Without Overclaiming

A defensible procedure has several layers. First create a baseline physical model and identify candidate residual forces. Second fit a sequence of sparsity levels using trajectory-level folds. Third measure not only pointwise derivative error, but integrated rollout accuracy, constraint violations, and stability. Fourth examine support stability under resampling, noise, derivative estimators, and library variants. Fifth test synthetic recovery where the truth is known and test negative controls with observationally equivalent structures. Sixth report whether the selected order is stable and what it means: active terms in a fixed library, not “true system dimension” unless additional evidence warrants that interpretation.

For hybrid motion, split continuous regimes or use a hybrid library with explicit mode indicators. A single polynomial approximation may approximate stance and flight over a narrow interval, but its coefficients can be uninterpretable and its transitions inaccurate. Impacts involve velocity reset maps and nonsmooth constraints; smooth regression of sampled acceleration is not a substitute for an impact law. The repository’s hybrid dynamics materials should be reviewed alongside this chapter before specific implementation recommendations are finalized.

Failure Analysis

A low training residual with many terms can signal an overcomplete library. A sparse model that changes support dramatically across seeds signals weakly determined structure. A small derivative error but poor rollout indicates accumulated integration error or noisy derivatives. Good rollouts in one task and failure under perturbation indicate policy/task confounding. Unexpected nonzero terms may expose missing variables, delay, or contact rather than new physical laws. Each symptom suggests a different correction; simply increasing the penalty is not a universal fix.

Limitations

Sparse regression is not invariant to feature transformations. A nonlinear change of coordinates changes which terms are sparse. Threshold values have no universal scale. High-dimensional systems with short trajectories can have too many candidate terms relative to informative data. Hybrid events violate smoothness assumptions. Missing inputs and hidden states lead to effective rather than mechanistic equations. Interpretability depends on grouping and units as much as on sparsity.

Residual Discovery Around A Known Model

A useful compromise is to preserve a trusted physical backbone and discover only the unexplained component. Let $\tau_{\text{known}}(q, \dot{q}, \ddot{q}; \theta)$ denote known or estimable mechanics, and let τ_{res} be the residual force. Then fit $\tau_{\text{res}} = \Phi(q, \dot{q}, u)\beta$ with group sparsity. This reduces the search space and makes a selected term easier to interpret. For example, if gravity and inertia are known, the candidate residual library might contain passive damping, joint elasticity, or a small set of actuator moment-arm corrections. The residual should be evaluated with its contribution plotted over the observed domain; a coefficient alone can hide a function that grows unbounded outside it.

But a residual is not automatically “small physics.” If the known model has wrong coordinate definitions, sensor alignment, or contact semantics, the residual may become large and structured. A diagnostic should compare residual magnitude with sensor uncertainty and with the scale of each known force. If the residual cancels the main model terms, revisit the backbone before adding more library elements. Use held-out trajectory residuals to determine if a missing effect repeats consistently.

Library design should be pre-registered or documented before selecting support. If many candidate terms are added after inspecting test errors, the test set has become part of training. Preserve an audit trail of feature transformations, normalization constants, group definitions, threshold search, and rejected candidates. The repository can make this easier by storing a machine-readable model-selection manifest alongside publication artifacts.

Exercises

1. For a pendulum observed only over $|q| < 0.1$, explain why a library containing both q and $\sin q$ can have unstable support.
2. Propose a grouped library design for a two-joint model with gravity, viscous damping, and two measured actuators.
3. A discovered model uses more terms when the sample rate is reduced. List possible explanations and diagnostics.

Solution Sketches

1. Over this range, $\sin q$ and q are almost collinear, so the data cannot reliably allocate the restoring effect. Expand the angle range if safe, remove redundant parameterization, or impose known gravity structure.
2. Keep known gravity features as one group; assign one damping feature per joint; represent each actuator through measured input channels and mechanically specified moment arms; group symmetric couplings if justified. Test whether the inputs excite both channels independently.
3. Reduced sampling may worsen derivative estimates, alias high-frequency modes, miss contact transitions, or change effective delay. Compare smoothing/integral estimates, inspect event timing, use a controlled simulator, and evaluate rollouts at matched conditions.

An Adaptive Model-Constrained Matching Design

Design Status and Research Claim

This chapter proposes an architecture for adaptive matching; it does not announce a completed UpstreamDrift implementation or a validated human model. The design builds on existing observation contracts, constrained fitting, model identities, ambiguity handling, and independent replay. Its new obligation is to decide which model changes the observations actually support.

The objective is not to let an optimizer freely change every dimension until the displayed motion agrees. That procedure can replace an observation error with a fictitious anatomical change, or replace a missing force with an extra state. The objective is to identify a defensible family of models, preserve uncertainty, and choose complexity using evidence that did not tune the candidate.

We propose separate adaptation loops for geometry, control representation, mechanical structure, and latent memory. Each loop has different admissible changes and different invalidation rules. A geometry change invalidates affected inertia, marker, contact, and replay evidence; a structural change also invalidates coordinate maps and dimension-specific learned checkpoints.

The Three-Level Estimation Problem

Let k index a registered structure, θ_s denote subject parameters shared across trials, and ζ_j denote trial-specific unknowns. A trial-specific vector includes initial state, control coefficients, camera perturbations where appropriate, and contact-event uncertainty. Define the constrained inner fit

$$\hat{\zeta}_j(\theta_s, k) \in \arg \min_{\zeta_j \in \mathcal{F}(\theta_s, k)} L_j(\zeta_j, \theta_s, k),$$

where $\mathcal{F}$ enforces the declared dynamics and physical constraints. The shared-parameter level solves

$$\hat{\theta}_s(k) \in \arg \min_{\theta_s \in \Theta_k} \sum_{j \in \mathcal{J}} L_j(\hat{\zeta}_j(\theta_s, k), \theta_s, k) + P_k(\theta_s).$$

The structure level compares candidates using a separate selection set $\mathcal{V}$:

$$\hat{k} \in \arg \min_k \left[\sum_{j \in \mathcal{V}} L_j^{\text{pred}}(\hat{\theta}_s(k), k) + \gamma \mathcal{C}(k) \right].$$

These equations specify roles, not a guarantee that nested optimization is tractable or globally optimal. A practical system may alternate blocks, approximate inner solves, and compare a small set of structures. Its receipt must record those approximations and the solver budgets.

The predictive loss needs a precise adaptation policy. For a held-out trial, estimating its initial state from a short prefix and then predicting a later interval is different from fitting controls to the entire trial. The latter tests transfer of geometry under new fitting, while the former tests conditional forward prediction. Both are useful, but their scores answer different questions.

A final test set should remain untouched after selecting structure and regularization. Repeatedly choosing dimensions on the same validation trials overfits the selection process even if each inner fit is properly separated. With few recordings, nested cross-validation or an explicitly exploratory designation is more honest than a single confirmatory claim.

Stage A: Establish the Observation and Model Identity

Every run begins with immutable target and model identities. The target identity binds source bytes, time units, coordinate frames, masks, interpolation policy, sensor calibration, and any derived quantities. The model identity binds topology, physical parameters, actuator ordering, contact laws, configuration representation, and integration settings.

The program first checks that each observed channel has a defined prediction. A positional club trajectory cannot be promoted into measured club orientation. An orientation reconstructed from ball flight is a derived channel with its own model uncertainty, not an independent direct measurement.

The program next defines nuisance parameters and gauge conventions. For image observations, this includes camera calibration, world scale, and permitted camera motion. For markers, it includes attachment offsets and rigid-cluster assumptions. For motion and force data, it includes synchronization and reference-point conventions.

Gauge fixing should be declared once and used by every solver. If absolute world translation is unobserved, fix one reference or quotient out that freedom. Adding a tight prior to all translations can stabilize optimization, but it does not establish that the world origin was observed.

Stage B: Calibrate Geometry Before Interpreting Dynamics

Start with a fixed topology and a restricted geometry vector. Segment scales, joint centers, and marker offsets should be grouped according to the observations available. A static trial can identify some attachment and alignment relationships, but generally does not identify dynamic inertia.

Use held-out marker clusters or calibration postures to test geometry. A segment scale that improves the fitting markers but degrades an independent cluster may be absorbing soft-tissue movement or a labeling error. The full observation model should be reviewed before the scale is accepted.

The scaling law must remain explicit. UpstreamDrift’s shared segment scaler preserves mass while scaling inertia with the square of length. That is a declared approximation, not a con-

sequence of learning subject length alone. Constant-density similarity would instead change mass with the cube and inertia with the fifth power of scale.

The geometric proposal must update every affected attachment consistently. Distal joint translations, centers of mass, markers, contact locations, and graphics need the same declared transformation. Updating only rendered geometry would create a visually convincing but mechanically inconsistent subject model.

A practical geometry loop can alternate marker-offset estimation with bounded segment scales. Each proposed step uses a trust region and is rejected if it worsens held-out geometric prediction or violates positive lengths and inertia consistency. The published result should preserve both the data-only sensitivity and the prior contribution.

Stage C: Fit Dynamics With Controlled Actuation Flexibility

After calibration, fit trial controls against the forward model with geometry fixed. This baseline reveals whether the declared plant can reproduce observations without co-adapting shape and actuation. It also establishes a comparison for later joint refinement.

Represent controls on normalized time $\sigma = (t - t_0)/T$. For a degree-six polynomial,

$$u(t) = \sum_{r=0}^6 a_r \sigma^r.$$

All a_r then have the physical units of the control channel. For a polynomial in physical seconds, the coefficient of t^r has units of control divided by time to power r . Sharing numerical bounds between these conventions is not physically equivalent.

Control basis changes require a represented-function comparison, not coefficient equality. A normalized monomial, Bernstein polynomial, and B-spline can encode the same input with different coefficients. Cross-engine parity should compare evaluated input histories, integrated work, and trajectories at common physical timestamps.

Multiple shooting is appropriate when one long rollout becomes difficult to optimize. Its intermediate states improve the numerical search, but final acceptance still requires an uninterrupted native replay. A defect penalty is not a hard continuity guarantee, and a transformed node state requires correctly transformed Jacobians.

After the fixed-geometry baseline, bounded joint refinement may be attempted. Accept a geometry update only if it improves independent prediction and remains supported after profiling control and calibration nuisance variables. Otherwise preserve the fixed geometry and report the unresolved residual.

The Crucial Test: Profile Out Nuisance Compensation

Let r denote whitened residuals, J_θ their local derivative with respect to candidate dimensions, and J_ζ their derivative with respect to nuisance variables. Linearizing gives

$$r(\theta + \delta\theta, \zeta + \delta\zeta) \approx r + J_\theta \delta\theta + J_\zeta \delta\zeta.$$

For a fixed dimension perturbation, nuisance variables can compensate within their local column space. In the unconstrained local least-squares case, remove that compensable component using

$$P_\zeta = J_\zeta J_\zeta^+, \quad \tilde{J}_\theta = (I - P_\zeta) J_\theta.$$

The profiled local information is

$$\mathcal{J}_{\text{data}} = \tilde{J}_\theta^T \tilde{J}_\theta.$$

If a candidate length sensitivity lies entirely in the nuisance column space, this information is zero even if the raw length Jacobian is large. The data can respond strongly to length while remaining unable to distinguish length from a camera or control change.

In practice, compute the projection through a rank-revealing factorization rather than constructing a dense observation-space projector. Column scales should represent meaningful parameter changes, and row whitening should reflect the declared observation model. Singular-value thresholds need a sensitivity analysis; they are numerical decisions rather than universal physical constants.

Equality constraints and active inequality bounds alter feasible nuisance directions. The correct local analysis restricts perturbations to the constraint tangent space or uses a reduced Karush-Kuhn-Tucker system. Treating every bounded nuisance parameter as freely compensating can understate information near a genuine physical boundary.

Prior precision contributes separately:

$$\mathcal{J}_{\text{posterior}} = \mathcal{J}_{\text{data}} + \Lambda_{\text{prior}}.$$

A full-rank posterior matrix can coexist with rank-deficient data information. A parameter supported only by Λ_{prior} should be reported as prior-dominated. Tightening the prior is a modeling decision, not new observational evidence.

The projection is local. Nonlinear profiles may be asymmetric, disconnected, or constrained by contact switches. Important dimensions should also receive finite perturbation profiles: re-optimize nuisance variables at several fixed dimension values and retain alternative feasible solutions.

Stage D: Adapt Representation Without Inventing Anatomy

Control representation should adapt before mechanical topology when the residual pattern suggests insufficient actuation resolution. Compare a small nested sequence of bases under the same observation and feasibility gates. Penalize complexity and assess held-out replay rather than rewarding training error alone.

An extra basis term is accepted only if its improvement survives uncertainty and numerical refinement. A high-order polynomial may fit one interval while oscillating outside it. A local spline may improve contact timing but introduce unrealistic control rate unless its boundary continuity and rate constraints are specified.

Adaptive mesh refinement belongs to this representation layer. Refine around large integration error, unresolved contact transitions, or high residual curvature. Changing the mesh changes the discrete optimization problem; optimizer termination on one mesh does not prove continuous-time convergence.

The system should preserve the simplest accepted candidate when improvements are indistinguishable. This policy is an engineering choice toward interpretability and reproducibility. It does not prove that the human mechanism uses the smallest representation.

Stage E: Compare Mechanical Structures

Mechanical changes should come from a finite registered hypothesis set. Examples include planar versus spatial motion, rigid versus flexible shaft, fixed versus compliant grip, and alternative wrist structures. Each candidate needs its own physically coherent state and control maps.

A structure proposal should name the residual it aims to explain. A missing out-of-plane coordinate cannot be repaired by more planar torque coefficients. Conversely, a camera synchronization error should not motivate a new anatomical joint.

Reject dimension candidates that reduce marker error by violating force feasibility or exploiting hidden state resets. Also reject candidates whose improvement disappears when observations are evaluated at the same timestamps and masks. Numerical and acquisition differences should not masquerade as model selection.

When several structures remain compatible with the data, retain them as an ensemble of hypotheses. Report the spread of decision-relevant quantities, including reaction loads and counterfactual predictions. A single best scalar score conceals structural uncertainty that can matter more than parameter covariance within one model.

Stage F: Add Latent Memory Only With a Testable Role

A latent state may capture unmodeled hysteresis, activation delays, or compliant memory. Its dimension should be chosen by future predictive performance across held-out histories. A decoder that sees the entire target trajectory can improve reconstruction without demonstrating a causal forward state.

Define how the latent state is initialized. If it is fitted separately to every complete held-out trial, the procedure measures reconstruction capacity. If it is inferred from a bounded prefix and then evolved causally, later prediction provides a stronger test of dynamical sufficiency.

The physical and latent components need a declared coupling. A residual generalized force can be bounded and reported separately from the physical actuation. A learned mass matrix requires symmetry, positive definiteness, and coherent velocity terms. Neither choice automatically guarantees conservation, stability, or correct contact behavior.

The latent learner should abstain outside its trained model and observation domains. Model topology, morphology, contact law, time grid, and sensor semantics belong in the checkpoint identity. Silent application across a changed dimension is a contract failure.

Decision Rules and Failure States

A candidate dimension update is promoted only after it passes physical validity, supported information, independent prediction, and uninterrupted replay. The thresholds should be preregistered for each benchmark family and justified by observation uncertainty or task requirements. This book proposes the gate structure, not universal numerical thresholds.

Failure states should distinguish insufficient observations, solver failure, physical rejection, unsupported runtime, out-of-domain learned model, and ambiguous selection. An ambiguous result is valuable: it identifies which additional measurement or experiment could resolve the choice.

The fit receipt stores all tried candidates, including rejected ones. Reported runtime includes failed starts and native replay costs. Accepted-only timing creates survivorship bias, particularly when a neural proposal produces many fast but infeasible suggestions.

Worked Synthetic Design Example

Suppose a synthetic two-link arm has known geometry, measured endpoint positions, and independently measured joint torques. Fit a baseline with the correct structure, then perturb one segment length while re-fitting initial state and small sensor offsets. This experiment tests whether the pipeline can recover an identifiable geometry perturbation under controlled conditions.

Now remove torque measurements and allow flexible controls. The geometry estimate may become much less certain because actuation can compensate. Compare raw and profiled sensitivity spectra, finite parameter profiles, and the effect of adding an independent elbow marker.

Finally introduce a delayed elastic component in the synthetic truth. Compare a rigid model, a physical spring state, and a latent residual state using the same prefixes and held-out future observations. The correct selection depends on excitation, noise, and recorded channels; recovery of the generating structure is not guaranteed under insufficient observations.

The benchmark should include a deliberately impossible task: infer absolute mass scale from purely zero-input motion when the equations admit common scaling. A trustworthy adaptive matcher reports nonidentifiability rather than returning a confident numerical estimate because its optimizer finished.

Exercises and Solution Sketches

1. **Raw Versus Profiled Information.** Let $J_\theta = J_\zeta$ be one nonzero column. What is profiled dimension information? **Solution:** the nuisance projection removes the entire column, giving zero data information.
2. **A Prior-Rank Trap.** Add a positive diagonal prior precision to a zero data information matrix. What becomes full rank? **Solution:** posterior precision becomes full rank; data identifiability remains absent.
3. **Scaling Laws.** Compare length scaling at constant mass with constant-density similarity. **Solution:** inertia scales quadratically in the first case; mass scales cubically and inertia to the fifth power in the second.
4. **Held-Out Adaptation.** Why can fitting controls over an entire held-out trial fail to test future prediction? **Solution:** future observations selected those controls; the experiment tests new-trial reconstruction under transferred parameters.
5. **Model Promotion.** A latent model lowers training residual but fails contact replay. What is its status? **Solution:** a rejected reconstruction candidate, not an accepted adaptive dynamics model.

Research Synthesis and Priority Gaps

A Review With Bounded Authority

This review combines mathematical exposition, an immutable UpstreamDrift source audit, and a proposed adaptive matching architecture. The audited source revision is d7c7d07add0d7407a5a3376d5fe4a97aee9112a0. Checked-in tests and receipts establish what their authors recorded; this review did not rerun native MATLAB, OpenSim, Drake, MuJoCo, or Pinocchio experiments.

The source audit covers the principal matching families discovered through targeted code, test, and documentation inspection. It does not certify every experimental checkpoint or every historical branch. Repository-wide words such as “complete” should therefore be avoided when a narrower evidence status is available.

The literature includes standard trajectory optimization, musculoskeletal optimal control, sparse discovery, and structured neural dynamics (Kelly 2017; Betts 2010; Dembia et al. 2020; Brunton et al. 2016; Cranmer et al. 2020; Greydanus et al. 2019). These sources motivate methods and limitations. They do not establish that the proposed combination is novel, nor that it will recover a golfer’s true internal dynamics.

Method Comparison

Method Family	Estimated Variables	Principal Benefit	Main Failure Mode	Required Independent Check
Framewise or trajectory inverse kinematics	Configuration and calibration	Efficient geometric initialization	Motion can be dynamically infeasible	Forward controls and physical replay
Inverse dynamics plus force allocation	Net loads and selected actuator/contact allocations	Mechanics-based initialization and diagnosis	Derivative noise, missing loads, nonunique allocation	Recorded forces and feasible forward replay
Single shooting	Initial state, controls, parameters	Direct integrated reproduction	Long-horizon sensitivity and local minima	Refined native uninterrupted integration

Method Family	Estimated Variables	Principal Benefit	Main Failure Mode	Required Independent Check
Multiple shooting	Controls, parameters, intermediate states	Better local conditioning and data-informed starts	Residual state defects hide discontinuities	Zero-reset native replay and defect gates
Direct collocation	States, controls, parameters at nodes	Sparse treatment of path constraints	Mesh-dependent defects and between-node violations	Mesh refinement and independent reintegration
Filtering and smoothing	State histories and sometimes parameters	Sequential uncertainty and partial observations	Process noise absorbs model mismatch	Predictive innovations and process-noise audit
Derivative-free and multistart search	Any bounded finite parameter vector	Handles black-box engines and difficult starts	Evaluation expense and weak global assurance	All-start accounting and native acceptance
Differentiable physics	Parameters and controls through sensitivities	Efficient local derivatives	Incorrect event, constraint, or discrete-map derivatives	Directional derivative convergence
Inverse neural proposals	Candidate controls or parameters	Fast amortized initialization	Averaging incompatible solutions or domain shift	Native reranking and rejection
Forward surrogate inversion	Parameters through a learned forward map	Cheap search over expensive plants	Optimizer exploits surrogate errors	Exact plant replay and uncertainty checks
Imitation and feedback tracking	Policy or controller parameters	Robust reference-following behavior	Feedback conceals open-loop mismatch	Separate policy and open-loop claims
Adaptive model selection	Geometry, order, structure, latent dimension	Tests adequacy beyond one fixed model	Confounding and selection overfit	Profiled information and held-out comparison

These families can form one pipeline. Inverse kinematics supplies initialization, inverse dynamics supplies a control guess, collocation or multiple shooting finds a candidate, and single shooting verifies the final result. Neural components can propose starts without replacing the physics acceptance authority.

Gap 1: A Canonical Represented-Control Contract

UpstreamDrift contains several polynomial, piecewise, Bernstein, and knot-based control paths. Some engine-specific fitters choose their own default bounds and regularizers.

This flexibility is useful experimentally, but numerical coefficient equality cannot establish identical represented controls.

The first priority is a versioned control contract binding actuator names, ordering, coefficient-to-power mapping, time normalization, units, bounds, saturation policy, and interpolation. A conversion between basis conventions should preserve the evaluated control function. Conversion tests should include endpoints, interior timestamps, derivatives where meaningful, and integrated work under a declared velocity history.

Polynomial coefficients in physical time have different units by degree. A constant coefficient has torque units, while the coefficient multiplying physical time squared has torque divided by time squared. A comment that gives every coefficient torque units is correct only for a dimensionless-time convention.

This gap is a parity risk, not a demonstrated defect in every engine path. Each adapter must be reviewed against its actual evaluator and tests before changing it. The research book should preserve historical conventions while recommending a canonical representation for future comparisons.

Gap 2: Data Information Must Be Separated From Prior Information

The existing identifiability module can augment a Jacobian with prior rows before singular-value analysis. That computation evaluates regularized local sensitivity. It should not be interpreted as the rank of information supplied by markers alone.

A future result should publish both data-only and prior-augmented spectra. It should also profile out nuisance parameters, because camera, control, and marker-offset changes can compensate for a dimension change. Parameter names implicated in a null vector help interpretation, but null directions are combinations rather than independent on/off labels for individual joints.

A thin singular-value decomposition needs special care when the number of observations is smaller than the number of parameters. The returned positive-length spectrum may omit right-null directions unless the decomposition or nullspace construction explicitly retains them. The book identifies this as an audit target for underdetermined cases, not a claim that every current output is wrong.

The falsifier is straightforward: construct an observation function with fewer independent outputs than parameters and verify that all missing directions are reported. Add a positive prior and verify that posterior precision changes while the data-only nullspace remains unchanged. These are meaningful TDD cases for a future implementation.

Gap 3: Geometry and Dynamic Parameters Need Different Evidence

Segment scaling and anthropometric model generation already exist. They provide necessary machinery for adapting a model to a subject, but a configurable scale is not automatically a learned scale. The estimation objective, information, uncertainty, and independent validation still need to be supplied.

Static calibration should estimate observable geometric combinations. Dynamic identification should then estimate only those inertial or passive parameters that measured inputs and informative motion support. A single high-speed trial with unknown actuation is generally a poor basis for assigning independent masses, inertias, damping, and strength.

Population anthropometric tables can supply a prior. Their uncertainty and segment definitions must remain visible. Tight demographic bounds can improve numerical conditioning while preserving a prior-dominated result; the report should describe that result accordingly.

A useful falsification benchmark deliberately allows dimension-control compensation. If the adaptive learner confidently recovers a chosen arm length from an observationally equivalent family, it is overclaiming. Adding an independent elbow marker or calibrated multi-view geometry should reduce the ambiguity when the observation model supports it.

Gap 4: Tracking, Feasibility, and Replay Need Independent Statuses

Several upstream systems already separate kinematic fit, force feasibility, continuous replay, runtime, and biological actuator qualification. Future reporting should reuse this ladder rather than introduce another scalar “matched” flag.

Feedback-assisted tracking tests the model-controller pair. It can be valuable for stable simulation and initialization, but it does not identify a feedforward control trajectory. A reconstructed actuator history can be replayed independently, provided its interpolation, initial state, contact law, and integration method are preserved.

Force feasibility is another separate status. Satisfying an inverse-dynamics equation at fitted nodes does not show that a forward solver follows the trajectory. A continuous trajectory can also match positions while using unrealistically large reserves or invalid contact reactions.

The highest-value computational gate is uninterrupted native replay with physical audits. It should measure observation residuals, constraint error, torque saturation, contact behavior, energy accounting, and event timing. A result may pass one gate and fail another; the receipt should retain both outcomes.

Gap 5: Surrogates Need Accepted-Quality Benefit, Not Only Prediction Error

Modern upstream source contains implemented forward surrogate and conditional inverse architectures. Older option directories still contain stubs, so status must be assigned to exact module paths and artifacts. A planning freeze that performed no training does not erase earlier training work elsewhere.

The next research question is whether these models improve time to an accepted native match. A small supervised trajectory error is insufficient if optimization drives the surrogate outside its reliable domain. A neural inverse can also suggest plausible coefficients that fail when replayed by the exact plant.

Benchmark total query cost, native evaluation count, failure rate, accepted quality, and the training cost needed to amortize speed. Include direct optimization and simple retrieval baselines. If the neural method only improves an easy synthetic fixture while degrading difficult observed targets, its promotion should remain restricted.

Split by source trial, subject, model identity, or generated family as appropriate. Adjacent frames from one trajectory should not appear on both sides of a split. Scaling changes and new contact laws require domain checks or renewed training evidence.

Gap 6: Model Dimension Selection Needs a Registered Hypothesis Space

An adaptive learner should distinguish changes to anatomy, mechanical freedom, latent memory, and representation. A single dimension integer is inadequate when these changes have different physical meanings and computational consequences.

Register a small model ladder with explicit embeddings and observation maps. Compare planar and spatial models, rigid and flexible equipment, and alternative grip or wrist hypotheses only where the recorded channels can distinguish them. An output outside a model's representational scope should trigger a model adequacy failure rather than extreme control fitting.

Use held-out prediction and physical replay to compare the ladder. Preserve alternative models when the score difference is smaller than the uncertainty. Structural uncertainty should propagate into reported loads and counterfactuals instead of disappearing after one selection.

A learned latent state can be included as a separate hypothesis. Its role should be causal prediction from a bounded observation prefix, not whole-trajectory reconstruction. The learner must specify initialization and abstention when observations cannot constrain the hidden memory.

Gap 7: Observation Uncertainty and Timing Need First-Class Treatment

Markers, inferred joint angles, silhouettes, club outcomes, and projected images have different error processes. Derived channels can share source uncertainty and should not automatically be treated as independent measurements. A force estimate computed from the same kinematics is not an independent force observation.

Synchronization deserves particular attention. A small time shift during rapid motion can create large positional and velocity residuals. Flexible actuation or extra latent state may absorb that shift, creating a false dynamic explanation.

Fit a bounded time offset only under an explicit calibration policy. Preserve the offset in the target identity and validate it against independent timing events if available. If a trial requires a large unexplained shift, investigate acquisition and preprocessing before changing the mechanical model.

Missing data should remain missing. Report observed-channel counts and residual coverage by phase, especially around delivery and contact. A full-trajectory average can conceal a fit that has no observations during its most important interval.

Gap 8: Numerical Accuracy Must Be a Model-Selection Control

Contact-rich replays can change when the integration step, constraint stabilization, or regularization changes. An adaptive method can therefore select dimensions that compensate for numerical error. Native refinement is necessary before assigning a residual to missing physics.

Compare candidate structures under commensurate numerical accuracy. Equal step sizes are not necessarily equal accuracy because stiffness and state dimension differ. Report convergence of decision-relevant metrics, not merely a common integrator setting.

Derivative checks should follow the actual discrete map used by optimization. Event-time and reset derivatives need their own tests. An accurate derivative of an inaccurate discretization can still optimize the wrong continuous behavior.

A Research Program With Explicit Outcomes

The first experiment is a synthetic identifiability suite with known truth and deliberately unidentifiable cases. It establishes whether the adaptive method distinguishes recoverable parameters from prior-driven choices. Recovery errors are model results, not empirical subject validation.

The second experiment compares canonical fixed-geometry controls across engines using shared observations and represented inputs. It measures adapter parity and numerical sensitivity before introducing adaptation. Unsupported engines remain explicit matrix cells rather than silently disappearing from the benchmark.

The third experiment estimates restricted geometry from independent calibration and then fits motion controls. It compares fixed, staged, and jointly refined parameter policies. Held-out markers and trials test whether adaptation improves reproducible prediction.

The fourth experiment compares a short structural ladder and optional latent memory. It tests whether more complex models improve independent outcomes while remaining physically feasible. The result may favor no adaptation, or preserve multiple compatible structures.

Physical subject validation requires new measurements, appropriate consent, and independent reference instruments. It remains deferred until collected; no amount of software testing supplies those observations. A maintained publication should list precisely which empirical questions remain open.

Exercises and Solution Sketches

1. **Coefficient Parity.** Two engines receive equal coefficients but use seconds and normalized time. Are controls equivalent? **Solution:** only if the coefficients are transformed by the appropriate duration powers and ordering; equal vectors alone are insufficient.
2. **A Prior-Dominated Dimension.** A scale estimate barely changes across trials but changes substantially when its prior mean changes. What should be reported? **Solution:** prior sensitivity and insufficient data support, rather than confident learned morphology.
3. **Neural Benefit.** A proposal generator is fast but most candidates fail native replay. Which metric is missing? **Solution:** total time and evaluations to accepted quality, including rejected candidates and amortization.
4. **Model Adequacy.** A planar model cannot reproduce observed spatial club orientation. What should the fitting system do? **Solution:** report representational inadequacy and compare a registered spatial structure; more planar coefficients do not supply the missing coordinate.

Latent Learned Dynamics and Their Physical Interpretation

Why Introduce A Latent State?

A Markov forward model predicts the next state from its current state and input. If the observed kinematics do not contain enough information to predict future motion, a latent state may carry memory of unobserved activation, control intent, tissue deformation, fatigue, contact mode, or sensor delay. A latent representation can improve prediction even when its coordinates have no direct anatomical interpretation.

A continuous-time latent model can be written

$$\dot{z} = f_\phi(z, u), \quad y = g_\psi(z),$$

or, for a mechanics-centered model, z may augment measured configuration and velocity. Neural ordinary differential equations parameterize the derivative function and use a numerical solver to obtain trajectories. Latent ODE methods infer a latent initial state from irregular observations and evolve it continuously. References: Chen et al., “Neural Ordinary Differential Equations”; Rubanova, Chen, and Duvenaud, “Latent ODEs for Irregularly-Sampled Time Series”. Related structured neural mechanics approaches include learned Lagrangian and Hamiltonian models (Cranmer et al. 2020; Greydanus et al. 2019; Lutter et al. 2019). Live link validation is pending under the repository’s current network restriction.

The latent dimension is a model capacity choice. It is not automatically the number of physical degrees of freedom. A state coordinate can be transformed by an invertible map while changing the vector field and decoder but preserving the same observations. Extra latent units can be unused, redundant, or encode nuisance history. Conversely, a compact latent model may compress a complex physical system over the limited domain of its training data.

The Identification Problem

Suppose observations satisfy $y_k = g_\psi(z_k) + \epsilon_k$, while latent states follow $z_{k+1} = F_\phi(z_k, u_k)$. If both F and g are flexible neural functions, many combinations can produce the same observed sequence. The learned state may be poorly determined even when predictions are excellent. This is a familiar realization ambiguity in system identification: observable input-output behavior does not uniquely label hidden coordinates.

The issue gets sharper when controls are unknown. An augmented latent state may reconstruct the recent history of observed motion and thereby infer an effective control signal. That can support forecasting, but it does not tell us whether the latent variable represents

neural activation, an omitted force, or a delayed sensor. Independent measurements and interventions are required for semantic interpretation.

To decide whether latent memory is warranted, compare a Markov mechanics baseline, a mechanics model with residual force, and a latent augmented model under identical data splits. If the latent model improves only one-step prediction but not multi-step rollouts, it may exploit local interpolation. If it improves long-horizon prediction across unseen inputs and subjects, it offers stronger evidence for useful memory but still not for anatomical identity.

Mechanics-First Augmentation

An unconstrained neural vector field can violate kinematic constraints or produce impossible accelerations. A mechanics-first candidate preserves known equations and lets learned components represent specific residuals:

$$M_\theta(q)\ddot{q} + C_\theta(q, \dot{q}) + g_\theta(q) = B_\theta(q)u + J(q)^T \lambda + r_\phi(q, \dot{q}, u, z).$$

The residual r_ϕ should be bounded or regularized and should not override positive-definite inertia, joint constraints, or contact rules. If a latent state is included, its observation relationship and update dynamics should be stated. For instance, it may represent an activation variable with a first-order lag, but such an interpretation should be supported by independent physiological or control data. Without that evidence, call it a learned memory state.

This architecture is a research proposal, not an implemented AffineDrift feature. It should be evaluated against simpler alternatives and should not be introduced as a proven improvement. Candidate constraints may include energy dissipation for passive components, bounded actuator force, smoothness within a mode, and separate reset dynamics at impacts. Each constraint must match the intended physical regime.

Worked Example: Hidden Actuator Activation

Let a measured joint angle satisfy

$$I\ddot{q} + b\dot{q} = \tau, \quad \dot{a} = (u - a)/T, \quad \tau = ka.$$

Here u is a command, a a delayed actuator activation, and T a time constant. A model using only $(q, \dot{q}, u)$ is not Markov because current torque depends on the history of u through a . Augmenting with a restores Markov structure. If only q is observed and u is unknown, a latent model may infer a variable that predicts the lag. But that variable cannot be called activation without measurements or a validated actuator model. If u is known and varied in independent pulses, lag parameters may become identifiable; a constant input provides little information about the time constant after transients settle.

A practical diagnostic is to compare predictions after abrupt input changes. The no-latent candidate predicts an immediate response; the lagged model predicts a transient. The input protocol is informative because the hypotheses produce distinguishable trajectories. The latent dimension earns support through a testable response signature, not merely a lower training loss.

Selecting Latent Dimension

One can fit a sequence of latent sizes and compare held-out sequence likelihood or rollout error with a complexity penalty. A Bayesian formulation may put shrinkage priors on latent coordinates or compare discrete dimensions, but posterior conclusions depend on prior structure and data. Architecture searches should include multiple random seeds and report variation. A dimension that changes with initialization while predictions remain stable is evidence that representation size is not uniquely identified.

Useful controls include: (1) simulate data from known latent dimension and test recovery under realistic noise; (2) simulate data with no latent state but unmodeled contact or delay to see if the estimator invents one; (3) perturb latent size while keeping the decoder and solver fixed; (4) evaluate on held-out input waveforms, not only neighboring timestamps; (5) inspect whether latent trajectories are stable and smooth under small observation perturbations; (6) test whether dimensions map consistently across subjects if population claims are intended.

Latent-state dimension selection should be separated from neural network width, hidden-layer count, solver tolerance, and observation-decoder capacity. These all affect fit complexity. If a model is given a larger decoder and a different integrator at the same time that latent dimension changes, comparisons are uninterpretable.

Numerical and Physical Considerations

The numerical solver is part of the learned model. Stiff latent dynamics may require implicit solvers; adaptive step-size and tolerances affect gradients and runtime. Contact discontinuities can cause solver difficulties and should use event or mode-aware handling. Long rollouts can reveal drift not visible in teacher-forced training. Keep train and evaluation solvers consistent enough to identify modeling errors, but also test robustness to reasonable solver changes.

Physical invariants can guide architecture. Hamiltonian neural networks parameterize an energy function and use canonical equations; Lagrangian networks parameterize a Lagrangian and derive dynamics. These approaches encode useful structure for suitable conservative or mechanical systems, but they do not automatically solve actuation, friction, contact, or identifiability. The local bibliography lists relevant papers; verify the entries and scope before relying on specific performance claims.

For contact-rich human movement, a single smooth latent ODE may hide discrete stance/flight transitions. Hybrid models provide continuous flows in each mode plus transition guards and reset maps. The latent state can augment those flows, but mode logic should remain explicit where impact physics matters. A learned latent variable that tracks “stance” is not a substitute for checking contact constraints and impulse balance.

Evaluation and Reporting

Report separate metrics for observed-coordinate reconstruction, one-step prediction, multi-step forward rollout, event timing, constraint violation, and physical plausibility. Use complete held-out trajectories and conditions; random frame splits create leakage. Compare models at equal tuning effort and document parameter counts, latent dimensions, solver settings, and priors. Include uncertainty intervals or ensembles. Check performance under small perturbations to initial conditions and inputs; fragile latent codes can create unstable extrapolation.

A model intended for a specific task may use a latent state without claiming physical discovery. A model intended for scientific interpretation needs stronger evidence: identifiable observa-

tion map, independent signal corresponding to the latent, intervention response consistent with the proposed mechanism, and replication under varied conditions.

Limitations

Latent models can be excellent predictors while remaining uninterpretable. Flexible decoders may compensate for incorrect dynamics. Latent dimension comparisons are sensitive to priors, architecture, optimizer, and training duration. ODE integration assumptions may fail at events or under discrete sampling. Population-level latent axes may rotate or permute across individuals. Good prediction within the training domain does not validate physical extrapolation.

When Memory Is A Better Explanation Than A New Coordinate

A measured configuration may fail to be Markov for several reasons. Actuator activation can lag a command. A viscoelastic tissue model may require an internal strain variable. A camera pipeline may introduce delay. A contact sensor may omit the current mode. Before increasing latent size, test these alternatives because each suggests a different state equation and different evidence. Delay can often be diagnosed by synchronization checks; actuator lag by measured command-response transients; contact mode by force sensing; viscoelasticity by loading-unloading hysteresis. A generic latent state can predict all of these while identifying none of them.

The Markov property can be probed empirically: find pairs of observations with similar measured $(q, \dot{q}, u)$ but different short-term futures. If the difference is systematic and exceeds noise, then the observed state is insufficient under the current input description. This does not prove that a latent coordinate is physically necessary; hidden control and observation error can also cause the effect. Add candidate history summaries and compare them with explicit physical states. Use held-out sequences where the relevant history differs while the current measured state is similar.

Representation alignment matters for subject-level studies. Independently trained latent spaces can be rotated, permuted, or sign-flipped. Averaging latent coordinates across subjects is not meaningful until a mapping or alignment criterion is established. A model may instead compare decoded physical predictions or invariant latent distances. If latent dimensions are claimed to correspond to common mechanisms, test that correspondence with shared perturbations and independent measurements.

Exercises

1. Explain why an invertible transformation of latent coordinates can preserve observed predictions while changing the learned dynamics.
2. Design an experiment to distinguish instantaneous actuation from a first-order activation lag.
3. A latent dimension of 6 outperforms dimension 4 on a random frame split but not on held-out trials. Interpret the result and propose next steps.

Solution Sketches

1. If $w = T(z)$ is invertible, transform the vector field by the chain rule and decoder by composition; outputs remain the same though coordinate meaning changes.
2. Apply independently timed input pulses, observe the response onset and decay, and include enough time before/after each pulse to estimate the lag. Measure input and output with synchronized clocks.
3. Random frame splits likely leak temporal neighbors. The larger model may interpolate trial-specific trajectories but not generalize. Use trial-level splits, examine uncertainty, add varied inputs, and test synthetic dimension recovery before interpreting dimension.

Experimental Design and Model Comparison

Model Choice Begins With the Question

An experiment cannot be designed without specifying the quantity it is meant to identify. The target might be a physical parameter such as damping, a discrete structure such as whether a joint is free, an effective model for predicting one task, or an anatomical claim. These targets require different evidence. A task predictor can be useful without identifying the true actuator anatomy; a physical parameter estimate requires force and input assumptions; anatomical dimension needs structure-specific measurements or interventions.

Write the candidate models before collecting data. For each candidate, define state, input, observation map, physical constraints, unknown parameters, contact behavior, and predicted differences. Then ask which safe and feasible input or sensor condition makes those predictions diverge most relative to expected measurement noise. This changes experiment design from “record more motion” to “record motion that distinguishes hypotheses.”

Information and Excitation

For parameter vector θ , the local information matrix is approximately $\mathcal{J} = S^T R^{-1} S$, with output sensitivity S . An experiment can be scored by a design criterion such as maximizing the smallest eigenvalue (E-optimality), the determinant (D-optimality), or reducing variance for a selected parameter combination. These criteria are approximations and need meaningful noise models, but they expose a core principle: repeated observations along an already-excited direction do not inform a parameter that changes predictions orthogonally to that direction.

For structure selection, compute predicted differences between candidates under input sequence u . If two models yield mean outputs $\mu_1(u)$ and $\mu_2(u)$, a simple separation score is

$$D(u) = (\mu_1(u) - \mu_2(u))^T R^{-1} (\mu_1(u) - \mu_2(u)).$$

A larger value suggests the candidate predictions are easier to distinguish, subject to nonlinear effects and parameter uncertainty. More principled Bayesian design averages this separation over parameter distributions, but can be computationally expensive. In human movement, safety, comfort, fatigue, and protocol feasibility constrain the design space; maximize useful information within those limits.

Worked Example: Locked Versus Weakly Free Joint

Suppose one candidate locks coordinate q_2 and another allows q_2 to move under an unknown stiffness. During normal motion q_2 remains nearly fixed, so both predict almost identical marker trajectories. A targeted, low-amplitude perturbation can reveal whether q_2 returns with a restoring response or remains constrained. A sensor placed near q_2 improves observation sensitivity. The protocol should include repeated perturbations at varied starting positions, because a free coordinate at one equilibrium can still look locked locally.

The trial must also control nuisance effects. If input force is not measured, a lack of movement could reflect weak perturbation rather than a locked constraint. Synchronization errors can mimic lag; marker noise can obscure small motion. Thus the design includes calibration, force measurement, and repeated conditions, not only a perturbation waveform.

Data Splits That Test the Intended Use

Time samples within one trajectory are highly dependent. Randomly assigning frames to train and test leaks nearly identical neighboring states. For forward dynamics, split by entire trial, initial condition, subject, input waveform, or task condition according to the deployment claim. A model for new conditions should be evaluated on conditions withheld during all tuning. A population model intended for unseen people needs subject-level splits and should report the range of subject variation.

Use nested validation if model order, regularization, preprocessing, or architecture are selected empirically. The outer split estimates generalization; inner folds tune choices. Keep the test set untouched until the model and candidate-selection procedure are frozen. If data are scarce, use repeated grouped splits and report uncertainty rather than a single best score.

Compare Multiple Dimensions of Quality

A comprehensive comparison includes:

- observation fit: marker or joint-angle likelihood with declared noise model;
- derivative fit: only when derivative estimation is justified and uncertainty is included;
- short and long rollout error under measured inputs;
- response to unseen perturbations and initial conditions;
- physical feasibility: positive inertia, joint limits, constraint residuals, force bounds;
- hybrid quality: event timing, mode classification, impulse or contact residuals;
- robustness: seed, preprocessing, solver, and resampling variation;
- complexity and uncertainty: parameter groups, active terms, latent size, posterior dispersion.

Do not collapse every metric into one score without retaining components. A weighted total can be useful for a predeclared engineering objective, but it should not erase the fact that one candidate is more accurate and another more physically feasible. Pareto comparisons often communicate these tradeoffs better.

Bayesian and Frequentist Model Comparison

Information criteria, cross-validation, marginal likelihood, and posterior model probabilities answer related but different questions. Cross-validation asks how well a fitting procedure predicts held-out data under a specified split. Marginal likelihood averages fit over parameter

priors and can penalize unused parameter volume; it is sensitive to prior scale. AIC-like criteria estimate expected predictive information under assumptions. None produces a prior-free truth ranking.

For different-dimensional candidate models, reversible-jump MCMC can sample across parameter spaces of changing dimensions. Green's foundational reference is "Reversible jump Markov chain Monte Carlo computation and Bayesian model determination," *Biometrika*, 1995, DOI <https://doi.org/10.1093/biomet/82.4.711>. Link validation is pending. Such methods require careful proposal design and convergence checks. Label switching, symmetric parameterizations, and poor mixing can make model probabilities unreliable. Use synthetic known-truth experiments and multiple chains before interpreting results.

A practical compromise is a candidate table. Fit each physically meaningful structure with identical observation models, comparable prior knowledge, and equal validation effort. Report held-out predictive performance and physical checks. Use Bayesian model comparison as a secondary audit on a manageable candidate set. Conduct sensitivity checks across prior scales. If plausible alternatives remain close, report uncertainty instead of forcing a winner.

Contacts and Hybrid Protocols

Walking, striking, and other contact-rich motion involve discrete events. Design trials to include clear event labels or force plate data when contact transitions are part of the scientific question. Distinguish rigid contact with impacts from compliant contact with sustained deformation; sample rate and sensor bandwidth must capture the relevant event timescale. A smooth model may fit samples away from impact while failing the impulse itself. Event windows should be scored separately.

If contact cannot be observed directly, include uncertainty in mode labels and compare explicit hybrid candidates. A model may otherwise infer a larger latent state that encodes unobserved stance status. That may help prediction, but it complicates physical interpretation and may fail when event patterns change.

Adaptive Dimension as A Sequential Decision

For a system that changes over time, dimension adaptation should be treated as sequential inference. A new residual may indicate a changed system, a changed task, fatigue, measurement drift, or out-of-distribution input. Maintain a posterior or score over a finite set of candidate structures, monitor calibrated residuals, and require sustained evidence before switching. A cost for topology changes prevents rapid dimension chatter. Keep a record of triggering conditions and re-evaluate old and new models on shared reference trials.

The adaptive model should have an abstention threshold. If posterior support is diffuse or candidate predictions are too similar under available data, continue with a conservative model and request informative observations. This is especially important for human systems, where an unnecessary structural change could be misleading even if the software remains operational.

A Reproducible Study Template

1. State the estimand and intended deployment domain.
2. Enumerate candidate models and mark assumptions that are fixed versus learned.
3. Specify sensors, calibration, sampling, noise, input measurements, and preprocessing.
4. Run simulation-based design to find informative and safe trajectories.

5. Collect training and independent validation trials with grouped splits.
6. Fit baselines before adaptive models: known mechanics, parameter estimation, sparse residual, and latent residual.
7. Test identifiability and synthetic recovery under realistic noise.
8. Compare prediction, physical feasibility, contact behavior, uncertainty, and complexity.
9. Perform prior, regularization, seed, and preprocessing sensitivity analyses.
10. Publish the complete candidate inventory, exclusions, validation limits, and scripts needed to reproduce model selection.

Limitations

Optimal experiment design depends on candidate models and assumed noise. An input that is informative mathematically may be unsafe or unnatural. Human subjects may respond differently to repeated perturbations, creating nonstationarity. Bayesian model probabilities depend on priors; held-out performance depends on the split representing future use. Sparse data may force a compromise between identification and practical prediction. These constraints should be described as limits of evidence, not concealed behind a single metric.

Sequential Allocation of New Trials

An adaptive research program can choose new trials based on the current uncertainty. After fitting candidate models, identify the input condition with the greatest expected reduction in uncertainty or candidate separation, then collect a small batch and update. This is active learning for system identification. It can reduce wasted trials when candidate hypotheses make similar predictions under routine motion. In practice, selection should include a safety filter, fatigue limits, and a stopping rule. A nominally informative experiment should be rejected if plausible parameter uncertainty predicts unsafe forces or motion.

A stopping criterion should be tied to the scientific claim. For parameter estimation, stop when confidence or credible intervals meet a predeclared tolerance over the relevant parameter combinations. For structure selection, stop when one candidate has robust predictive and physical support across prior scales and validation conditions, or when the remaining candidates are observationally equivalent in the feasible domain. Do not stop merely because the optimizer converged or an information criterion changed little on repeated copies of the same trajectory.

Sequential protocols can themselves change the system: participants learn, fatigue, or adapt their strategy. Randomize trial order where feasible, include reference trials throughout the session, and model time/order effects. If a new perturbation changes behavior, that response can be part of the system under study rather than mere noise. Record protocol order and participant state so the analysis can separate stationary mechanics from adaptation.

Exercises

1. For two candidate models, propose an input sequence that maximizes prediction separation while respecting a bounded torque constraint.
2. Explain why random frame-level cross-validation overestimates generalization for a trajectory model.
3. A Bayesian model comparison assigns 0.92 probability to a larger model, but changing the prior scale lowers it to 0.54. What should be reported?

Solution Sketches

1. Simulate each model across feasible input sequences, compute noise-weighted separation, and optimize under the torque bound. Validate the chosen input with parameter uncertainty and safety constraints; do not use a nominal-only score if parameters are uncertain.
2. Neighboring frames share nearly identical states and noise conditions, so test frames are effectively represented in training. Group by whole trajectories or trials.
3. Report strong prior sensitivity and treat the data as insufficient for decisive structure selection. Show held-out predictive performance and the prior assumptions, and seek more discriminating observations.

Learning-Based Tracking and Iterative Control

Several Meanings of “Matching”

Learning-based motion matching can mean at least three different tasks. **Imitation learning** fits a policy or trajectory to demonstrations. **Tracking control** computes actions online from current state error. **Iterative learning control (ILC)** updates an entire feedforward command between repeated trials. They use related data but answer different questions. A policy can imitate demonstrated state-action pairs yet fail to track under disturbances; a tracking controller can follow a reference that was never generated by a human; ILC can improve repeatable performance without learning a general policy.

In forward model matching, the physical dynamics remain explicit: $x_{t+1} = f(x_t, u_t; \theta)$, and controls or policy parameters are chosen so the forward rollout explains measured motion. In pure kinematic fitting, the system may directly assign poses without requiring dynamic feasibility. In reinforcement learning, an agent maximizes expected return through interaction or simulation; imitation objectives may be auxiliary rewards or initializers. Keep these labels distinct. A visually convincing simulated character does not establish that its controls are human, uniquely identified, or physically consistent with measured forces.

No inverse is generally unique. Many control sequences can produce similar observed trajectories, particularly with partial markers, hidden states, unknown initial conditions, and flexible dynamics. Equally, different physical parameters and control policies can compensate for one another. A learned policy therefore represents one solution under a chosen model, objective, data, and regularizer. It should not be presented as the unique recovered neural command or causal mechanism.

Open-Loop and Closed-Loop Control

An open-loop command is a time-indexed sequence $u_t = \bar{u}_t$ chosen without using current tracking error. It can reproduce a nominal trajectory under the same initial state and model, but disturbances accumulate. A closed-loop controller uses observations or state estimates: $u_t = \pi(x_t, t)$, often with feedforward plus feedback, $u_t = \bar{u}_t + K_t(x_t - \bar{x}_t)$. Feedback can correct model mismatch and perturbations, but its apparent trajectory fit then depends on the estimator, gains, sensor delay, and disturbance assumptions.

This distinction matters for scientific inference. If the question is whether a candidate forward model can generate the observed trajectory from a plausible control, open-loop replay is a strong and interpretable test. If feedback is allowed, report the feedback law, information available to it, latency, bandwidth, and whether it uses measured future data. A controller that reads the reference at the current time is legitimate tracking; a controller that uses future motion can unrealistically correct errors and must be described as preview control. A

policy trained against the full reference may encode timing in its state or reward and should not be mislabeled as an autonomous explanation.

A tracking loss can combine state and effort errors:

$$J = \sum_{t=0}^{T-1} (x_t - \bar{x}_t)^T Q_t (x_t - \bar{x}_t) + u_t^T R_t u_t + (u_t - \bar{u}_t)^T S_t (u_t - \bar{u}_t) + J_T.$$

The last control term regularizes deviation from nominal feedforward. A time-varying linear feedback gain can be obtained from a Riccati recursion around a nominal trajectory. For nonlinear models, iLQR or DDP repeatedly linearizes/quadratizes and updates the nominal. These methods can provide stabilizing local corrections, but their guarantees depend on model validity, constraints, and local convergence.

Imitation Learning Objectives

Behavioral cloning minimizes action prediction error on demonstrations: $\mathbb{E}_{(s,a) \sim D} [\ell(\pi(s), a)]$. It is simple but suffers covariate shift: at deployment, small errors visit states absent from the demonstrations, where the policy is unconstrained. Dataset aggregation addresses this by collecting expert labels at states induced by the current policy. In biomechanics, acquiring expert labels for perturbations may be impractical, so robust simulators or carefully justified proxy controllers are used; this introduces model dependence.

Trajectory imitation can instead match state distributions, key events, end-effector paths, or task outcomes. A pose loss may use markers and orientation on manifolds; a velocity or acceleration loss amplifies measurement noise. A reward can combine imitation with balance, effort, task success, and contact constraints. Reward design determines what behaviors are considered equivalent. A policy may exploit a weakly specified reward or simulation artifact. Include held-out motion segments, perturbation tests, and independent dynamics evaluation.

DeepMimic is relevant as an example of physics-based character skill learning from reference motion. It uses deep reinforcement learning to imitate motion while solving a task, demonstrating that a simulated character can learn robust skills that combine reference tracking and task rewards. Its relevance is conceptual and methodological: it shows a possible policy-learning architecture and training pattern. It does not establish that a learned policy uniquely recovers human intent, that its rigid-body or contact model matches a particular person's mechanics, or that a motion-matching inverse problem has a unique solution. These distinctions are essential when citing reinforcement-learning character animation in a scientific forward-dynamics study.

ILC for Repeated Motions

ILC assumes repeated execution of a task with repeatable dynamics and updates the feedforward command using prior-trial error. For trial index j , let output error be $e_j(t) = r(t) - y_j(t)$. A basic update is

$$u_{j+1}(t) = u_j(t) + L e_j(t),$$

where L is a learning operator. A filtered update might be $u_{j+1} = Q(u_j + L e_j)$ to limit high-frequency amplification. Under a linear time-invariant trial model $y = G u$, a sufficient norm condition for monotone error reduction in a simple setting is $\|I - GL\| < 1$. Real systems require robust analysis for model uncertainty, delays, and nonminimum-phase behavior.

ILC is useful when each swing or robot trajectory is repeated under nearly the same conditions. It learns a trajectory-specific feedforward command, not necessarily a general state-feedback

strategy. Trial-to-trial variability, changing fatigue, equipment changes, and nonrepeatable disturbances violate assumptions. If one human demonstration is available, ILC cannot be run on that individual without repeated trials; applying its logic to multiple athletes assumes meaningful alignment and a shared task. Distinguish subject-specific adaptation from population learning.

A practical ILC pipeline aligns trial phase, estimates output error with uncertainty, applies a bounded update, and checks safety before the next attempt. Phase alignment errors can make a correct command appear wrong and lead the update in a harmful direction. Time warping should be constrained and recorded; otherwise it can erase real timing differences. Use trust regions on command updates, limit actuator demand, and keep a baseline command available for rollback.

Combining Model Matching and Learning

A useful research architecture separates identification from policy adaptation. First estimate a family of plausible dynamics parameters from kinematic and force data. Next optimize feedforward controls or a policy over that uncertainty set. Finally replay candidates in an independent evaluator and compare prediction. This avoids treating the controller as a hidden correction term that masks dynamics error. Robust control can optimize worst-case or expected performance over parameter uncertainty; a posterior over models is more honest than a single best-fit model when identifiability is weak.

A differentiable model can support policy gradients or model-based planning, but differentiation through a learned surrogate can exploit errors. Use trust regions that keep simulated states near training data, uncertainty penalties, periodic real-engine rollouts, and final validation in the target dynamics. For contact-rich motion, gradients through a smooth surrogate may help initialize a candidate, while hybrid replay checks mode sequence and event timing. Report the surrogate and target model separately.

Evaluation should separate demonstration fidelity, task performance, robustness, and plausibility. Demonstration metrics include marker error and event timing. Task metrics include success and terminal constraints. Robustness metrics perturb initial conditions, dynamics, observations, and timing. Plausibility metrics may include actuator limits, contact forces, energy, and smoothness, but must have declared thresholds. No single weighted reward can replace a table of interpretable outcomes.

Proposed Adaptive Architecture

A proposed dimension-adaptive learner would maintain uncertainty over both physical model dimension and policy capacity. Use separate selection variables: d for generalized coordinates or structural modes, m for policy/control basis complexity. Candidate changes should be accepted only when they improve held-out prediction or task return beyond uncertainty and pass physical qualification. Synthetic tests should distinguish a missing state dimension from inadequate policy capacity by generating cases where each is independently necessary.

The policy should not be allowed to compensate invisibly for structural model inadequacy. Log residuals before and after feedback, policy action magnitude, and uncertainty. If feedback effort rises sharply in one phase, investigate missing coordinates, contact discrepancy, or sensor bias before increasing network size. Population generalization requires subject-level holdout; frame-level random splits leak adjacent temporal information. This entire adaptive architecture is a proposal, not an implemented upstream feature.

Repository Context and Limits

Upstream contains a MyoSuite surrogate training path under `src/engines/physics_engines/myosuite/python/`, an inverse-surrogate module, MuJoCo MJX tracking components, and a separate tracking plant. These are evidence of learning/tracking-related code pathways, not proof that all are integrated with the same calibrated forward model or qualify human motor intent. The pinned source revision for references in this book is `d7c7d07add0d7407a5a3376d5fe4a97aee9112a0`; claims about a specific algorithm should link to the exact immutable file and any candidate receipt.

Worked Illustrative Example

Consider an illustrative discrete system $x_{t+1} = x_t + u_t + w_t$ with reference r_t . Open-loop commands u_t chosen from a nominal disturbance-free model leave accumulated error when $w_t \neq 0$. A feedback law $u_t = \bar{u}_t - K(x_t - r_t)$ corrects measured deviation but depends on valid state estimates and cannot correct unobserved disturbances immediately. Across repeated trials, ILC can update $\bar{u}_t$ from the previous error. The three methods use different information and should be compared under distinct evaluation conditions; this toy example is not empirical evidence about a human swing.

Exercises and Solutions

Exercise 1. A policy reproduces all demonstration poses in simulation. Does that show it recovered the demonstrator’s unique control? **Solution.** No. Multiple controls, policies, and model parameters can produce similar observed poses. Additional force, perturbation, and physiological evidence may constrain the explanation but rarely guarantee uniqueness.

Exercise 2. Why can feedback tracking conceal an incorrect forward model? **Solution.** The controller can compensate online for prediction error. Report control effort and open-loop replay alongside closed-loop performance to reveal the correction burden.

Exercise 3. When is ILC inappropriate? **Solution.** When trials are not repeatable, phase alignment is unreliable, the environment changes substantially, or updates cannot be safely tested. ILC also cannot infer a general policy from one unrepeated trajectory.

Further Reading

The AffineDrift bibliography entry (Peng et al. 2018) provides the DeepMimic publication reference. Use it for the specific character-skill learning method, while keeping claims about model validity and inverse uniqueness separate. For trajectory optimization and local feedback context, see (Jacobson and Mayne 1970).

Validation Protocol for Learned Policies

A reproducible policy comparison fixes the state estimator, control frequency, action scaling, initial-state distribution, episode termination rules, reward weights, training budget, and random seeds. Report both training and evaluation environments; randomize only the quantities intended to represent uncertainty. Use multiple independent training seeds because deep reinforcement learning can vary substantially with initialization and exploration. A single successful video is a demonstration, not a performance distribution.

For model-based policy learning, compare predicted and actual state transitions on held-out states before trusting long-horizon rollouts. Report one-step error and rollout error separately: a small one-step bias may compound over time. Track the distance of evaluation states from the model’s training distribution. When using hardware or human trials, safety constraints and oversight define the feasible exploration protocol; offline replay cannot reveal all effects of interactive adaptation.

Keep reference tracking reward distinct from task reward. If a character receives both, report each term and ablate them independently. A task reward may appropriately permit deviations that improve the outcome, so a policy can be successful without reproducing the reference exactly. Conversely, a high imitation score may fail a task or violate contact constraints. Interpret the objective according to its construction and show the measurements that support each claim.

Software Architecture and Test Contracts

Scientific Software Is Part of the Model

A fitted trajectory is produced by code, configuration, data, and numerical libraries. Therefore software architecture determines which scientific claims can be audited. The principles TDD, design by contract (DbC), Law of Demeter (LoD), and DRY are useful when applied as engineering constraints rather than slogans. TDD means that a behavior is specified by a failing test before production implementation; DbC makes assumptions and guarantees explicit; LoD limits a component to collaborators it owns or is directly given; DRY centralizes one authoritative definition of each scientific rule. Applied carelessly, these principles can create brittle tests, overabstract engine differences, or hide meaningful duplicate formulations. This chapter gives a concrete contract-first approach.

A robust matching system separates five responsibilities: (1) data and observation semantics; (2) model and parameter schema; (3) engine adapter and discrete dynamics; (4) optimizer; and (5) independent candidate evaluation and receipt writing. The optimizer should not reach through a provider into engine internals, and a receipt writer should not reconstruct scientific semantics by parsing a filename. Components exchange typed values with units, frames, time grids, and provenance.

Contracts at Boundaries

A trajectory contract should state configuration representation and tangent convention, velocity dimensions, monotone timestamps, finite values, coordinate units, body/frame identifiers, and observation mask shape. An engine contract should state accepted model identity, integration step policy, contact formulation, interpolation semantics, initial state, and output dimensions. An optimizer contract should state parameter bounds/transforms, deterministic seed, termination criteria, and whether returned values are feasible. An evaluator contract should state which metrics are independent, which use training data, and the thresholds for promotion.

Preconditions reject invalid requests before expensive simulation: for example, timestamps are strictly increasing, a quaternion is normalized within tolerance, active masks align with observation arrays, and positive inertial parameters remain positive after transform. Postconditions check returned state shape, finiteness, timestamp coverage, and physical feasibility. Invariants include conserved identities for idealized tests, model dimensions, and stable parameter ordering. API checks should raise explicit errors that remain enabled under optimized execution; internal assertions should carry actionable context and should not silently repair data. If a repair such as quaternion normalization is intended, make it an explicit preprocessing stage and record it.

A useful API returns either a typed result or a typed failure. Do not convert solver failure into a zero trajectory or infinite penalty without preserving its cause; optimizers may exploit discontinuous sentinel values. Separate expected infeasibility from programming errors and numerical breakdown. Each failure receipt should identify phase, state/time, solver status, residual, and retry policy.

Law of Demeter and Engine Adapters

LoD discourages deep navigation such as `fit.provider.model.data.qpos[...]`. Such expressions couple fit logic to provider internals and engine storage. Pass an explicit `DynamicsProblem` or protocol with operations such as `step(state, control, dt)`, `observe(state)`, and `linearize(state, control)`. The adapter owns engine-specific details. However, this protocol must not erase scientific differences: include a capability/semantics record declaring supported constraints, contact, differentiability, and solver behavior. An adapter that pretends all engines expose identical contact physics creates false abstraction.

For ownership and testability, the objective can depend on injected functions and immutable arrays. The test should be able to replace an engine with a small deterministic fake to verify orchestration, while a separate engine-specific suite checks physical integration. Avoid test doubles that merely repeat production calculations; they prove call choreography but not the physics.

DRY Without Destroying Provenance

Centralize units conversions, parameter bounds, observation residual definitions, and schema validation when they are scientifically identical. Do not centralize code that only looks similar but has different conventions. For example, two engine adapters may both compute a Jacobian but use different tangent coordinate ordering; sharing the numerical routine is valid only after tests establish the same contract. Explicit conversion functions are better than hidden implicit conversion.

DRY also applies to evidence generation. A single metric implementation should produce both optimizer diagnostics and independent evaluation metrics where definitions are genuinely shared, but fit-time and held-out metrics must not share data selection by accident. Store metric version and input hashes. Avoid duplicating a threshold in code, YAML, and documentation; define one machine-readable policy and render its documentation from that source.

TDD Workflow for Scientific Changes

Start with a failing test that describes an observable behavior. For a contact law, test signs, units, zero-force cases, normal force nonnegativity, limiting behavior, and energy behavior in a one-body setup. For an adapter, test exact mapping of state order and units. For gradients, test a toy analytic fixture and production directional checks. For a receipt, test schema, hashes, and required fields. Then implement the smallest production change that passes; refactor while preserving tests; finally add an independent scientific check where the original tests could share the same bug.

Test layers should be deliberate:

Layer	Contract	Example evidence
Pure function	Algebra and boundary values	Force law sign, transform inverse

Layer	Contract	Example evidence
Adapter	Mapping and units	State round trip, frame labels
Integrator	Discrete dynamics	Analytic oscillator convergence
Optimizer	Decision semantics	Bounds, deterministic termination
End-to-end	Data-to-receipt behavior	Small fixture with hashes
Independent validation	Scientific claim	Fresh engine replay, held-out trial

Property-based tests are effective for invariants across random valid states, but generated states must lie on the configuration manifold and obey physical bounds. Metamorphic tests can check that changing a global coordinate frame transforms markers consistently, that duplicating an observation with correct covariance scaling preserves the estimate, or that reducing timestep converges toward a reference. These do not replace real data validation.

Test/Contract Matrix for a Forward Matching Implementation

Component	Preconditions	Postconditions / invariant	RED test before implementation	Independent check
Data loader	Declared units, unique times, valid mask	No silent unit/frame change	Reject duplicate/nonfinite timestamps	Compare selected rows to source record
Model schema	Known coordinate and actuator names	Stable ordered dimensions	Reject missing/duplicate IDs	Export/import parity against engine
Parameter transform	Bounds and positivity declared	Round-trip within tolerance	Property test inverse transform	Physical unit and range review
Observation map	Frame and marker map explicit	Shape and frame identities preserved	Known pose gives known marker points	Engine-native FK comparison
Contact law	Convention, units, regularization fixed	Feasible force and declared energy behavior	Zero-gap/separation and friction edge cases	Independent contact fixture/replay
Integrator	Step policy, initial state, controls	Exact endpoint coverage, finite outputs	Analytic ODE convergence test	Alternate solver convergence study
Gradient	Smooth region declared	Directional discrepancy bounded	Analytic scalar/vector gradient	Multi-epsilon production directional test
Optimizer	Seed, limits, termination declared	Feasible candidate or typed failure	Bounds and deterministic toy fit	Re-run in fresh process
Evaluator	Holdout split and metric policy fixed	Metrics trace to exact input hashes	Wrong split/hash is rejected	Fresh independent replay

Component	Preconditions	Postconditions / invariant	RED test before implementation	Independent check
Receipt	Required provenance fields	Valid schema and reproducible references	Missing field fails schema test	Hash recomputation and replay

This matrix is a proposed design artifact, not a statement that all rows currently have complete coverage upstream. Existing named evidence includes tests/test_drake_fit_swing_autodiff.py and tests/unit/motion_matching/test_contact_law.py, cited in upstream implementation/evidence. Coverage must be inspected before asserting that a specific behavior is tested.

Contracts for Adaptive Model Dimension

An adaptive model introduces a second layer of contract: the candidate set and selection rule. A dimension proposal must specify a valid embedding from (d) to $(d+1)$, the meaning and prior of the new coordinate, an initialization procedure, and a rule for rejecting unsupported dimensions. For genuinely nested hypotheses, the nested-model property should hold when the new coordinate is zero or inactive, up to numerical tolerance. Alternative nonnested structures instead need a declared comparison and mapping policy. A test should verify that inactive expansion reproduces the lower-dimensional outputs. Selection must be evaluated on held-out data; code should make it impossible for training and test masks to alias silently.

Dimension selection is not a single gradient step. It combines discrete model choice and continuous fitting, so test the selection policy with synthetic data where the true dimension is known, under identifiable and deliberately nonidentifiable conditions. A model should retain a set or posterior over plausible dimensions when evidence is weak. The contract should guard against adding coordinates to compensate for mislabeled units, contact mismatch, or poor marker calibration.

Worked Example: Contract-First Timestep Change

Suppose an integrator currently returns samples at a fixed rate. A request changes the total horizon to 1.0 s at 60 Hz. Before implementation, specify endpoint policy: include both endpoints, giving 61 states at spacing $1/60$ s; or return 60 samples excluding the final endpoint. Write a failing test that asserts exact returned timestamps and controls interpolated at those timestamps. This catches off-by-one differences before they contaminate optimizer comparisons. Then implement time-grid generation once and have all engine adapters consume it. A provenance record stores the grid itself or a hash plus rule, not only the requested rate.

Exercises and Solutions

Exercise 1. A test asserts that private function `_assemble_force_vector` is called once. Is that sufficient? **Solution.** No. It couples the test to implementation and proves little about physical behavior. Prefer a public contract such as equal-and-opposite action/reaction forces, plus a focused unit test if the helper itself is stable and important.

Exercise 2. Two engine adapters contain the same formula but use different velocity tangent conventions. Should they call one shared function? **Solution.** Only after an explicit conversion makes the conventions identical and property tests verify the mapping. Otherwise the

apparent duplication represents different mathematics and should remain separate with clear documentation.

Exercise 3. What RED test guards nested adaptive dimensions? **Solution.** Generate a trajectory from a (d) -coordinate model, embed it in dimension $(d+1)$ with the added coordinate inactive, and assert predictions agree within declared tolerance. Separately synthesize data requiring the new coordinate and verify held-out predictive improvement under a predeclared selection rule.

Further Reading

Use (Geburu et al. 2021) for structured provenance and (Olson 2026a) for local tangent modeling context in the AffineDrift bibliography. The repository's maintained engineering contracts and tests are primary evidence for actual behavior; architectural proposals above require implementation and verification before they become claims.

Keeping Test Suites Scientifically Useful

A test suite should be fast enough that developers run it during every change, while slower engine and replay tests remain available as explicit qualification jobs. Tag tests by cost and claim: unit, adapter, integration, replay, and scientific validation. A skipped test due to missing optional engine dependencies is not a pass; report skipped coverage. Mutation testing can reveal whether assertions detect meaningful defects, especially for signs, units, and bounds. Review flaky tests as evidence of hidden nondeterminism instead of merely increasing retry counts. Keep fixture generation deterministic and version the generated artifact so changes to a generator do not silently redefine the baseline.

References and Verification Ledger

How References Were Reviewed

This publication distinguishes bibliographic identity, successful page retrieval, full-text inspection, and validation of a scientific claim. These are separate checks. A DOI string or a search-index record does not establish that a live publisher page is reachable, and successful retrieval does not establish that a paper supports every application claim.

Before the network stop, the literature helper retrieved the official Moco overview and archived repository README. It also obtained indexed PubMed, publisher, and official documentation records. Some publisher opens returned access errors, while some PubMed and PMC opens returned shell pages or access challenges. Those sources support bounded metadata and abstract-level statements rather than unrestricted full-text review.

After GitHub returned an API quota error, the repository's network policy required stopping all network activity. New external URLs therefore remain pending live validation. Local repository links were checked against the audited checkout and are pinned to its commit. The machine-readable link ledger and source manifest in this directory preserve the distinction.

Core Mechanics and Trajectory Optimization

The existing AffineDrift bibliography supplies mechanics and numerical foundations. Featherstone develops rigid-body algorithms; Betts treats nonlinear programming for optimal control and estimation; Kelly presents practical direct collocation (Featherstone 2008; Betts 2010; Kelly 2017). These references establish method context rather than success on the current golf models.

- Rigid Body Dynamics Algorithms: canonical mechanics reference; external endpoint validation pending in this session.
- Kelly's Direct Collocation Tutorial: method tutorial; external endpoint validation pending.
- Porsa, Lin, and Pandy's OpenSim Direct-Methods Study: indexed metadata and abstract reviewed before the network stop; independent full-text and endpoint validation incomplete (Porsa et al. 2016).

The comparative study should not be used as a universal solver ranking. Performance depends on model size, transcription, constraints, derivatives, tolerances, and initialization. This book derives the relevant numerical distinctions and asks for fair comparisons on identical tasks.

OpenSim and Muscle-Driven Matching

OpenSim provides the musculoskeletal framework; Moco supplies direct-collocation optimal control within that framework (Seth et al. 2018; Dembia et al. 2020). Computed Muscle Control is an earlier forward tracking method with static optimization and feedforward/feedback components (Thelen et al. 2003; Thelen and Anderson 2006).

Reference	Stable Link	Review Status	Scope Used Here
Thelen, Anderson, and Delp, 2003	DOI	Indexed metadata and method description; full-text retrieval incomplete	Computed Muscle Control architecture
Thelen and Anderson, 2006	PubMed	Indexed metadata and abstract; publisher open blocked	Walking simulation and residual-elimination context
Dembia and Colleagues, 2020	PLOS	Indexed PLOS/PubMed metadata; full-text review incomplete	Moco formulation and supported problem families
OpenSim Moco Overview	Official Site	Successfully opened by helper before network stop	Software overview
Archived Moco Repository	Official Repository	README successfully opened by helper before network stop	Development-location notice; API versions still require review
MocoTrack Documentation	Version 1.1 Documentation	Indexed official documentation; endpoint validation incomplete	Tracking semantics; historical API reference

The muscle chapter also links broader reviews with bounded scope. References whose authors, dates, or publication details were uncertain in helper notes were excluded from the formal bibliography. They should be added only after authoritative metadata and relevant text are inspected.

Identification, Model Selection, and Learned Dynamics

Hermann and Krener provide classical nonlinear observability context; Green supplies reversible-jump Bayesian model determination; MacKay develops evidence-based regularization (Hermann and Krener 1977; Green 1995; MacKay 1992). The URLs for these records remain pending live endpoint validation.

Sparse discovery selects terms from a declared library rather than directly discovering anatomy (Brunton et al. 2016). Neural ODE and latent ODE methods define continuous learned states whose interpretation depends on their observation and initialization maps (Chen et al. 2018; Rubanova et al. 2019). Structured Hamiltonian and Lagrangian networks encode useful assumptions within their supported physical regimes (Greydanus et al. 2019; Cranmer et al. 2020; Lutter et al. 2019).

- SINDy: stable DOI; helper reported blocked automated publisher access.

- Neural Ordinary Differential Equations: stable preprint identifier; current endpoint validation pending.
- Latent ODEs for Irregularly-Sampled Time Series: stable preprint identifier; current endpoint validation pending.
- Hamiltonian Neural Networks: existing bibliography identity; current endpoint validation pending.
- Lagrangian Neural Networks: existing bibliography identity; current endpoint validation pending.
- Profile Likelihood and Practical Identifiability: supplementary method link; full metadata and endpoint validation pending.

No source is cited as proving the adaptive design proposed here. The synthesis uses these method families to formulate a testable engineering program and explicitly preserves identifiability limitations.

Repository Evidence

The immutable source anchor is UpstreamDrift Revision d7c7d07. The inventory and historical chapters link individual modules, tests, handoffs, and receipts at this revision. Their local targets were inspected without additional network requests.

The source audit separates older neural stubs from modern shared implementations. It also preserves conflicts between model-card promotion prose and diagnostic artifact receipts. Resolving those conflicts requires checkpoint hashes, original logs, native replay evidence, and repeatable benchmark commands. This manuscript does not manufacture that missing evidence.

Stored experimental values are reported as historical model results. They were not rerun in this session and do not establish empirical human validity. A stage marked not evaluated, unsupported, diagnostic, or rejected retains that status even when neighboring stages passed.

Reference Maintenance Procedure

For each new source, record its exact identity, stable URL, access date, retrieved content level, relevant sections, and the claim it supports. Separate inaccessible full text from a broken endpoint. Publisher access restrictions can coexist with correct metadata and a valid DOI.

When live access resumes, validate every unique external URL and preserve redirects and failures in the ledger. Then inspect the relevant primary text for substantive method claims. Do not replace an inaccessible primary source with an unsupported paraphrase from a search result.

Before publication, run citation-key, bibliography metadata, internal-link, title, syntax, render, and visual checks. Review all equations in the rendered output. The reference list below is generated from the configured bibliography, while direct software and receipt links preserve version-specific evidence.

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Benchmarks, Reproducibility, and Publication Maintenance

A Benchmark Is a Protocol, Not a Score

Forward matching benchmarks compare a complete inference procedure: data preparation, model, numerical integration, optimization, and independent evaluation. A table containing only marker RMSE and runtime is not enough to determine what was compared. Two runs can have the same engine label yet differ in contact law, interpolation, actuator mapping, initialization, control regularization, solver tolerance, and observation mask. A benchmark record must make these choices explicit and bind them to exact artifacts.

Separate three questions. **Optimization performance:** did the method find a lower objective under a stated training protocol? **Predictive performance:** does the candidate predict held-out kinematics or forces? **Physical qualification:** does an independent forward replay satisfy dynamics, contacts, actuator limits, and event timing? A candidate may succeed at one and fail another. Publication should label each claim directly. Video is useful for gross visual review but cannot establish torque validity, hidden contact forces, or continuous-time accuracy.

Reproducible Protocol

A run manifest should include source revision and dirty-tree status; code and configuration hashes; input file hashes; model asset hashes; engine and library versions; hardware; random seeds; thread counts; deterministic flags; time grid; integration method and tolerances; contact law and parameters; objective terms and weights; optimizer settings; parameter bounds; start state; observation/holdout split; and termination reason. Store the actual parameter vector and candidate state/control arrays, not only a path to transient optimizer memory.

Reproducibility has levels. **Artifact reproducibility** means the exact files and code can be identified. **Computational reproducibility** means rerunning produces the same or tolerance-equivalent outputs. **Scientific reproducibility** means an independent implementation or evaluator supports the conclusion. A hash proves identity, not correctness; a deterministic rerun proves consistency, not model validity. Receipts need all three types of evidence without conflating them.

Use immutable raw data and derived-data manifests. Record all filters, gap filling, coordinate transforms, resampling, and rejected samples. A useful transformation entry records code version, parameters, input/output hashes, and reason. If a marker gap is interpolated, the estimator's assumptions and uncertainty belong with the derived series. Leakage can occur when filtering or normalization uses the full trial before a train/test split; split by trial/person/session at the level appropriate to the generalization claim.

Benchmark Design

Use layered benchmark suites. A synthetic suite has known dynamics and parameters, enabling recovery and identifiability checks. A canonical small physical fixture exercises contacts and engine adapters. A representative motion trial evaluates realistic complexity. A held-out cohort evaluates generalization. Every suite should report which parameters are identifiable under its excitation and sensors.

Compare methods with matched conditions: same data and splits, equivalent objectives, comparable compute budgets, and identical endpoint conventions. If a method requires a different objective, provide both its native score and a common evaluation score. Repeat stochastic methods across seeds and report distributions with confidence intervals. Do not select the best seed as the headline number. Record failures and timeouts; excluding failed fits biases comparisons.

Metrics should include position and orientation errors with physical units, normalized residuals, temporal alignment, peak and RMS error, event-time error, constraint violations, energy residuals, actuator effort, optimizer evaluations, wall and CPU time, memory, and convergence status. Report performance by body region and motion phase as well as pooled metrics. A low whole-body mean can hide a major wrist or contact failure. Include parameter uncertainty and sensitivity where parameter interpretation is claimed.

A fair baseline suite should include kinematic playback (as a lower-bound fit reference, clearly labeled non-dynamic), inverse-dynamics or trajectory-optimization baselines where applicable, and multiple forward engines when the claim concerns engine robustness. For cross-engine parity, compare same-state forward kinematics first, then dynamics under matched forces and contact; do not attribute all discrepancy to the optimizer if the mechanics differ.

Ablations and Uncertainty

Ablation removes one method component while preserving all else: contact model, residual weighting, adaptive dimension, regularization, control knots, or initialization. An ablation is informative only if the changed component is isolated and the objective is comparable. For adaptive dimension, compare fixed dimension against the selector using held-out predictive performance, parameter stability, and complexity. Also include an oracle dimension synthetic case to distinguish selector limitations from data nonidentifiability.

Uncertainty can be approximated through bootstrap over trials or participants, profile likelihood, posterior sampling, or local curvature, depending on assumptions. Hessian-based intervals are unreliable near bounds, in multimodal fits, and across contact mode changes. Report that limitation. A confidence interval on marker error is not an uncertainty interval on mass or friction. Parameter uncertainty and prediction uncertainty are different quantities.

Repository Evidence and Interpretation

The upstream repository stores many matched candidates beneath evidence/matched, with receipts, candidate arrays, replay results, parity reports, and visual artifacts. This is a strong foundation for traceable comparison, but the existence of an evidence directory does not imply every candidate is equally qualified. For example, evidence/matched/driver_full_mujoco_replay/README.md explicitly reports a large marker change when the integrator is refined and documents the contact law used. It also notes that an earlier merged source receipt omits information such as armature, contact configuration, interpolation, and other replay details. This is direct evidence that provenance completeness and numerical sensitivity remain review dimensions.

AGENT_HANDOFF.md distinguishes kinematic preview artifacts from forward-dynamics qualification and records open gaps such as all-axis torque coverage, moving contact, continuous interpolation, and independent replay for certain workstreams. That is important scientific discipline: selected pose validity or a visually plausible animation is not sufficient to claim a complete forward match. The book should link to exact receipts and handoff notes when making case-specific claims and date-stamp evolving evidence.

A proposed benchmark receipt schema should make missing fields explicit rather than silently defaulting. It should include a qualification level and gates passed, plus warnings for absent force measurements, event mismatch, unverified interpolation, or unqualified solver exit. The report site should render a comparison table from receipts, not a hand-edited result table. This is a proposal; do not infer that upstream has this unified publication pipeline unless implementation is found and audited.

Publication Lifecycle

A research review and textbook should separate durable derivations from changing repository findings. Put core equations and methods in chapters; place implementation status in a dated case study or evidence index. Every source claim should have a local repository citation or stable external reference. Include access/verification date for dynamic web references. A citation validator can confirm links resolve, but a live hyperlink's resolution does not prove that it supports the sentence.

Maintain a claim ledger: claim text, evidence type, source location, version/date, independent reviewer, and status (supported, proposed, open, superseded). Keep hypotheses visibly labeled. When an implementation changes, update receipts and affected claims rather than rewriting history. Preserve previous candidate hashes and mark superseded records. A release should pin source revision, bibliography version, input hashes, and render toolchain.

A publication quality gate can check Quarto rendering, code examples, internal anchors, bibliography keys, HTTP/local link resolution, title conventions, and stale evidence dates. Scientific review should separately inspect equations, assumptions, units, and whether empirical conclusions follow from the data. Formatting automation cannot replace scientific peer review. Code snippets in a textbook should either execute in a pinned environment or be labeled pseudocode; do not imply unrun code is tested.

Proposed Adaptive-Dimension Benchmark

To evaluate an adaptive model that learns its dimensions, construct synthetic trials with known (d), varying excitation richness and observation noise. Include confounders: marker dropout, timing offset, contact-law mismatch, and unmodeled compliance. Measure dimension-selection accuracy, false expansion rate, held-out prediction, parameter coverage, runtime, and calibration of model probabilities. Test at least two kinds of split: held-out segment within a trial for interpolation and held-out trial/person for generalization. A geometric dimension chosen using one trial may reflect anatomy, calibration, or prior assumptions and needs independent validation; a population model should model participant variation separately.

Define a compute-matched comparison against (a) a fixed low-dimensional model, (b) a fixed high-dimensional model with shrinkage, and (c) a model-selection baseline. Freeze the selection rule before looking at test results. Evaluate robustness across engine/integrator refinements: a dimension that appears only at coarse timestep is likely compensating numerical error. Likewise, if a dimension vanishes when contact parameters are calibrated, it may represent contact confounding. Report selection uncertainty and candidate dimensions, not just

the winner.

Worked Benchmark Interpretation

In this illustrative synthetic comparison, suppose method A has 8 mm training RMSE and method B has 9 mm, but on held-out trials A has 15 mm and B has 10 mm. A also requires 3x runtime and has 4% contact-constraint violation; B has 0.2%. The training score favors A, while predictive and physical evidence favors B. The right conclusion is not a single ranking: A optimizes the training objective more closely, B generalizes better and satisfies the stated contact gate. The intended use determines which method is preferable.

Exercises and Solutions

Exercise 1. Why is an input hash insufficient for reproducibility? **Solution.** It identifies bytes but not the transformation, runtime, solver settings, randomness, or interpretation. Reproduction requires the full run manifest and recoverable environment.

Exercise 2. A candidate has excellent marker RMSE and no measured ground reaction force. Can it claim accurate contact force? **Solution.** No. Kinematics constrain force only through assumed dynamics and parameters; force may be nonidentifiable. Report it as a model estimate and seek independent force measurement or sensitivity/uncertainty analysis.

Exercise 3. An adaptive coordinate improves fit only under one integrator tolerance. What is the next test? **Solution.** Repeat with converged timestep/tolerance studies and identical contact settings, then use held-out data. Until stable under numerical refinement, treat the apparent dimension as unresolved discretization compensation.

Further Reading

The AffineDrift bibliography includes (Gebu et al. 2021) on dataset documentation, (Olson 2026c) on residual-aware approximation, and (Olson 2026b) on hybrid events. The upstream receipts and replay notes cited above serve as evolving case evidence; they should be rechecked at each publication release.

Statistical Reporting and Stopping Rules

Choose sample size and evaluation units before comparing methods. Frames within one trial are serially correlated and are not independent replicates; confidence intervals based on frame count can be dramatically too narrow. Bootstrap at the trial or participant level, preserving the hierarchy of repeated trials. For paired method comparisons, resample paired units so each method sees the same subjects. Report the number of participants, trials, and valid frames after masking, not only the frame total.

Predeclare the promotion rule. For example, a candidate may require held-out normalized error below a threshold, no constraint violation beyond tolerance, successful fresh-process replay, and complete receipt validation. If gates are tuned after seeing a candidate, label the result exploratory. A run that reaches an iteration limit is not converged merely because its final score is favorable. Record stopping reason, gradient norm, constraint residual, and whether the result satisfies KKT or other solver criteria.

Benchmark maintenance should include a small smoke suite that runs on every supported environment and a heavier scheduled suite for full replay. The smoke test checks interfaces and

numerical sanity, not publication claims. The release suite should preserve golden artifacts with versioned tolerances; avoid bitwise equality across hardware where floating-point reduction order differs. Use physical tolerances grounded in measurement precision and solver convergence, and retain the observed error distribution so tolerances do not expand without review.

Forward Surrogates and Multimodal Inverse Proposals

Two Different Learned Maps

A forward surrogate approximates the expensive simulator’s map from inputs to observations. An inverse proposal approximates a distribution of inputs compatible with observations. These maps have different training labels, uncertainty, and validation requirements, even when both produce candidates for the same native fit.

Write the exact forward observation map as $F(\vartheta; \xi)$, where ϑ contains the fitted controls and parameters. The context ξ binds model identity, initial state, observation convention, and time grid. A surrogate $\widehat{F}_\phi$ is trained on native evaluations:

$$\min_{\phi} \sum_{j \in \mathcal{T}} \|W_j(\widehat{F}_\phi(\vartheta_j; \xi_j) - F(\vartheta_j; \xi_j))\|^2.$$

This supervised objective teaches simulator approximation. It does not by itself teach matching to measured motion, identify human controls, or certify feasible dynamics outside the training domain. The native simulator used for labels must already have a declared mechanical meaning.

At query time, surrogate inversion solves

$$\hat{\vartheta} \in \arg \min_{\vartheta \in \Theta} \|W(\widehat{F}_\phi(\vartheta; \xi) - y)\|^2 + R(\vartheta).$$

The optimizer may discover inputs where surrogate error is much larger than on ordinary test examples. Average supervised accuracy is therefore insufficient. Validation must include the optimized proposals and their exact native rollouts.

An inverse proposal instead learns $p_\omega(\vartheta \mid y, \xi)$. It generates starts for native search or samples a compatible family. A point estimate may be fast, but an underdetermined matching task generally requires alternatives and uncertainty rather than one unique inverse.

Why Averaging Can Produce an Invalid Inverse

Consider the illustrative map $y = \vartheta^2$ and an observation $y = 1$. Two exact inverse solutions are $\vartheta = -1$ and $\vartheta = 1$. A squared-error regressor trained equally on both labels can return their conditional mean, zero. Its prediction is between the valid solutions, yet forward evaluation gives zero rather than the observed one.

The example captures a general problem in articulated matching. Different posture branches, torque allocations, or contact strategies can explain similar outputs. Their mean may violate closure, bounds, or the target itself. Label diversity is useful only if the model preserves the modes rather than averages them away.

A deterministic regressor can still be a valuable initializer when training labels follow a consistent selection policy. That policy might select minimum effort within a particular posture branch. The model then learns the selected inverse convention, not the entire physical inverse relation.

Conditional Variational Proposals

A conditional variational autoencoder introduces a latent proposal variable z . Let $q_\varphi(z \mid \vartheta, y, \xi)$ be a training posterior, $p_\eta(z \mid y, \xi)$ a conditional prior, and $p_\omega(\vartheta \mid z, y, \xi)$ a decoder. A conditional evidence lower bound is

$$\mathcal{L} = \mathbb{E}_{q_\varphi} [\log p_\omega(\vartheta \mid z, y, \xi)] - \text{KL}(q_\varphi(z \mid \vartheta, y, \xi) \parallel p_\eta(z \mid y, \xi)).$$

At inference, sample from the conditional prior and decode candidate parameters. True coefficient labels are unavailable then, so the training posterior must not be used as if it were the inference prior. This distinction belongs in both tests and model reproduction documentation.

A latent dimension controls proposal capacity; it is not a discovered mechanical degree of freedom. The decoder can ignore the latent variable, producing posterior collapse and nearly identical proposals. Diversity diagnostics should evaluate decoded physical behaviors as well as latent statistics.

Bounded decoder transforms enforce coefficient bounds, but they do not establish forward feasibility. A coefficient box may still permit extreme torques between timestamps, unstable contact, or invalid parameter coupling. Saturating transforms can also suppress gradients near bounds and distort uncertainty.

Train and evaluate the proposal on explicit masks when observations are partial. A missing orientation should not be filled with an identity quaternion and interpreted as measured. The conditioning representation must distinguish measured, derived, and unobserved channels, with uncertainty available to the model.

Other Proposal Families

Mixture-density models represent several conditional parameter modes directly. They can be useful when alternatives have a manageable structure, but selecting the number of components and preventing unused components requires care. Mixture components should be validated through native outputs rather than interpreted as anatomical categories.

Conditional normalizing flows map a reference distribution to a parameter distribution through invertible transforms. They support density evaluation, but the parameter manifold may contain constraints or disconnected branches that are difficult to express with one unconstrained map. A flow over a redundant coordinate vector can assign probability to invalid candidates.

Diffusion models generate candidates through repeated denoising steps. Their flexibility can represent complex distributions, but sampling cost and physical constraint enforcement matter for interactive matching. Native reranking remains necessary, and a large sample count should be included in query cost.

Retrieval is a strong baseline. A library of native feasible episodes can propose nearby controls or trajectories without training an inverse network. Interpolation between retrieved episodes needs its own feasibility checks; valid endpoints do not guarantee that their averaged parameters remain valid.

Forward Surrogate Architectures

A sequence surrogate predicts a complete observation history from a control vector and context. Time-conditioned networks evaluate outputs at requested times, while encoder-decoder architectures may emit a fixed grid. A fixed-grid model cannot silently be reused at a different duration or sampling interval.

UpstreamDrift’s shared FiLM surrogate conditions a time-dependent network on control coefficients. It emits club-related trajectories and auxiliary joint coordinates. Its quaternion output is normalized and sign-canonicalized, which supports representation consistency but introduces care around the sign boundary.

A per-step surrogate approximates transitions or derivatives instead of complete sequences. It can support new horizons through repeated integration, but errors accumulate and state distribution shifts under rollout. Teacher-forced one-step validation is therefore distinct from uninterrupted prediction.

A physical residual surrogate preserves a known dynamics backbone and learns a correction. This can reduce the approximation task, but the correction may absorb calibration errors, missing contact, or wrong topology. Report its magnitude relative to the mechanical terms and test behavior under perturbations.

Value Accuracy Does Not Guarantee Gradient Accuracy

Surrogate inversion relies on derivatives with respect to fitted variables. A model can have small value error and poor derivatives. For the illustrative scalar function $F(\vartheta) = 0$, define

$$\widehat{F}(\vartheta) = \epsilon \sin(\vartheta/\epsilon^2).$$

Its value error is bounded by ϵ , but its derivative has amplitude $1/\epsilon$. As ϵ decreases, values improve while gradients become more misleading. This is a mathematical counterexample, not an empirical claim about a particular network.

A gradient-aware surrogate benchmark should compare directional derivatives where native sensitivities are available. Where they are unavailable, use a step sweep of native finite differences in smooth regions. Contact transitions need mode-aware treatment; a central difference across two different event sequences does not certify one smooth derivative.

The inference optimizer can exploit gradient errors even if ordinary held-out prediction is good. Trust regions, periodic native evaluations, and final native polishing constrain that exploitation. Their thresholds and evaluation frequency should be tuned on selection data and frozen before test evaluation.

A Safe Acceleration Pipeline

The proposed pipeline begins with a checked observation contract and compatible checkpoint identity. The model generates a bounded set of candidates, preserving proposal modes and

rejection reasons. Each candidate receives a native evaluation before it can be presented as a dynamics match.

Native polishing then refines promising starts using the authoritative objective. A common budget allows fair comparison with direct multistart and retrieval. The final candidate is replayed from its original initial state without learned state injection or shooting-node resets.

Acceptance evaluates the same physical and observation gates used by the classical baseline. A surrogate score is diagnostic evidence for search; it is not an acceptance score. If every proposal fails, the result should say so and invoke a declared fallback rather than quietly return the least invalid sample.

Out-of-domain checks should compare model fingerprints, parameter ranges, time grids, and observation masks. Ensemble disagreement can indicate risk but is not guaranteed to detect every error. The model should abstain on unsupported dimensions and contacts even when its neural output is finite.

Native Dataset Design

A training episode binds control representation, initial state, model parameters, integration settings, outputs, and failure status. Training data should preserve unsuccessful simulations with explicit labels rather than silently removing them. Their inclusion allows learning failure boundaries and honest acceptance-rate accounting.

Split episodes by generating family, model identity, subject, or capture according to the claim. Nearby controls and trajectories can leak across random splits, producing deceptively easy interpolation. Keep the final test population independent of hyperparameter and architecture selection.

The dataset should sample the physical domain intentionally. Uniform coefficient sampling can generate mostly implausible or numerically unstable swings and leave realistic motion poorly covered. A staged sampling policy can combine broad exploration, feasible baselines, local perturbations, and adversarial proposals, with all selection rules recorded.

Adaptive geometry expands the input context and invalidates older checkpoints unless they were trained for that domain. A surrogate trained on one fixed morphology is not automatically differentiable with respect to segment scales. Parameter conditioning must be present in both labels and architecture before those derivatives have meaning.

Measuring Useful Benefit

Let C_{train} be the one-time native generation and training cost, C_{direct} the baseline query cost, and $C_{\text{accelerated}}$ the total accelerated query cost. If per-query savings are positive, the illustrative amortization threshold is

$$N_{\text{break-even}} = \frac{C_{\text{train}}}{C_{\text{direct}} - C_{\text{accelerated}}}.$$

If the denominator is nonpositive, no positive query count amortizes training under these assumptions. The accelerated cost includes inference, rejected proposals, native polish, replay, and loading overhead. Comparing only neural inference with a complete baseline fit is unfair.

Measure accepted-quality rate as well as speed. A faster method that returns fewer physically accepted matches can be inferior for the intended task. Compare distributions across targets, difficult strata, and independent seeds; report tail latency and unsupported cases.

Stored upstream promotion prose and diagnostic receipts are not consistent enough to establish one universal speedup. The historical chapter preserves that discrepancy. A future promotion should be computed from immutable benchmark and checkpoint evidence rather than a static claim in a model card.

Relation to Adaptive Dimension Learning

A proposal model can help explore alternative structures, but structure identity must remain explicit. A shared network needs dimension-aware inputs and outputs, or a registered model-specific decoder. Padding coordinates with zeros without a mask can confuse absence with a measured zero state.

Use learned proposals after establishing which dimensions are physically meaningful and observationally supported. A model that proposes many different lengths may represent uncertainty, poor training, or genuine equivalence; native fit and identifiability analysis must distinguish these cases.

The proposal's latent dimension, network width, and output parameter count should be reported separately from the plant's mechanical dimension. Improved prediction does not establish that any of those neural capacity choices correspond to anatomy.

Exercises and Solution Sketches

1. **Conditional Mean Failure.** Why does the mean of two exact inverse solutions to $y = \vartheta^2$ fail for $y = 1$? **Solution:** the inverse set is $\{-1, 1\}$ and its mean is zero, outside that set.
2. **Gradient Accuracy.** Compute the derivative of $\epsilon \sin(\vartheta/\epsilon^2)$. **Solution:** it is $\epsilon^{-1} \cos(\vartheta/\epsilon^2)$, so uniformly small value error need not imply small derivative error.
3. **Amortization.** What happens if accelerated total query cost exceeds the baseline? **Solution:** training cannot be amortized through those queries under the declared cost model.
4. **Dimension Identity.** Can a fixed-morphology surrogate certify gradients with respect to a morphology it never receives? **Solution:** no; the parameter conditioning and training domain do not define that derivative.
5. **Qualification.** A CVAE produces bounded coefficients with good surrogate agreement. What is still required? **Solution:** exact native fitting or evaluation, uninterrupted replay, physical gates, and independent prediction where claimed.